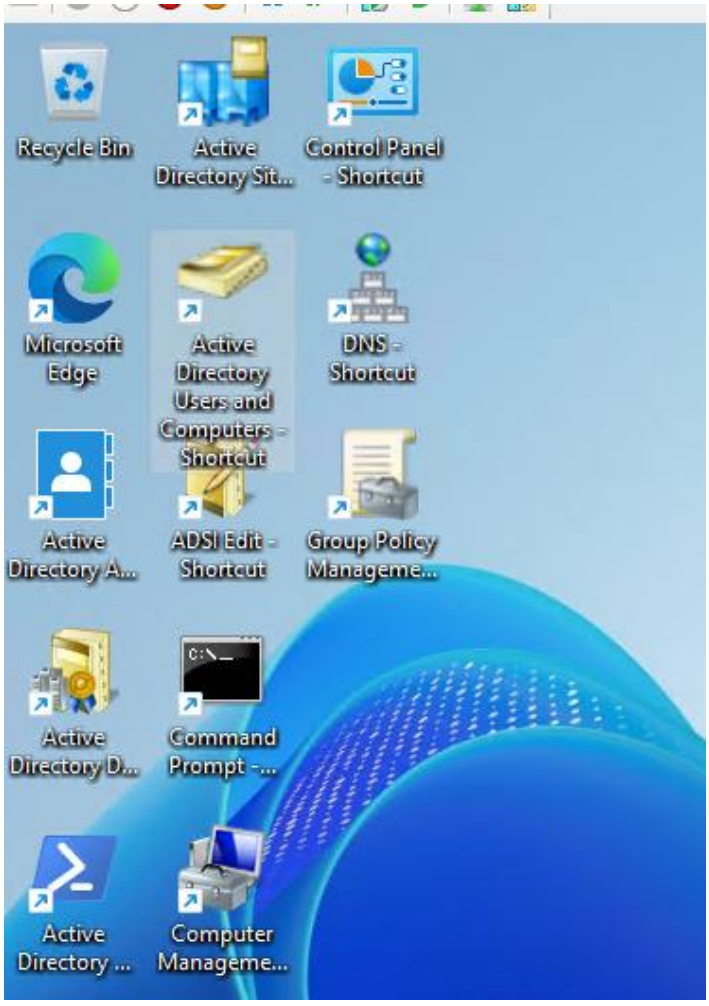
How DNSSEC Works (The Cryptographic Layer)

- **Data Origin Authentication:** DNSSEC makes sure that the DNS response received by the client genuinely came from the authoritative DNS server, and not from a rogue, spoofing source.
- **Data Integrity Verification:** It verifies that the digital data was not altered or tampered with, while it was in transit across the network.
- **Public Key Cryptography:** The DNS server digitally signs every resource record set (RRSet) using a private Zone Signing Key (ZSK), which creates something called an RRSIG record. The client's validation engine then fetches the public key, via a DNSKEY record, to dynamically decode and verify the signature. If the signature matches, the information is trusted; if the validation fails, the connection is blocked.

In simple terms, DNSSEC is like a digital seal, applied to DNS. It protects the IP information sent from the server, so that a hacker cannot secretly change it while it is in transit.

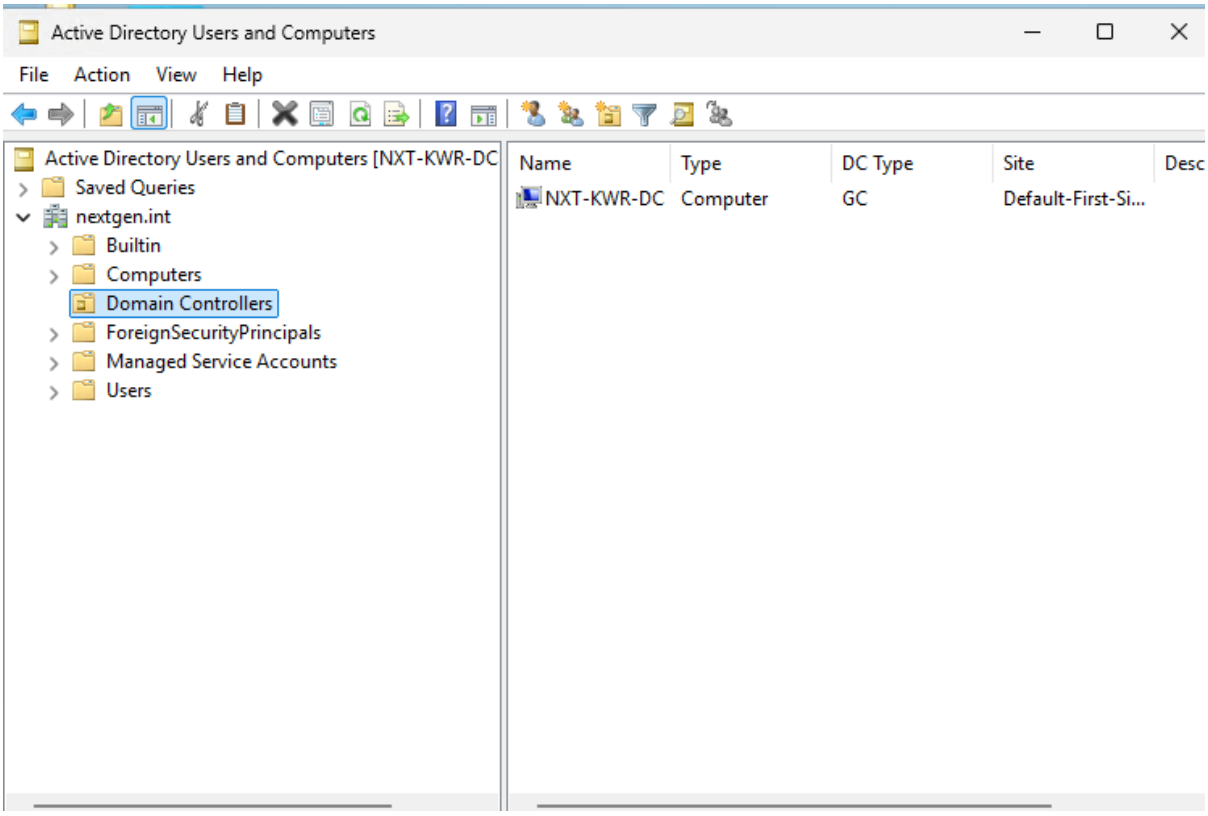
Active Directory Users and Computers

Within a company, we can manage computers, users, and organizational structures, using the Active Directory Users and Computers tool.



This tool lets us do things such as: create users, manage computers, create groups and OUs, and reset passwords.

Organizational Units (OUs)



The containers shown above, such as Builtin, Computers, and Domain Controllers, are default OUs that already exist when Active Directory is first installed. We generally do not change these particular default containers.

An Organizational Unit (OU) is a subdivision within an Active Directory domain, which can contain users, groups, computers, and even other OUs inside it.

1. Logical Organization: Structuring the Company

Imagine our company has 500 employees. If we placed every single one of them inside the same default "Users" folder, it would become almost impossible to tell who belongs to HR, and who belongs to Sales.

Solution: We create separate OUs, such as Sales, HR, IT, and Finance, and then move the respective employees and computers into their correct OUs. This makes the whole network much easier to manage.

2. Group Policy Scoping: Applying Different Rules

This is actually the biggest and most important benefit of creating OUs. We are able to link specific rules, known as Group Policies, so that they apply only to one particular OU.

For example, we could create a rule for the Sales OU, to "Block USB Drives (Pen Drives)." At the same time, we could create a completely different rule for the IT OU, to "Allow installing new software on computers." Doing this allows us to target only the necessary department, without mixing up rules, or causing confusion across the entire domain.

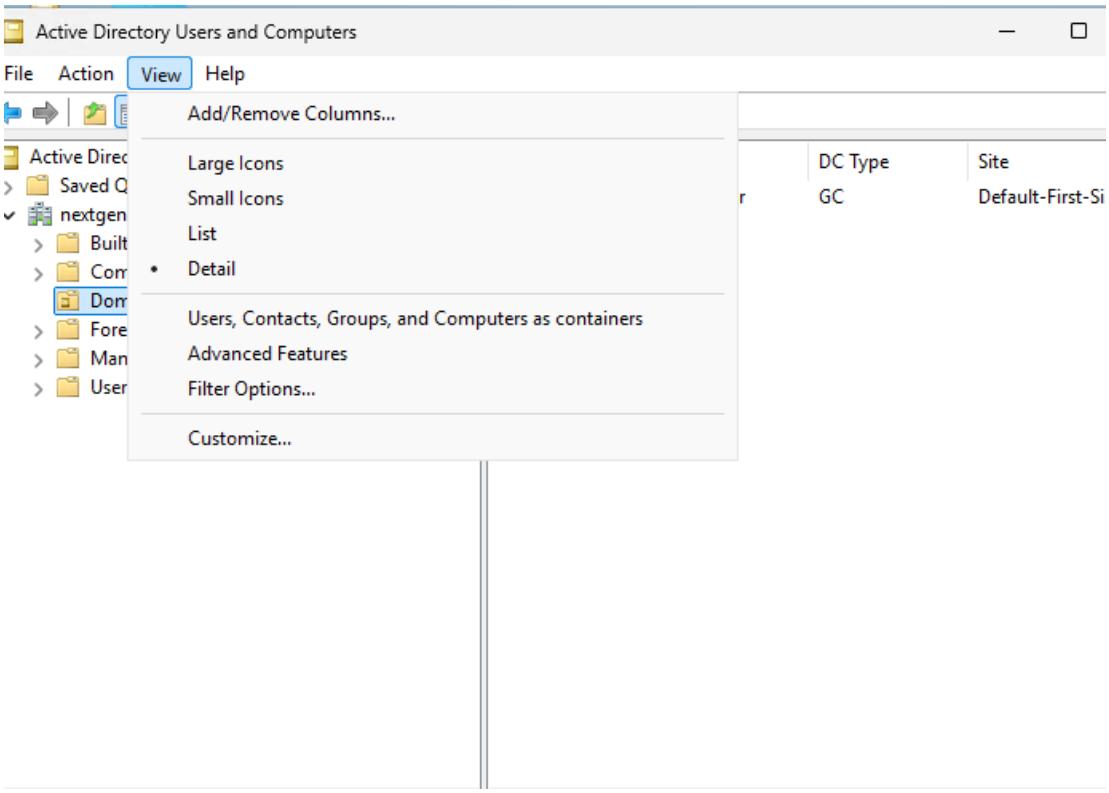
3. Delegation of Administrative Control: Sharing the Workload

As the main Administrator (also known as the Domain Admin), we will not always have enough time to personally handle every small task ourselves.

Solution: We can hand over, or "delegate," the management power of the Sales OU, to a supervisor within the Sales department. This would allow that supervisor to reset passwords for Sales employees, whenever they forget them, without needing to bother the main IT administrator. However, that same supervisor would have absolutely no permission to touch or change anything inside the HR or IT OUs.

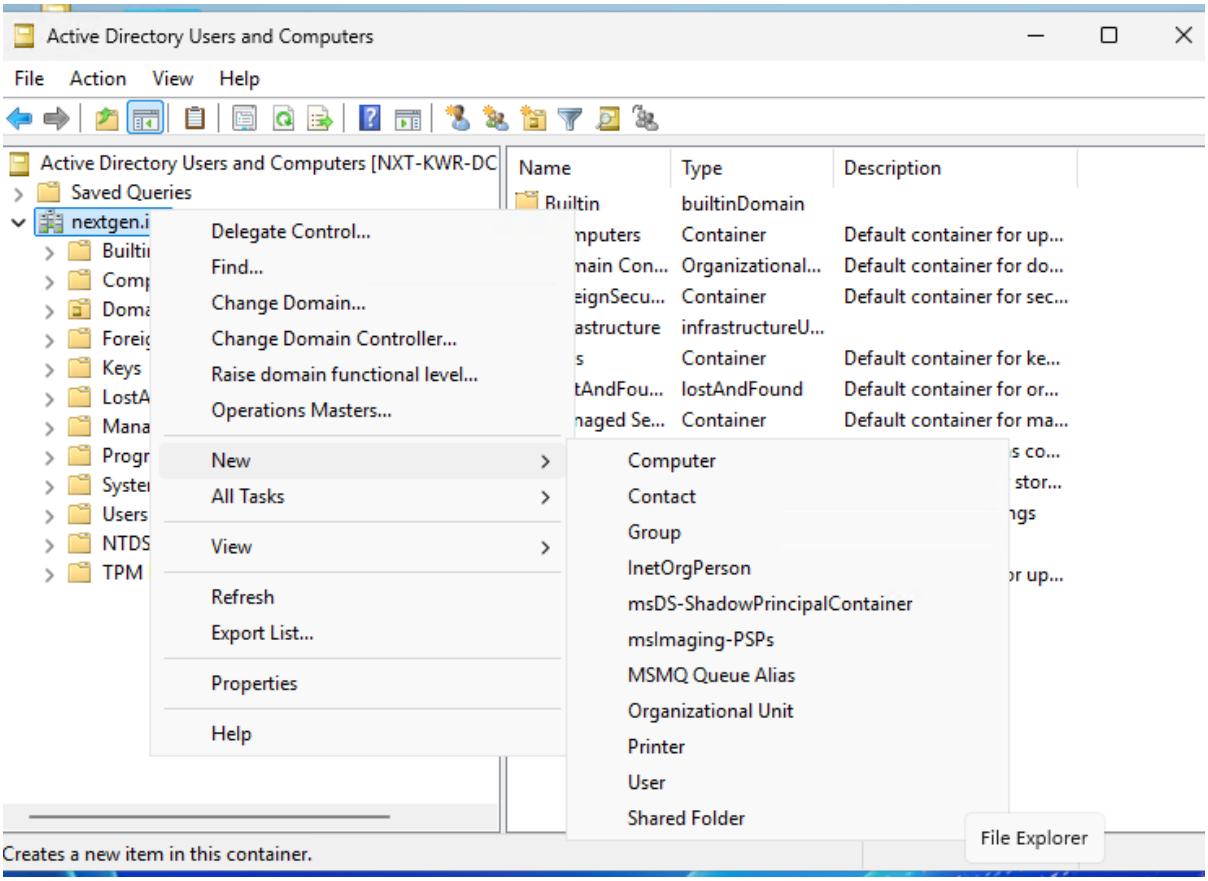
Enabling Advanced Features

We can access more organizational unit options, and additional built-in containers, by enabling the "Advanced Features" option.

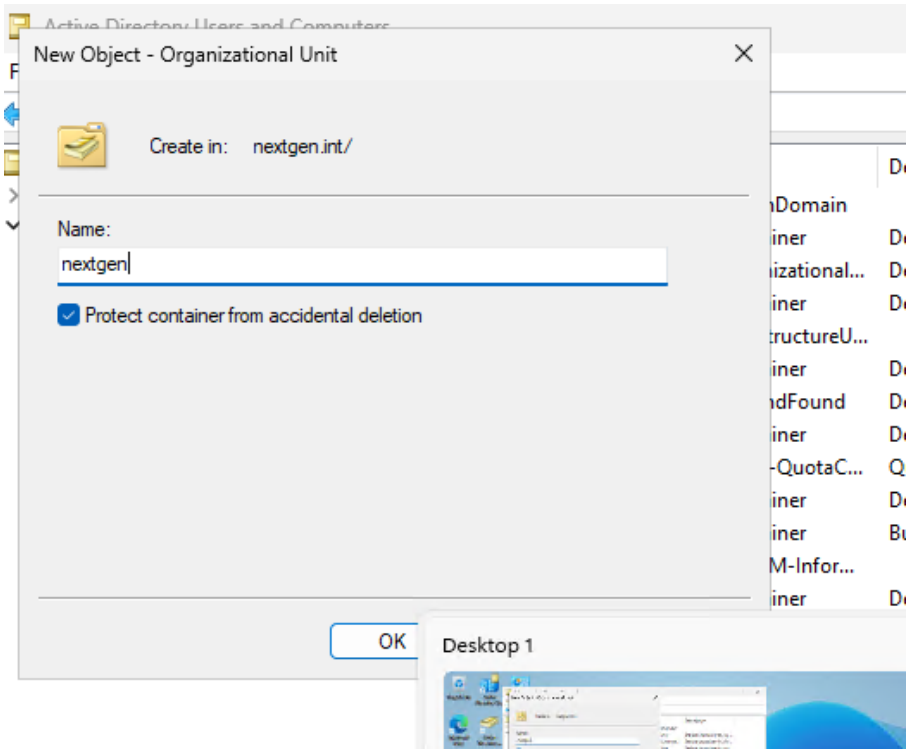


This option is found under the View menu, inside Active Directory Users and Computers.

Creating an Organizational Unit



To create a new OU, we right-click on our domain, select New, and then choose "Organizational Unit" from the list.

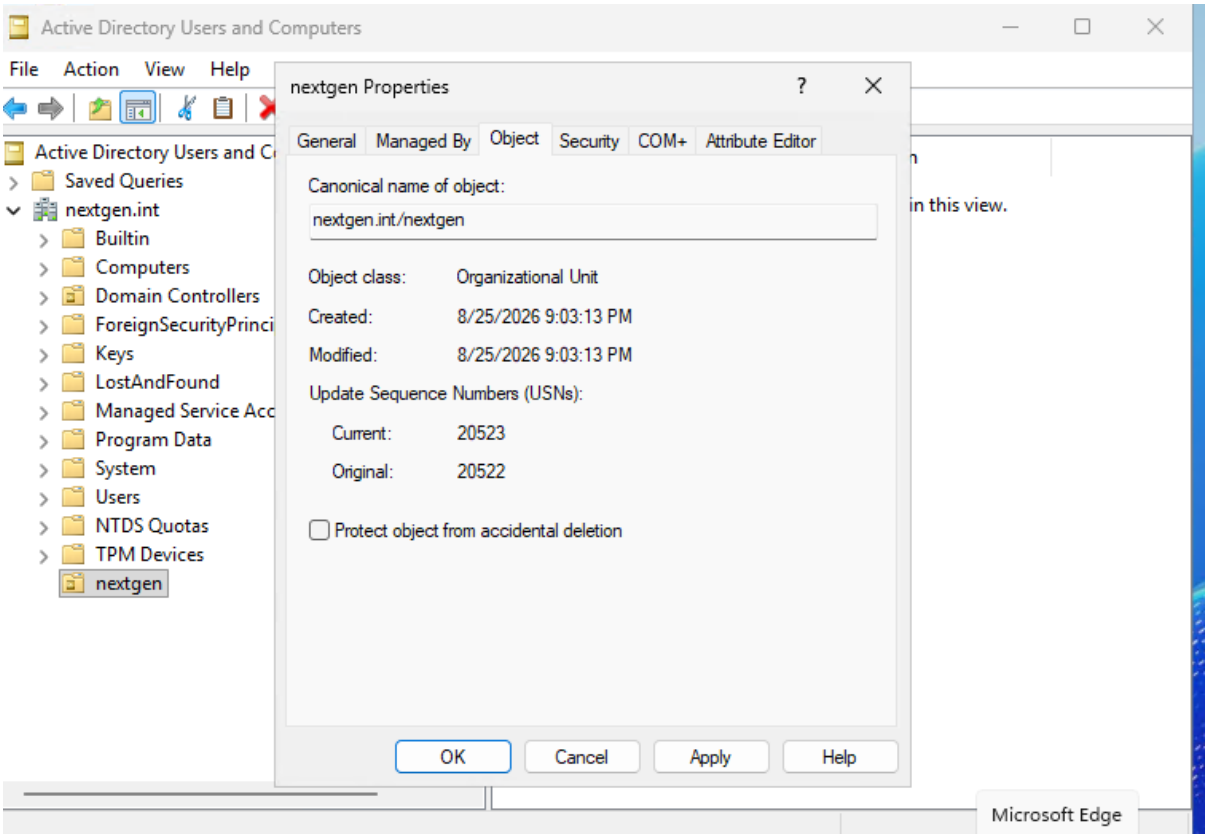


We are then asked to type a name for our new OU. Notice the checkbox labeled "Protect container from accidental deletion." It is very important to keep this checkbox ticked, since this locks the object down, preventing it from being accidentally deleted through standard deletion actions.

How to Delete a Protected OU

Suppose we now want to delete an OU (for example, one named "nextgen"), but we find that we cannot delete it automatically, because we had previously ticked "Protect container from accidental deletion." So, how do we actually delete it?

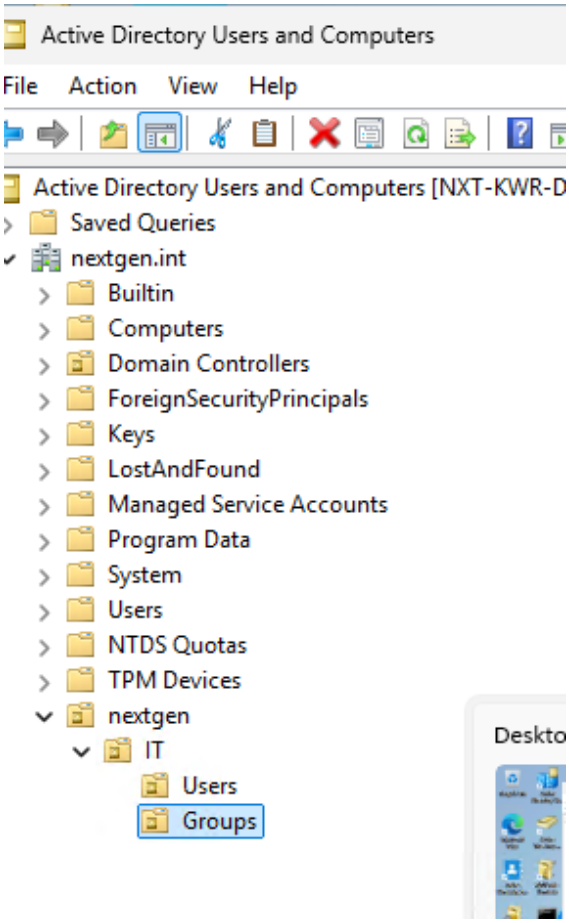
Here is the path to follow: First, enable "Advanced Features" from the View menu (as shown earlier). Then, right-click on the nextgen OU, and select Properties. Go to the "Object" tab, and untick "Protect container from accidental deletion." Once this box is unticked, we are now able to delete the OU normally.



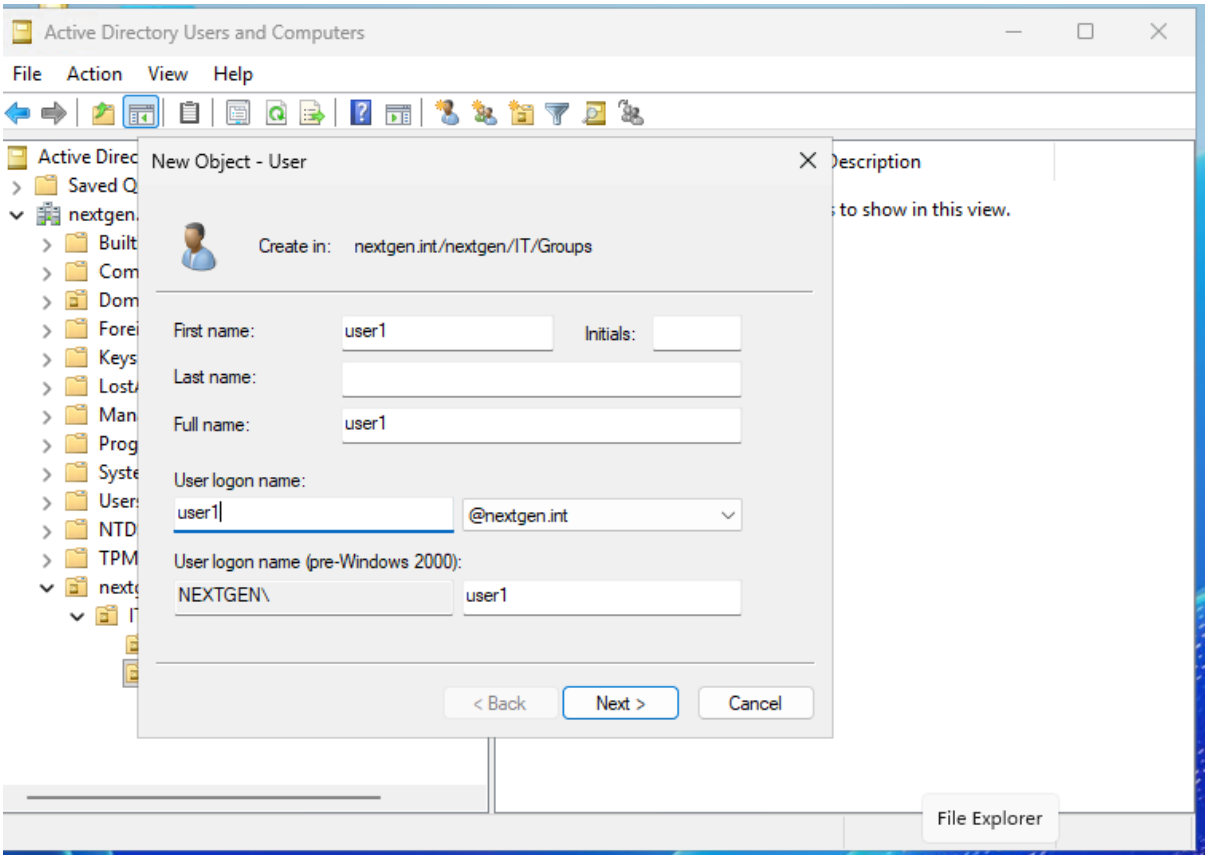
The screenshot above shows the Object tab, where this checkbox can be found and unticked.

Adding Users and Groups

Now that we have created some OUs, let's look at how to add users, and how to add those users into a group.

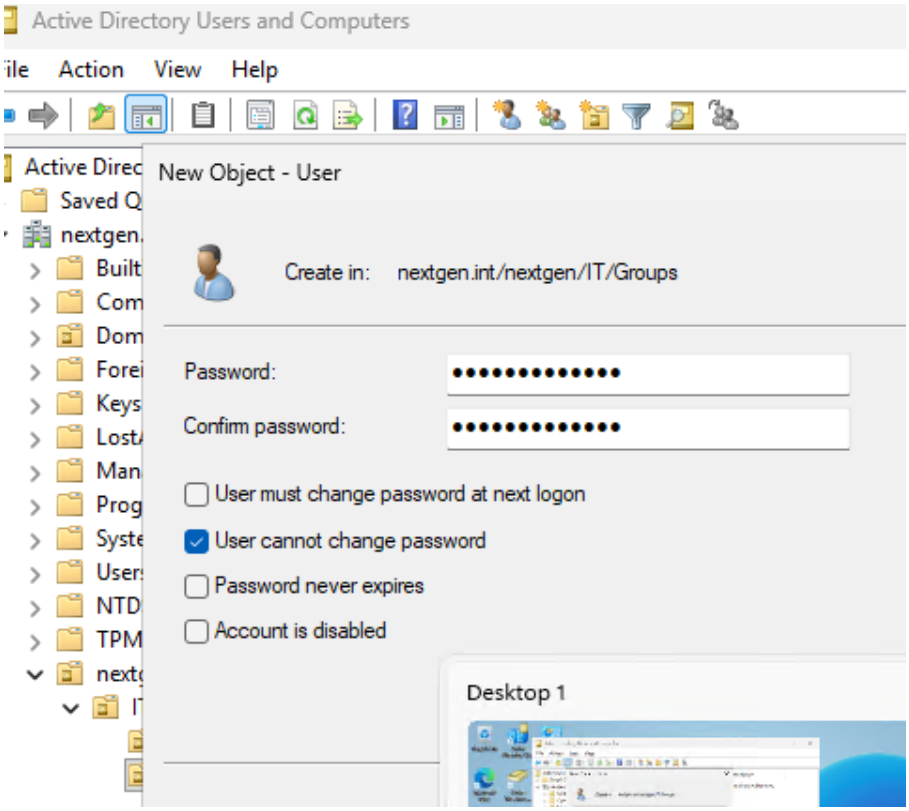


In this example, we have created an OU structure: nextgen → IT, which then contains two further OUs inside it, named Users and Groups.



To create a new user, we right-click on the correct OU (in this case, nextgen.int/nextgen/IT/Groups), select New, and then User. We then fill in the person's First name, Last name, Full name, and their User logon name, which will be combined with the domain name (for example, user1@nextgen.int).

Next, we click Next, and we are taken to the password options screen.

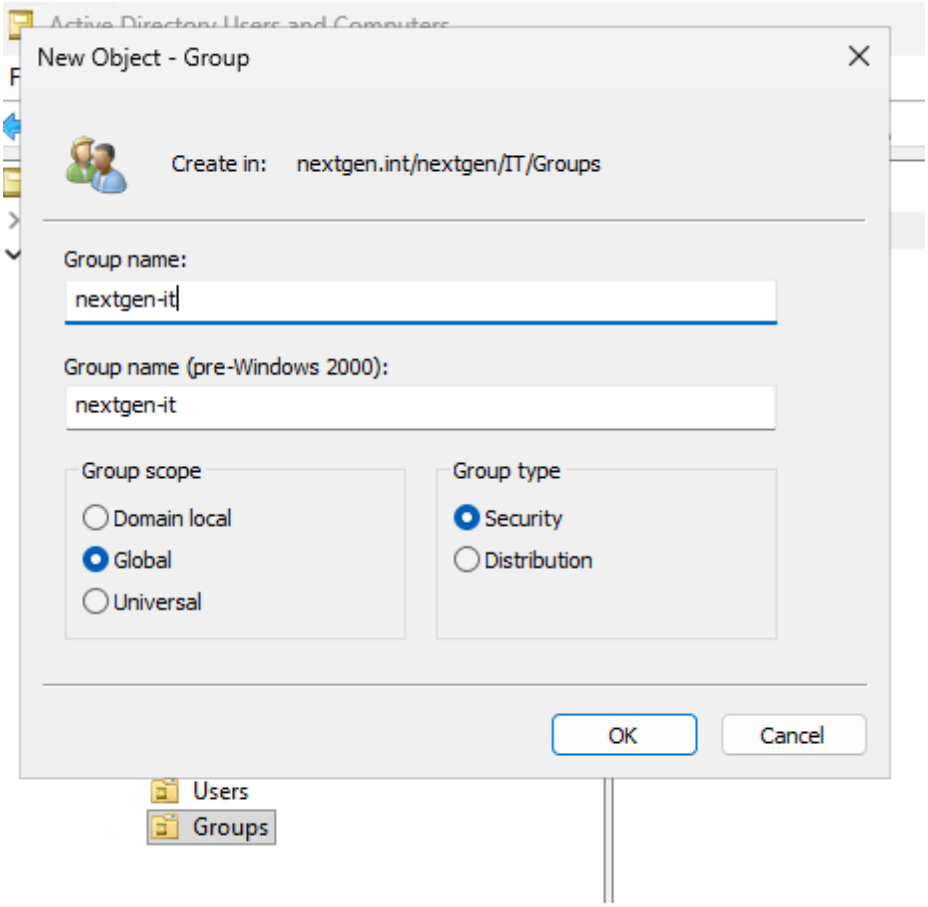


On this screen, we set a password for the new user, and we also choose from a few important options:

- User must change password at next logon: Forces the user to set their own new password, the very first time they log in.
- User cannot change password: Prevents the user from ever changing their own password themselves.
- Password never expires: Stops the password from automatically expiring after a set period of time.
- Account is disabled: Creates the account, but keeps it disabled, until we are ready to activate it.

How to Add a User to a Group

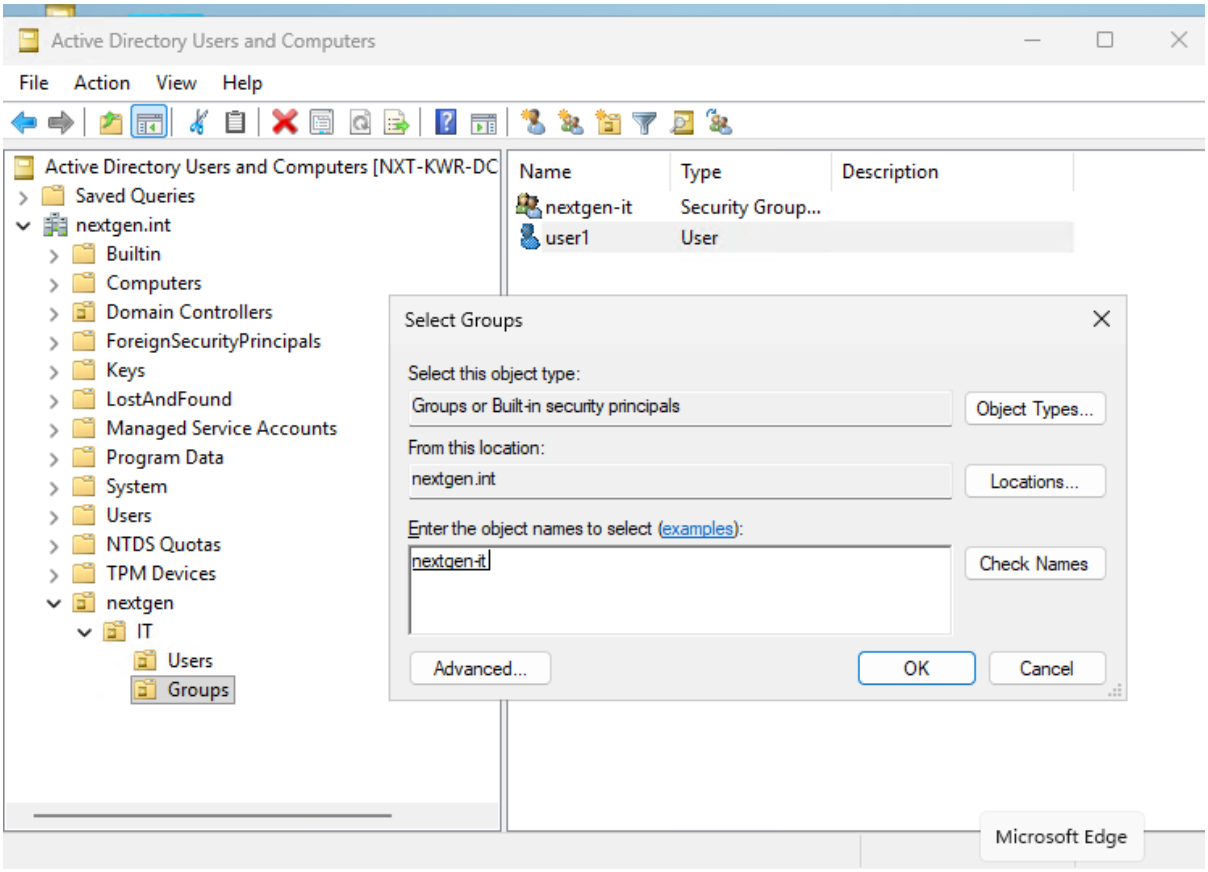
Before we can add a user into a group, we must first create the group itself.



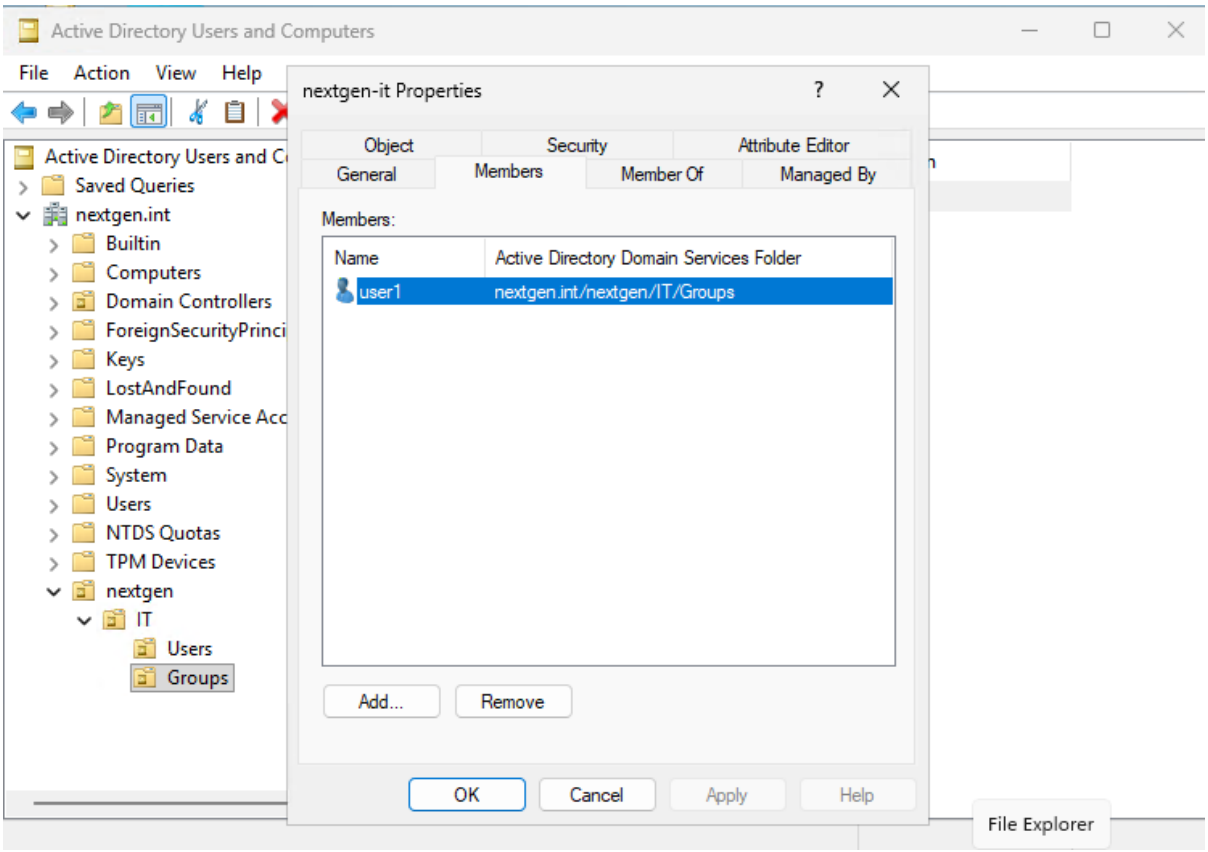
When creating a new group, we are asked to choose a Group Type:

- **Security (the option we selected):** This is the type we use about 95% of the time. We use this type to give permissions to shared folders on the network, or to apply Group Policy rules. For example: "Allow only the members of this nextgen-it group to read the shared folder inside the IT Server Room."
- **Distribution:** This type cannot be used to give any security permissions at all. It is strictly used for sending emails. For example, in large companies using Microsoft Exchange or Outlook, if we want to send a single email to the entire IT team at once, we would create a Distribution group like this.

Once the group is created (in our example, named "nextgen-it"), we can add our user into it.



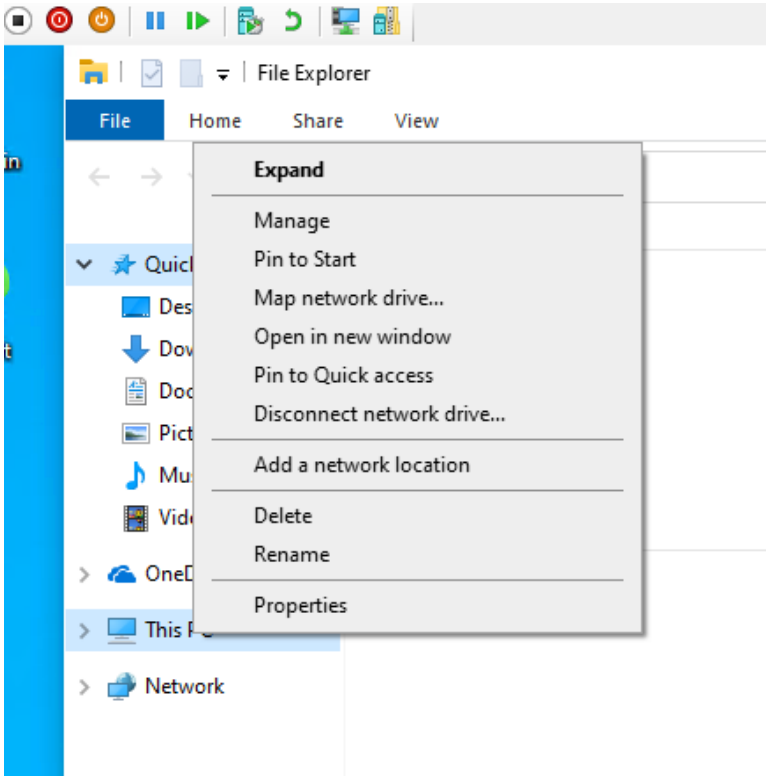
We do this by opening the group's properties, going to the Members tab, and clicking Add, then typing in the name of the user we want to add (in this case, user1).

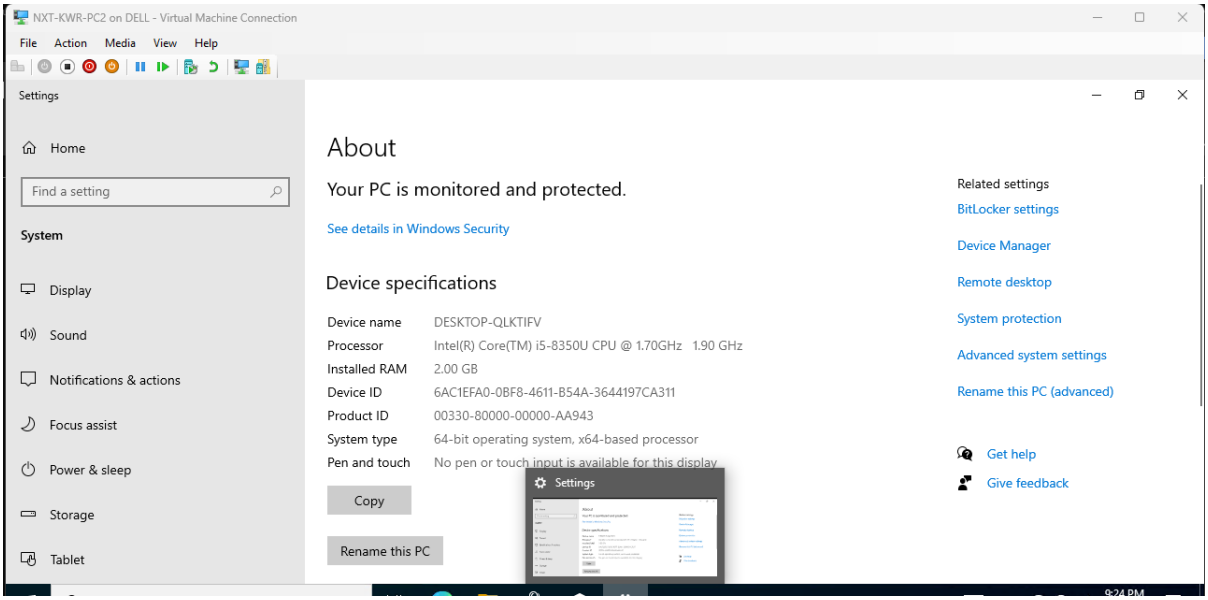


The screenshot above confirms that user1 has now been successfully added as a member of the nextgen-it group.

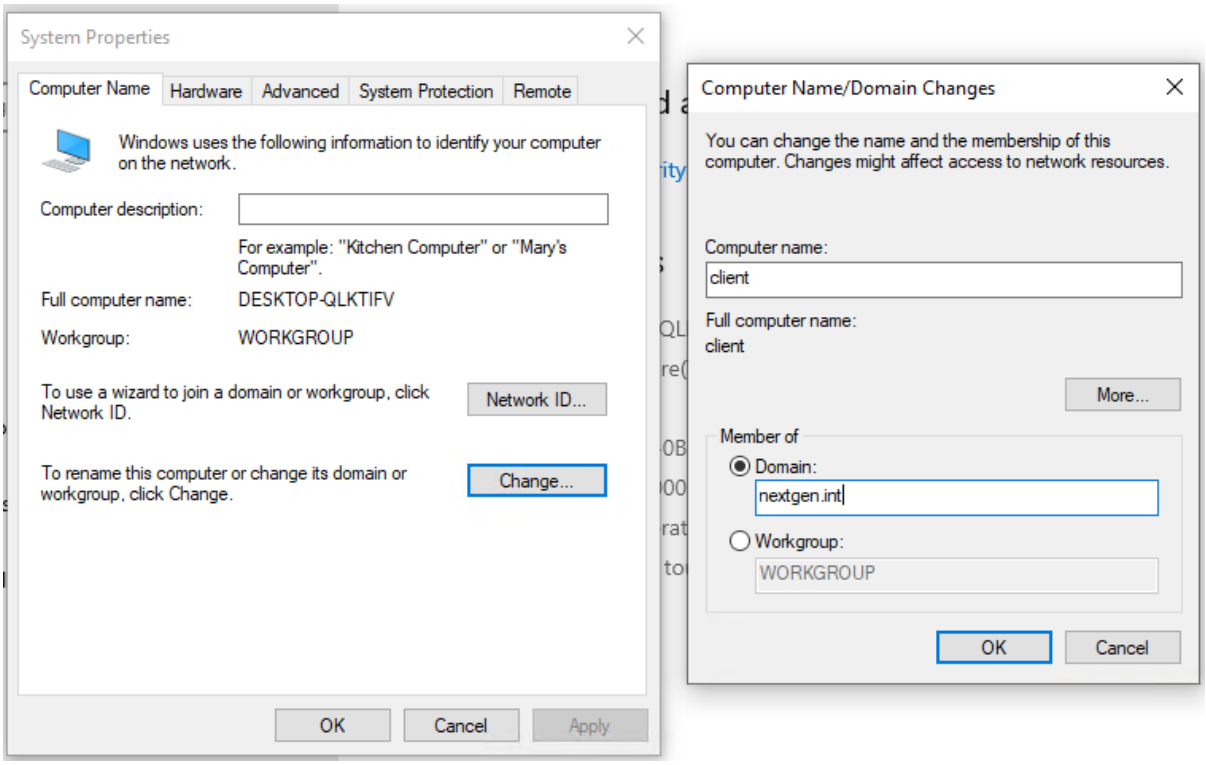
Client PC Domain Join

Now, we are ready to join our client computer to the domain, meaning we are adding the client PC to be managed by our Active Directory Domain Controller.





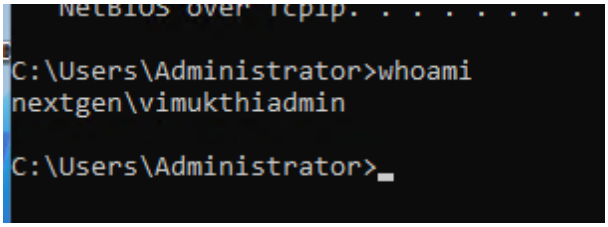
First, we open the client computer's System settings, to check its current computer name and workgroup status.



To actually join the domain, we go to System Properties, select the Computer Name tab, and click the "Change" button. In the window that appears, we select the "Domain" option, and type in the name of our AD server's domain, for example, nextgen.int.

After clicking OK, a login box will appear, asking us for the AD username and password. As a best practice, when typing the username, we should type it in the format "nextgen\next-gen-ad", meaning we include the domain's NetBIOS name ("nextgen") before the username itself. This clearly shows which specific administrator account is being used to join this domain.

If we ever want to check exactly which username we are currently logged in as, on the AD server, we can open Command Prompt, and type the command "whoami".



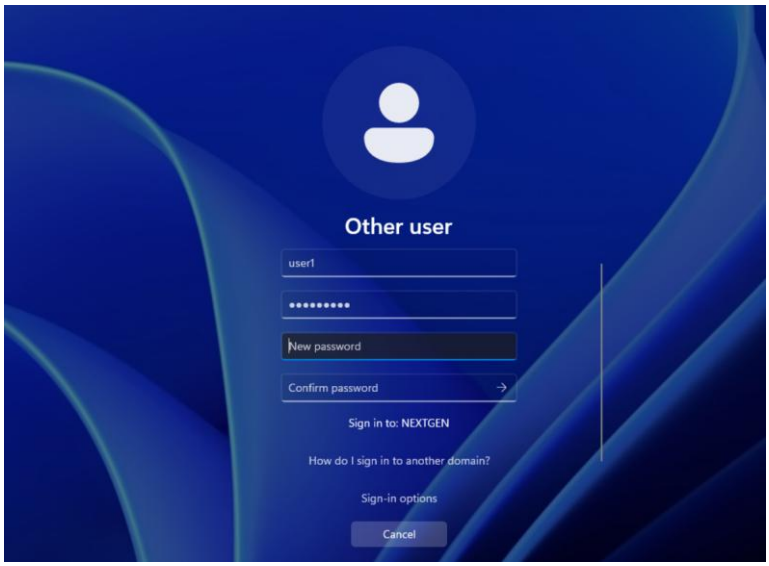
As we can see in the example above, the command confirms the currently logged-in user as "nextgen\vimukthiadmin", clearly showing both the domain name and the specific username together.

After entering the correct username and password, and restarting the computer, the client machine will now be successfully joined to the AD server.

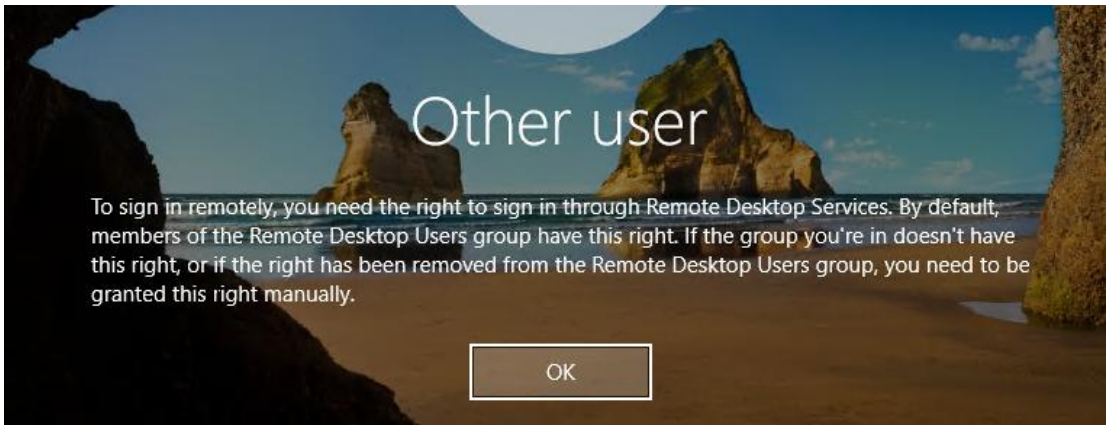
Important note: Every single time we need to join a domain, we should always use the credentials of an Administrator account for this process. We should never use the username and password of a regular, standard user account for joining a computer to the domain.

Logging In With a Domain User Account

Now, we can try signing in using the user account we created earlier inside Active Directory Users and Computers, called "user1". Since we selected the option "User must change password at next logon" when we originally created this account, the user will be asked to change their password, the very first time they successfully connect to the AD domain.



As shown above, when user1 logs in for the very first time, they are prompted to enter a brand-new password, and confirm it, before they are allowed to continue logging in.



Understanding This Error, and How to Fix It

Sometimes, when trying to log in as a standard domain user (such as user1) inside a Hyper-V virtual machine window, we might see an error message like the one shown above.

Here is the explanation: We have opened this client workstation inside Hyper-V. Hyper-V includes a feature known as Enhanced Session Mode. When this mode is active, Hyper-V actually operates very similarly to a Remote Desktop Protocol (RDP) connection.

While Administrator accounts naturally have global privileges to log into any network endpoint remotely, a standard domain user (such as user1) does not automatically inherit these same remote access rights, by default. This particular security restriction is exactly why Windows displayed this error message, when we attempted to log in using the standard user account instead of an administrator account.

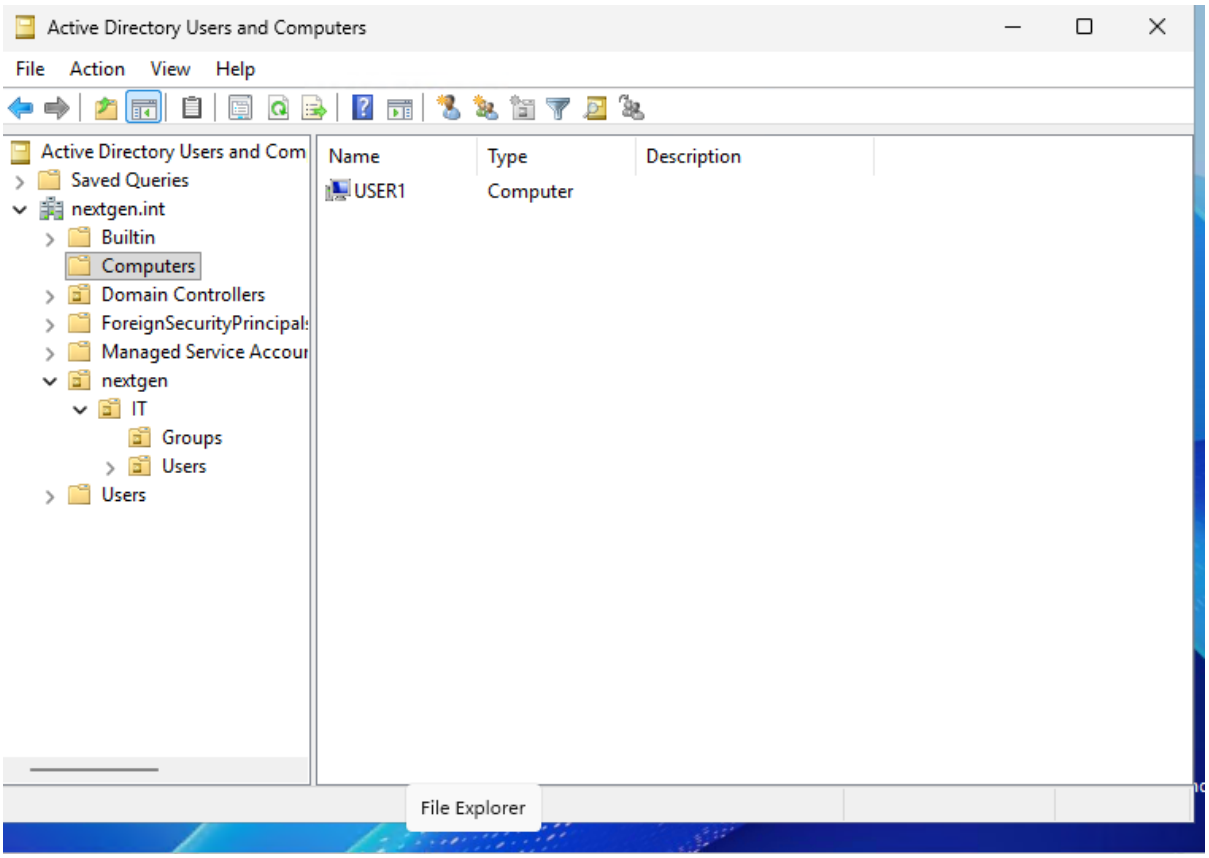
How This Error Fixes Itself (Mechanics of the Auto-Fix)

This particular issue can often be resolved simply by closing the connection window, and reconnecting again. Here is what actually happens behind the scenes, when we do this:

- **Session State Disruption:** Closing the connection window terminates the active Remote Desktop Protocol (RDP) pipeline that was running over something called the VMBus, which is what powers Enhanced Session Mode.
- **Basic Session Initialization:** Upon reconnecting, if the host operating system cannot immediately re-establish the enhanced user profile connection, it automatically falls back to something called Basic Session Mode, which behaves more like a raw, direct video output.
- **Security Context Shift:** Because Basic Session Mode behaves like directly accessing the physical hardware console (similar to a Local Interactive Logon), rather than a network-based remote terminal session, the more restrictive Remote Desktop security rules no longer apply in the same way. This allows standard domain user accounts to successfully pass through the authentication process.

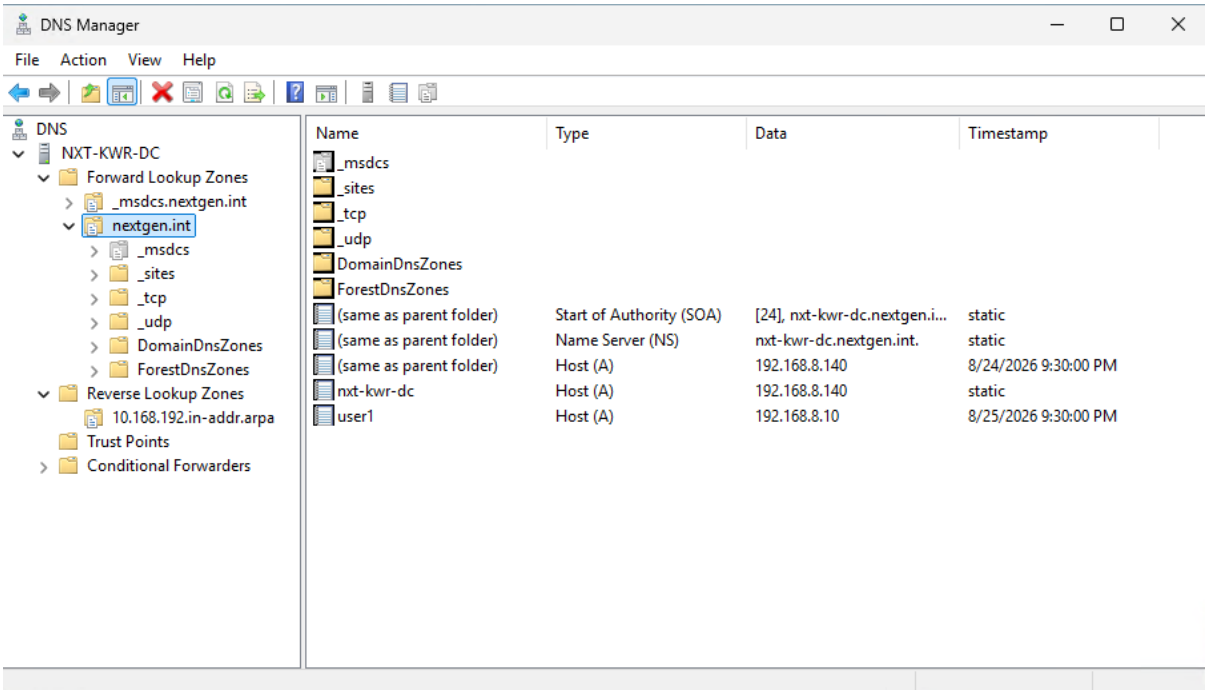
Confirming the Client Appears in Active Directory

Once a user's computer successfully joins the Domain Controller, we can usually find that computer listed inside the Computers section, within Active Directory Users and Computers.



As shown above, our client computer now appears under the "Computers" container, confirming that it has successfully joined the nextgen.int domain.

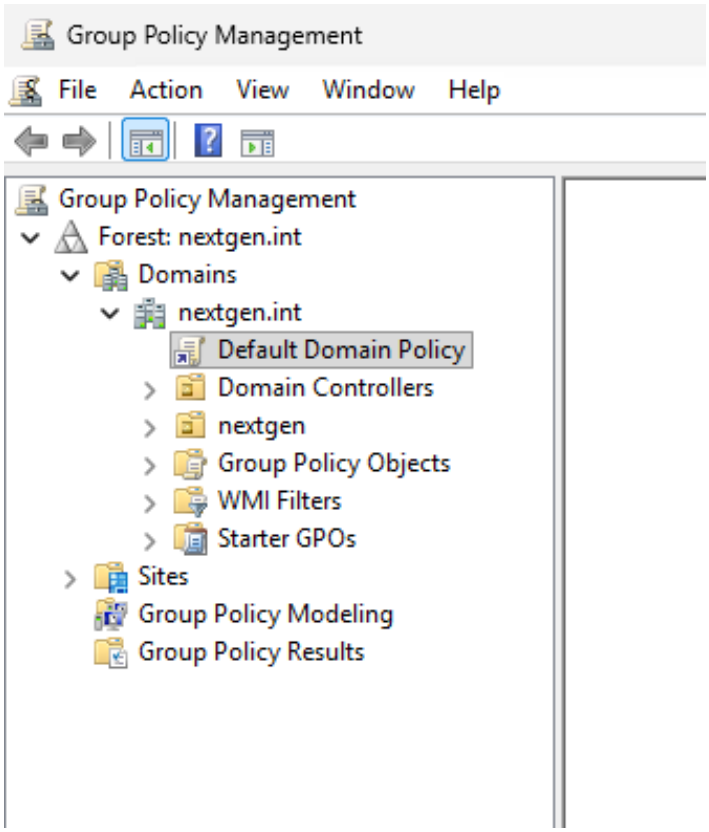
Checking the Automatically Created DNS Record



Finally, if we check back inside DNS Manager, we can see that a brand-new Host (A) record, named "user1", has now been automatically created, with an IP address of 192.168.8.10. This confirms that our earlier explanation of the Automated Registration Flow worked exactly as expected: once the client authenticated successfully against the domain, it automatically registered its own hostname and IP address into DNS, with absolutely no manual entry required from the administrator.

Group Policy Management: Assigning a Desktop Wallpaper Policy

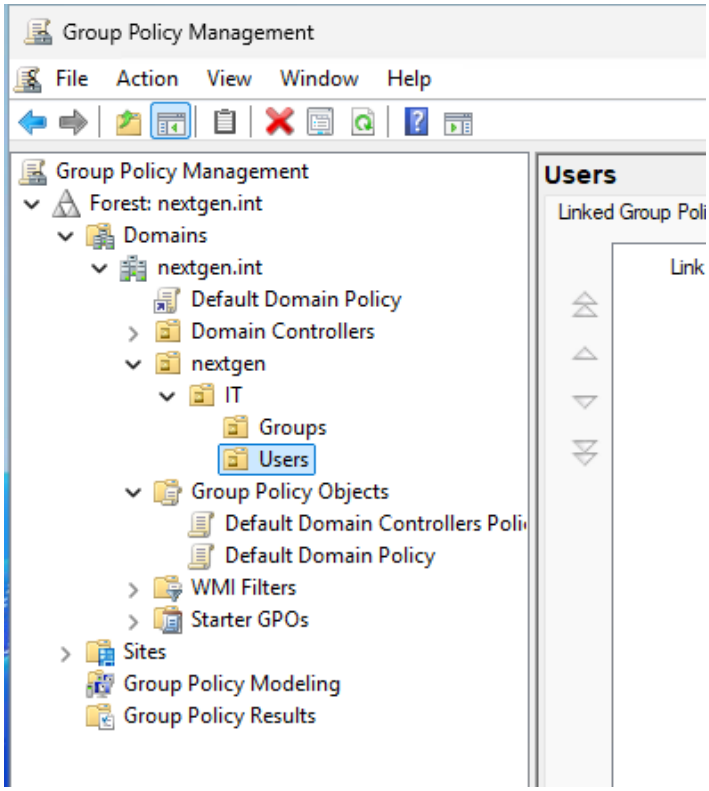
Group Policy Management: The Console Overview



The screenshot above shows the main Group Policy Management console. Here, we can see the overall structure: the Forest (nextgen.int), the Domain (nextgen.int), the Default Domain Policy, the Domain Controllers container, our own custom OU (nextgen), and the Group Policy Objects folder.

By default, the Default Domain Policy already exists, and it manages some baseline policy settings for the entire domain automatically.

An important rule to remember: we are able to link and apply policies to the OUs that we ourselves have created (such as our "nextgen" OU). However, we are not able to add or link new policies directly onto the default containers that come built into Active Directory, such as the default "Users" or "Computers" containers, since these are not true Organizational Units, and do not support Group Policy linking in the same way.



The screenshot above shows our custom OU structure in more detail: nextgen → IT, containing both Groups and Users. This is the kind of custom, self-created OU structure that we are able to link Group Policies onto.

Group Policy Objects (GPOs): The Most Important Folder

The Group Policy Objects folder is the single most important part of this entire structure. Every single new Group Policy that we create, for example, one that changes desktop wallpapers, or one that blocks USB drives, is actually created, saved, and stored inside this specific folder.

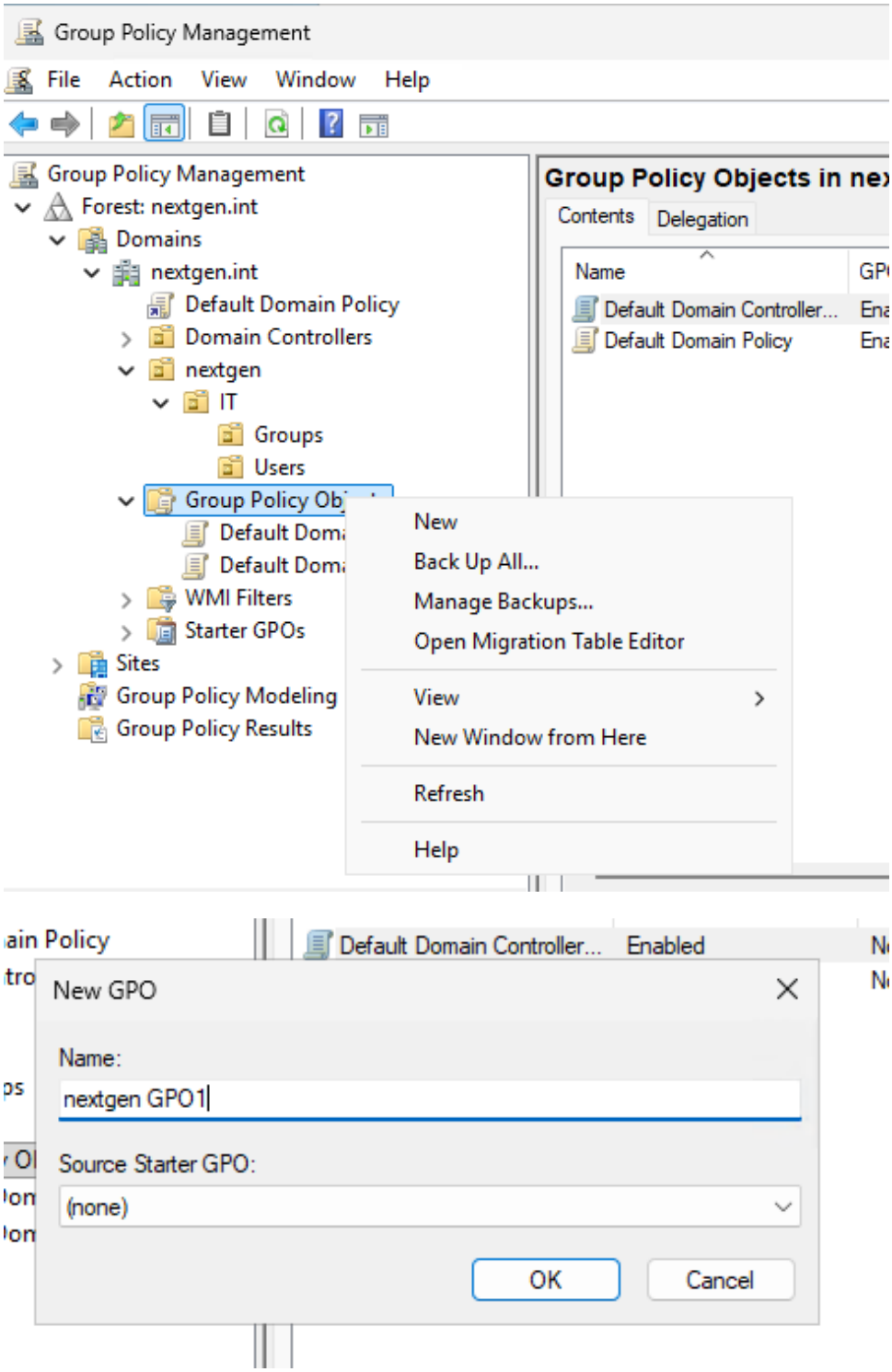
A simple way to picture this is to think of this folder like a bookshelf full of books. The individual policies stored here are like the "books" on that shelf. In order to actually apply a policy, we take a "book" (a Policy) off of this shelf, and we link it directly to our target department folder, such as IT or Management.

We always create our policies inside this Group Policy Objects folder first, and only afterward do we go and assign, or "link," those policies to the specific departments (OUs) where we want them to apply.

It is also worth noting that we do not create brand new policies directly inside the Default Domain Policy. Instead, we are only meant to change specific existing settings within it, such as adjusting the minimum password length, or the minimum password age requirements.

How to Create a New Group Policy Object

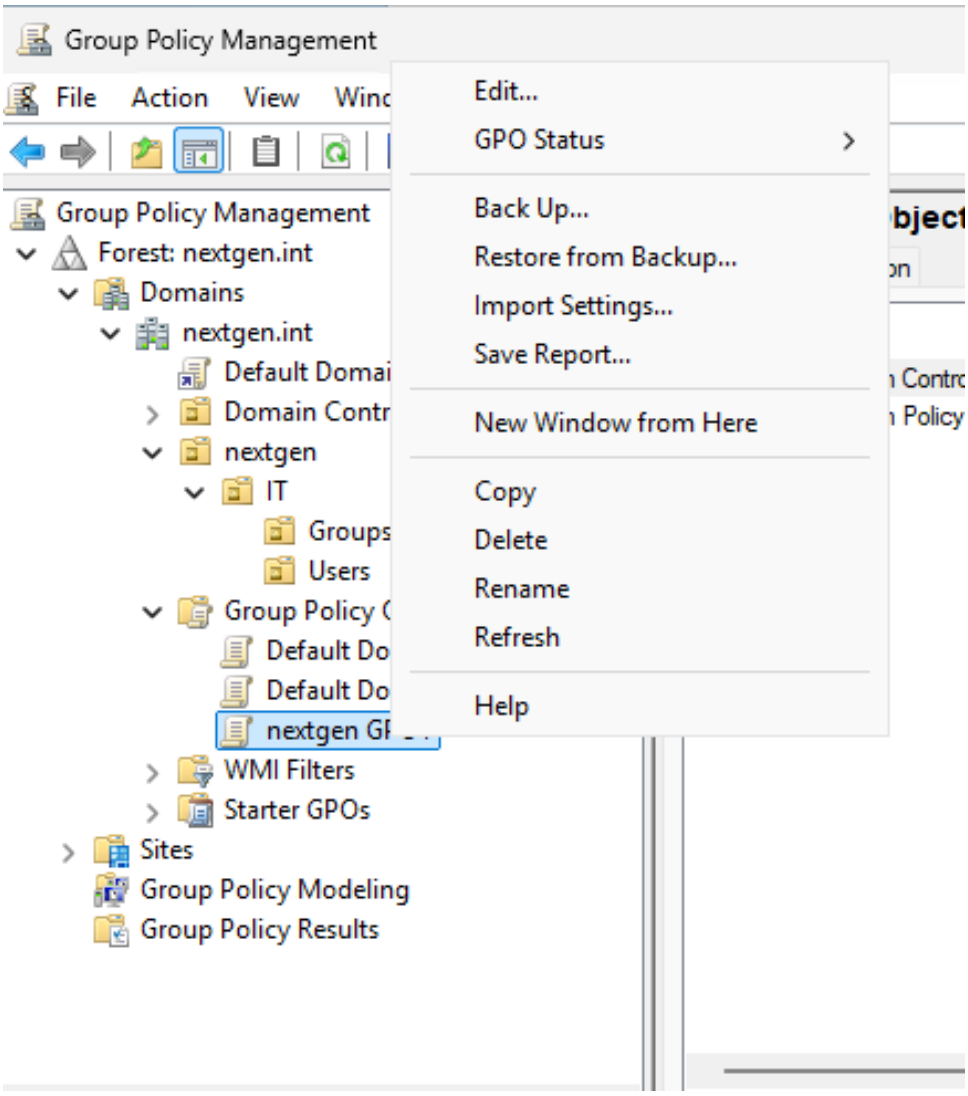
To create a new policy, we simply right-click on the Group Policy Objects folder, and select New.



A New GPO window then appears, asking us to type in a Name for our new policy, and also asking us about the Source Starter GPO.

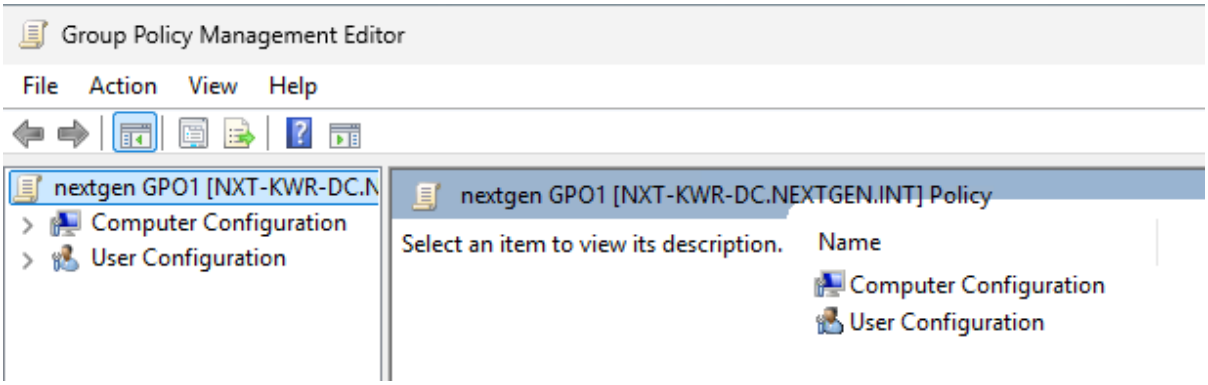
This particular Source Starter GPO option is essentially asking us: "Do you want to create this new policy completely from scratch, as a blank template, or do you instead want to use a previously saved Policy Template, to copy and duplicate its existing settings into this brand-new one?" In our lab, we simply leave this option set to "(none)," since we are creating our policy entirely from scratch.

Once a policy has been created, we can always go back and edit it later, simply by right-clicking on that policy, and selecting Edit.



Computer Configuration vs User Configuration

Whenever we open a policy for editing, inside the Group Policy Management Editor, we can see that policies are always separated into two main sections: Computer Configuration, and User Configuration.



1. Computer Configuration (Machine-Based Policies)

These are policies that are applied strictly to the local computer system itself, regardless of which particular user account happens to log into that machine. These settings are enforced during the system's startup process.

A few real-world, enterprise examples of this include:

- Restricting local USB storage access, across all corporate endpoints in the company.
- Enforcing specific Windows Update schedules, and other related WSUS settings.
- Hardening overall network security, by making sure Windows Defender Firewall is enabled by default.

2. User Configuration (Identity-Based Policies)

These are policies that are applied directly to a specific user's Active Directory account. Because of this, these particular rules follow that specific user around, to whichever workstation they happen to log into, anywhere across the domain. These settings are enforced specifically during the user's login phase.

A few real-world, enterprise examples of this include:

- Standardizing one uniform, corporate background wallpaper, across every single staff member's desktop.
- Disabling access to the Control Panel and Command Prompt, specifically for non-administrative, standard users.
- Automatically mapping shared departmental network drives, for example, automatically mapping an H:\ drive, specifically for HR department personnel.

Implementing a Desktop Wallpaper Policy via Group Policy

Now, let's walk through a complete, real example: setting up a Group Policy that forces a specific desktop wallpaper onto every user's computer.

Step 1: Sharing the Wallpaper Folder

Firstly, we need to save our chosen wallpaper image somewhere on our Active Directory server. Next, we need to share the folder where that wallpaper image is stored, so that every single user on the network is able to properly access it.

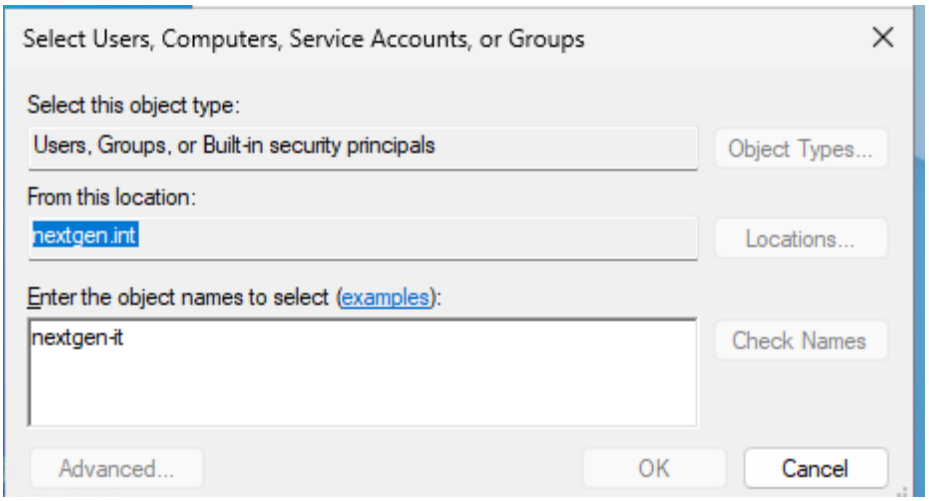
Here is exactly why this sharing step is so important: whenever a user turns on their computer and logs in, the Windows operating system reads the Group Policy that we have created. This policy essentially tells the system: "Go to this specific file path, and use this particular picture as the desktop wallpaper."

However, if that picture file is only saved locally on our server's own hard drive (for example, stored inside the server's private C:\Wallpapers\bg.jpg location), then the various client computers on the network (such as User 1's own machine) will not actually be able to access

the server's private C: drive directly. Because of this restriction, the client system will not be able to find the wallpaper picture at all, and the wallpaper will simply fail to display correctly on their desktop.

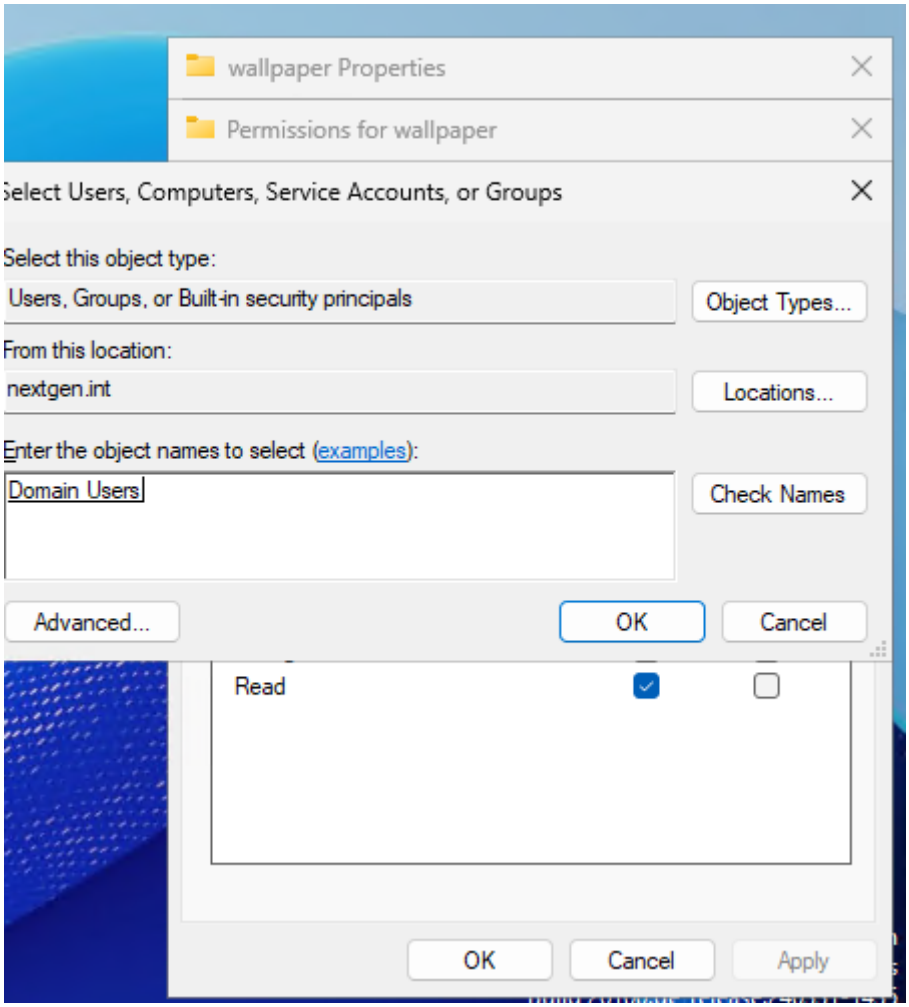
How to Share a Folder

To share our wallpaper folder, we follow this simple path: right-click on the folder, select Properties, switch to the Sharing tab, click on Advanced Sharing, and then tick the box labeled "Share this folder." From there, we click on Permissions, and then click Add.

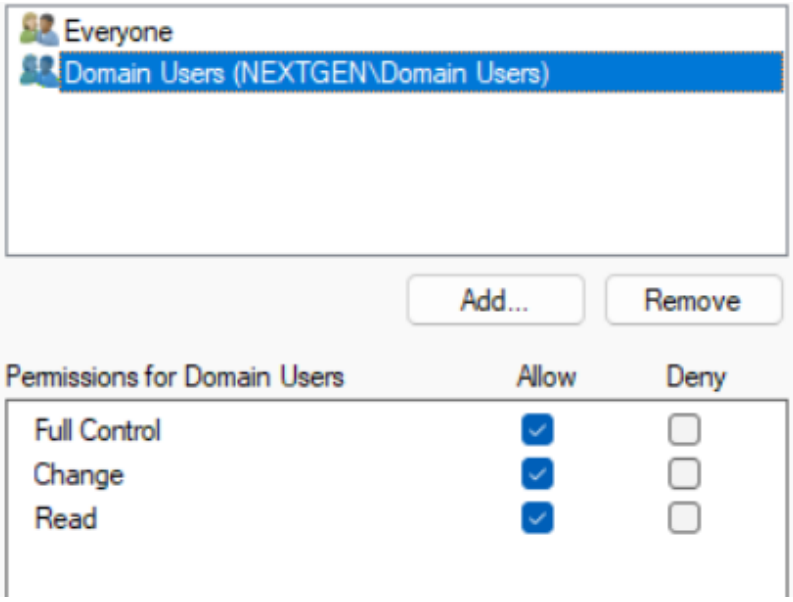


In the window that appears, we can now type in the name of the specific OU, or the specific group, that we want to share this folder with, and then click OK.

If we want to share this folder with absolutely every single user in the entire domain, we can simply type "Domain Users" instead, into this same field. It often helps to type just a simple part of the name first, and then click the "Check Names" button, which will then automatically show us the full, correctly formatted name.

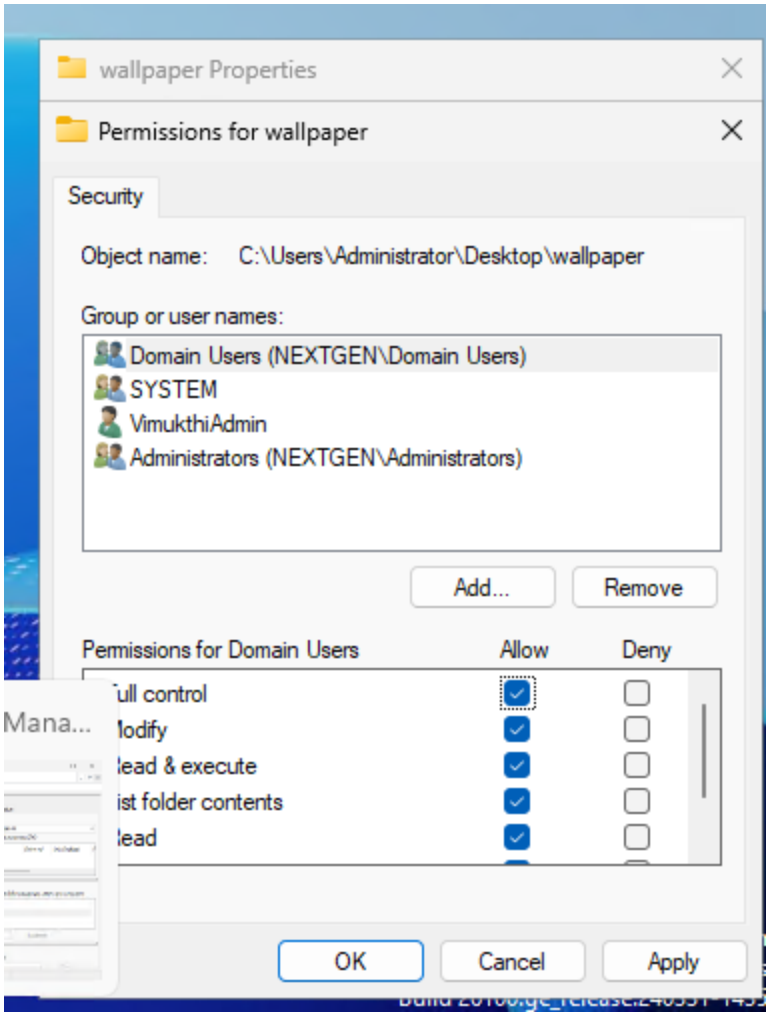


After adding "Domain Users," we can then select the exact permission level we want to grant them, and click Apply, and then OK.



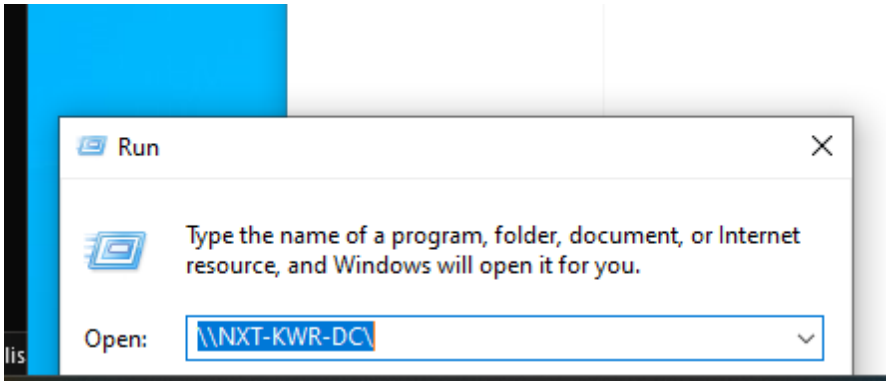
Setting Security Permissions

After configuring the basic sharing permissions, we should also go to the Security tab on that same folder's Properties window, and give similar permissions there as well. In our example, we gave the "Domain Users" group full permission access, on this Security tab as well.

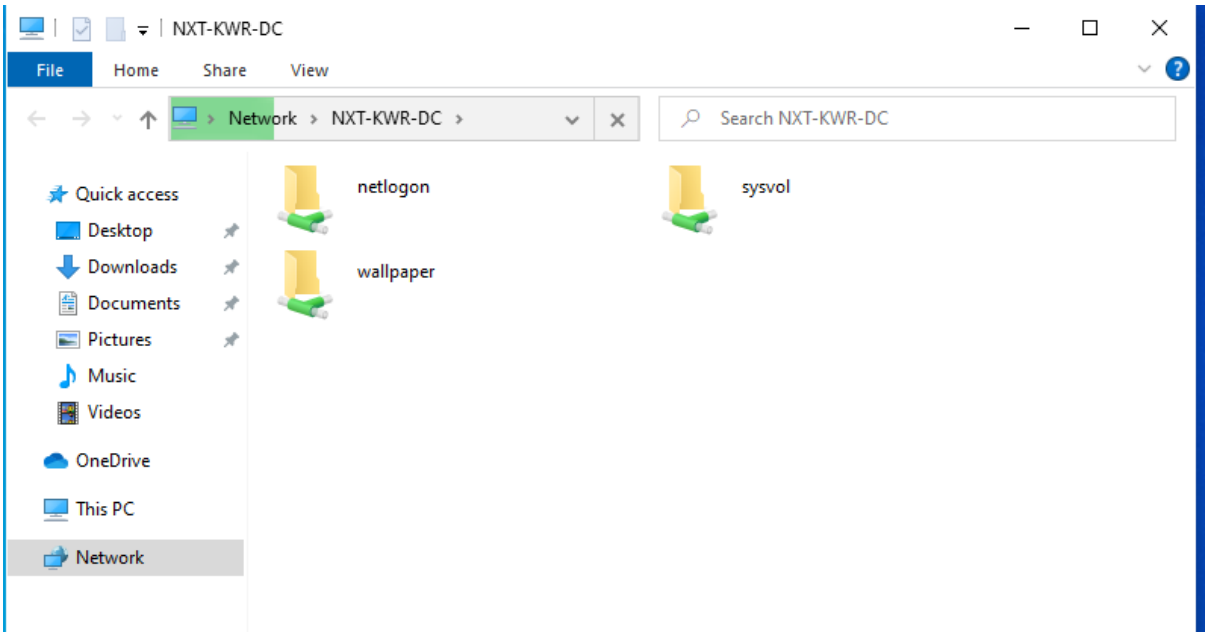


How to Check That Sharing Is Working

To confirm that our folder sharing is actually working correctly, we can go over to a client PC, press the Windows key together with the "R" key, to open the Run dialog box, and then type in the server's name, using this format: \\server_name\ (for example, in our lab, we typed \\NXT-KWR-DC\).



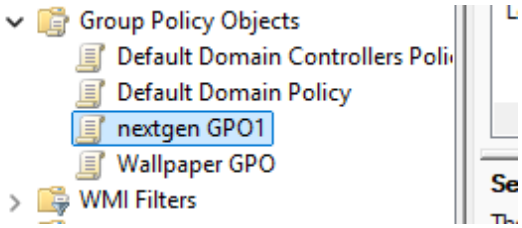
After pressing Enter, we should now be able to see the shared folder appear in the list of available shares, confirming that our sharing configuration was successful.



As shown in the screenshot above, our newly shared "wallpaper" folder now correctly appears alongside the other default shared folders, such as netlogon and sysvol.

Step 2: Create and Configure the GPO

Next, we create a new Group Policy Object, and we name it "Wallpaper GPO."



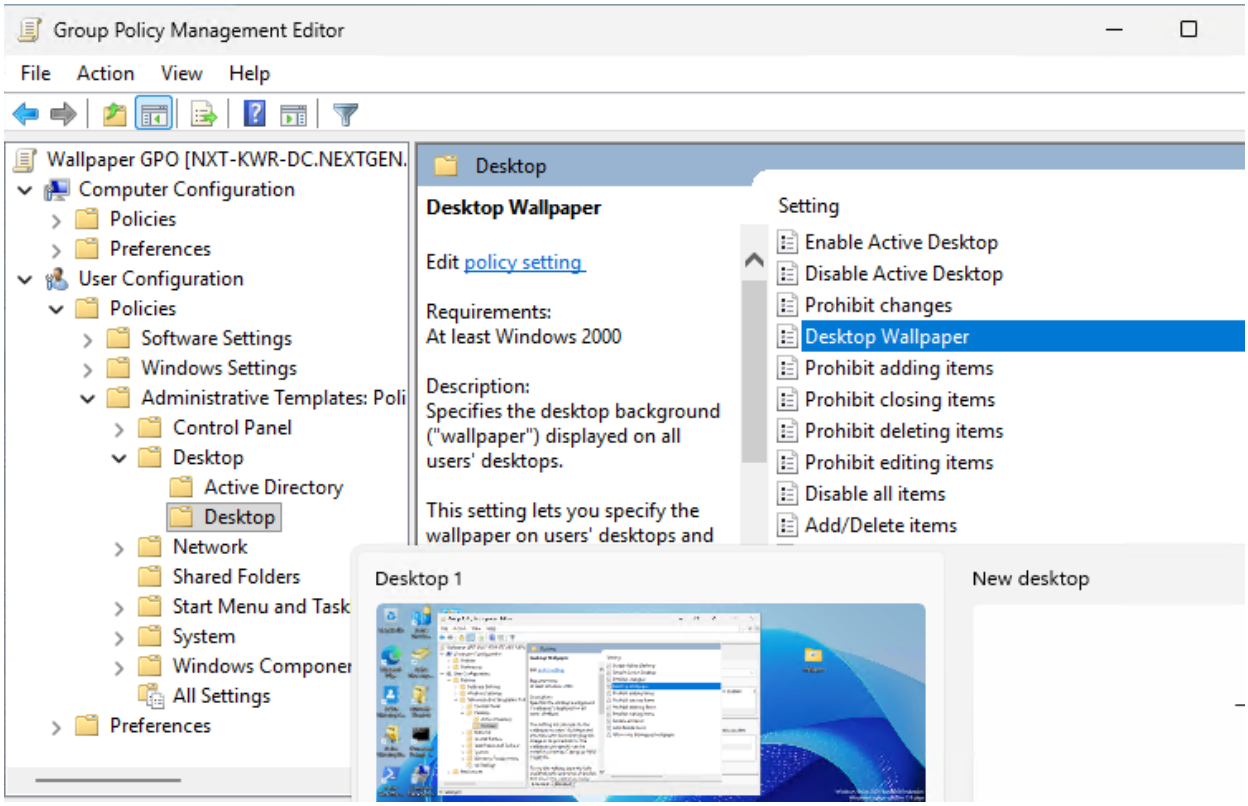
We then right-click on our new "Wallpaper GPO," and select Edit. From there, we need to navigate to this specific path, inside the policy editor:

```
User Configuration > Policies > Administrative Templates > Desktop > Desktop > Desktop Wallpaper
```

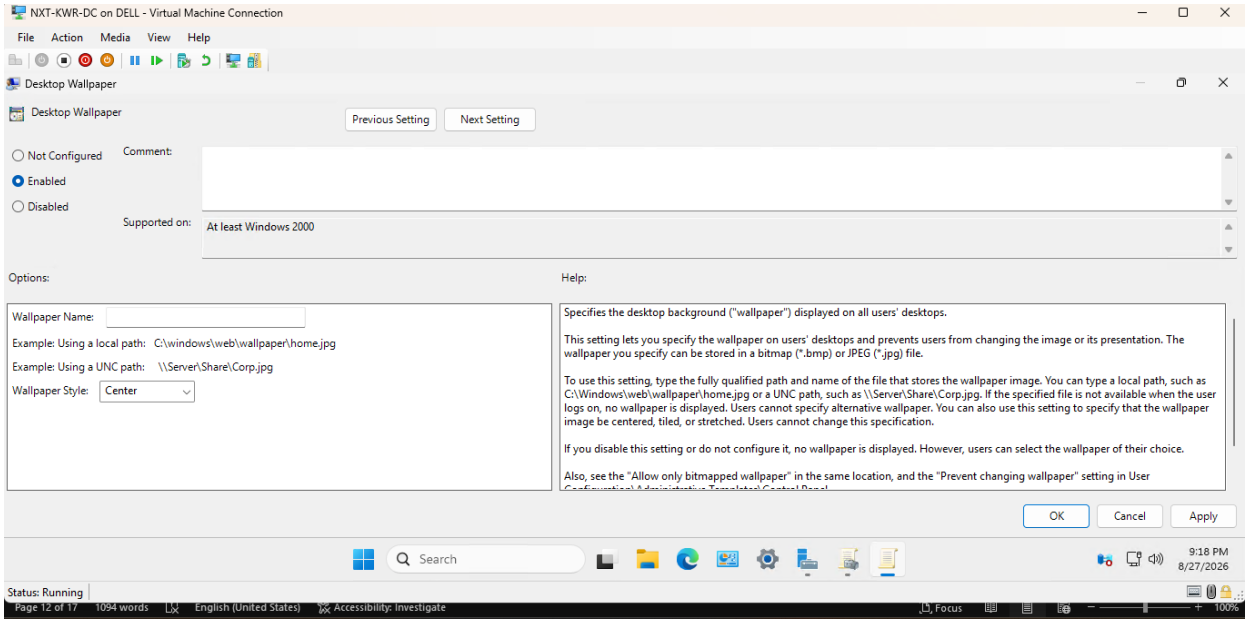
Why Is This Policy Placed Under User Configuration?

The reasoning behind this placement is actually very simple: multiple different people can potentially use the exact same physical computer (for example, think of a shared computer lab, where many different students or staff members use the same machine, at different times).

If we wanted an IT department employee to see one particular, normal wallpaper when they personally log in, but we wanted a Management department employee to instead see the official corporate company wallpaper, on that exact same shared physical machine, then this particular rule absolutely must depend specifically on which user is currently logging in, rather than which physical machine is being used. This is exactly why this policy setting is placed under User Configuration, rather than under Computer Configuration.



Configuring the Wallpaper Setting

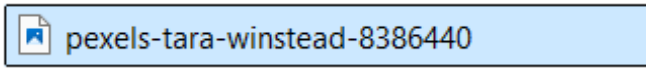


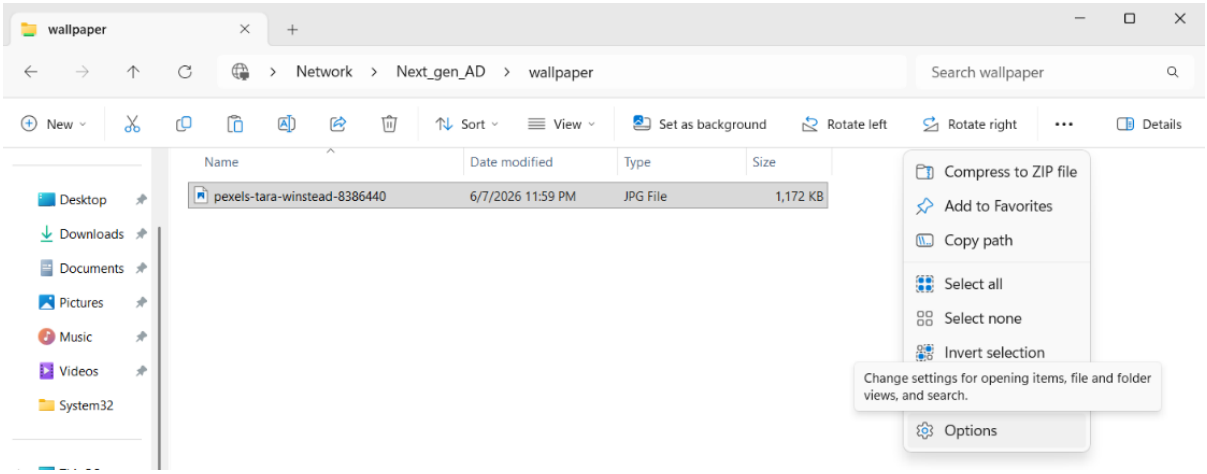
Once we open the Desktop Wallpaper setting, we select "Enabled." We then need to type in the wallpaper's name, which requires us to first provide the shared file path, followed by the file name, complete with its file extension (for example, .jpg).

To find our exact shared path, we can press the Windows key together with "R," and search for \\NXT-KWR-DC\ once again, just as we did earlier.

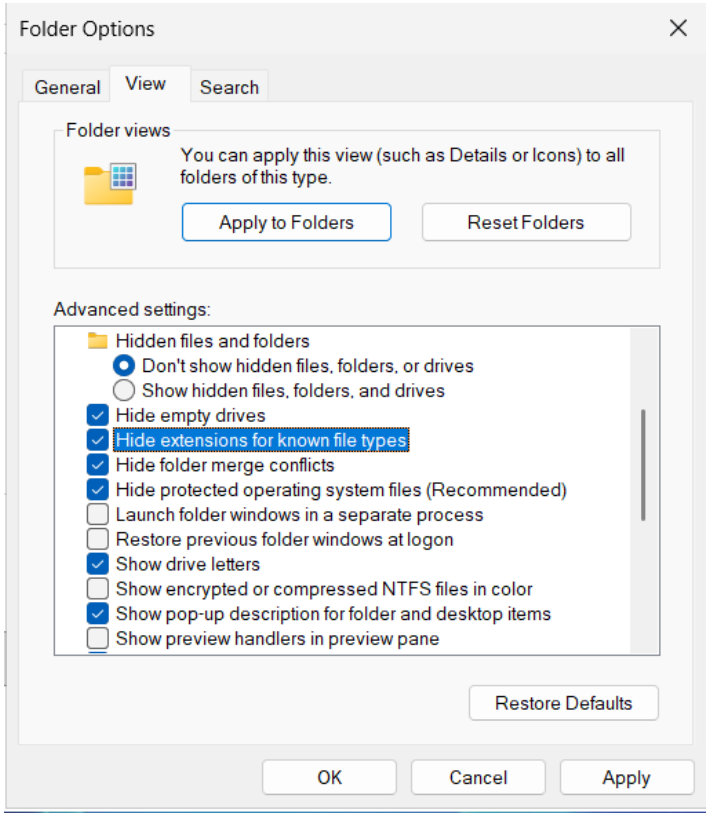
An Important Detail: Showing File Extensions

By default, Windows is often configured to hide file extensions for well-known file types. This means that a picture file might simply be displayed as "pexels-tara-winstead-8386440," without the important ".jpg" ending actually being visible to us.





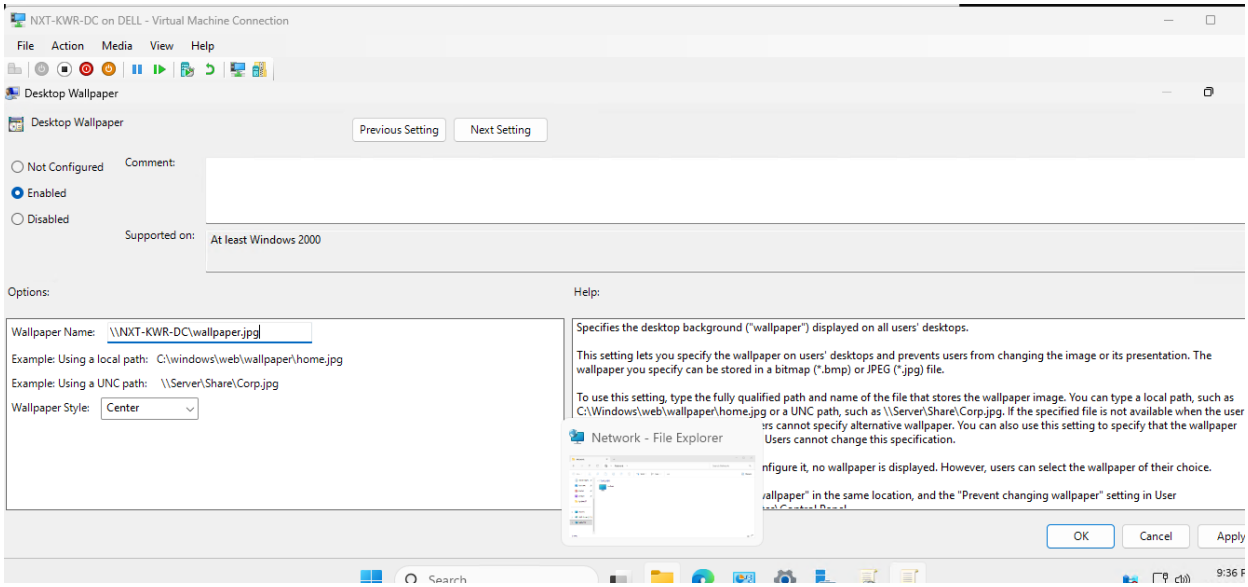
In order for our Group Policy setting to work correctly, we absolutely need to know and correctly type this exact file extension. So, let's make Windows show us the extension instead.



To do this, we open the Folder Options window, go to the View tab, and then, under Advanced Settings, we find and untick the specific checkbox labeled "Hide extensions for known file types." After unticking this box, we click Apply, and then OK, and now we are able to correctly see the full file extension.

Entering the Final Wallpaper Path

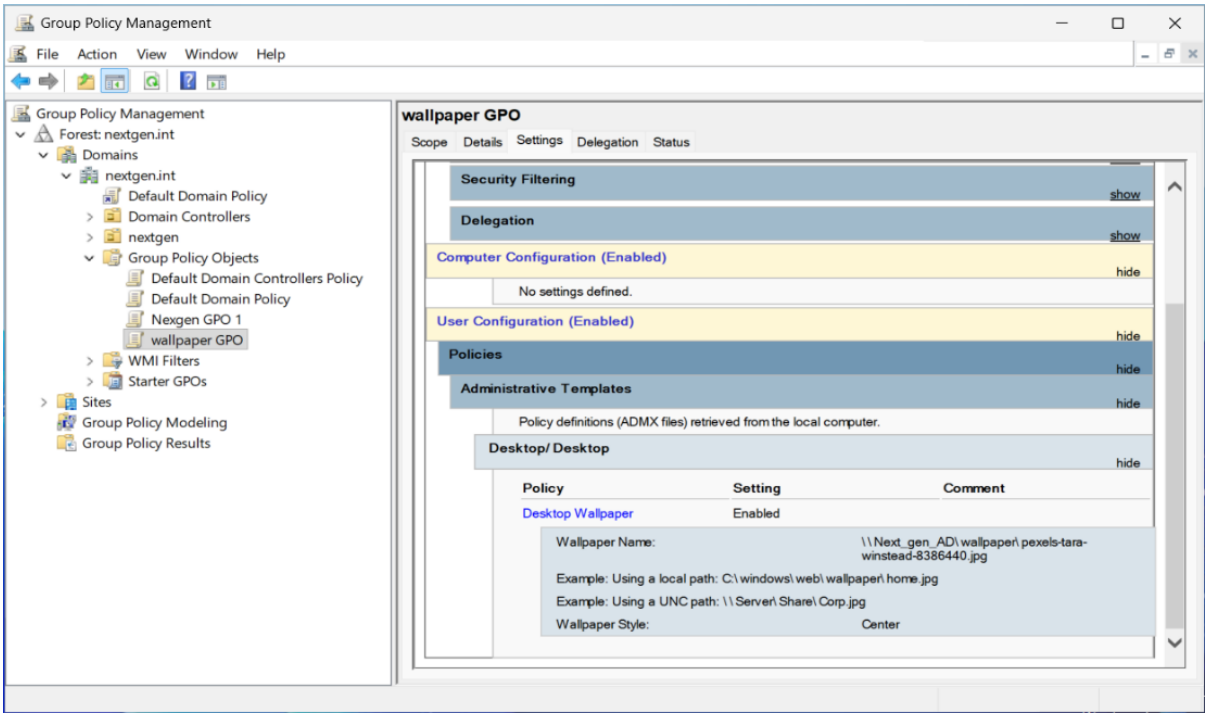
Now that we can clearly see the correct file extension, we can go ahead and correctly type in the full file path and file name, into the Wallpaper Name field.



In our specific example, we entered the path as \\NXT-KWR-DC\wallpaper.jpg. After this, we also set the desired Wallpaper Style, choosing from options such as Center, Fit, or other similar display options. Once everything looks correct, we click Apply, and then OK, to save our new policy settings.

Confirming the Policy Settings Were Saved

We can double check that our policy settings were correctly saved, by looking at the policy's Settings report tab, within Group Policy Management.

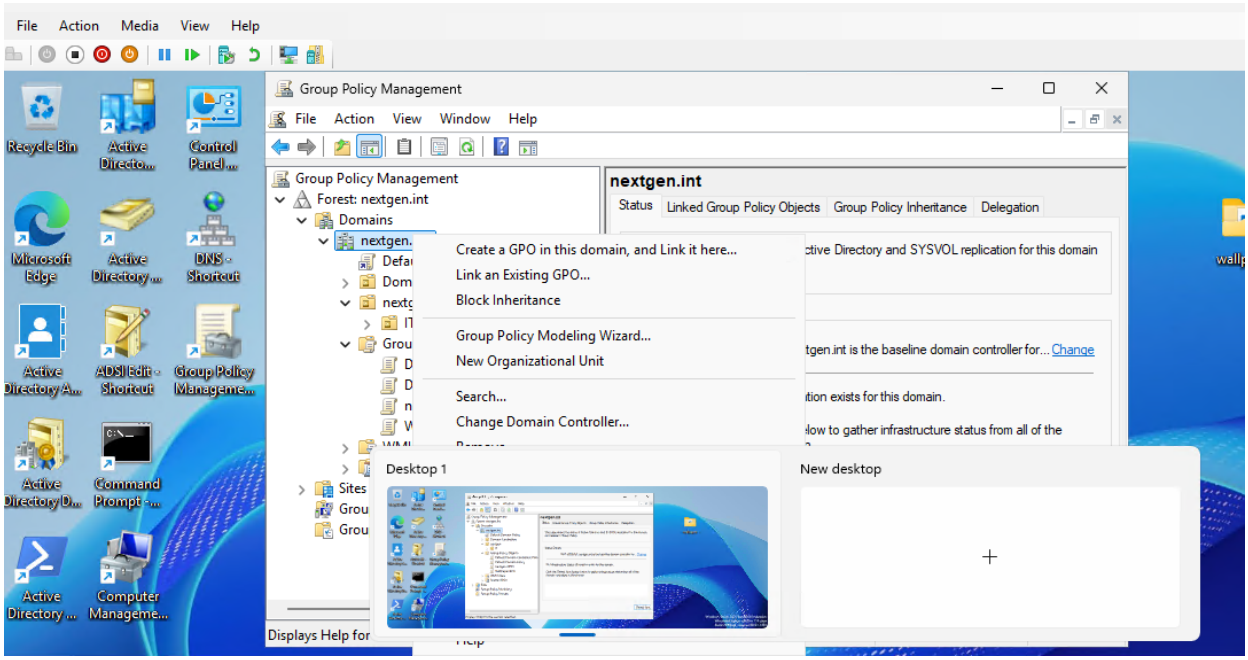


As shown in the screenshot above, this settings report confirms that our Desktop Wallpaper setting is indeed Enabled, with the correct Wallpaper Name path and Wallpaper Style clearly listed and saved correctly.

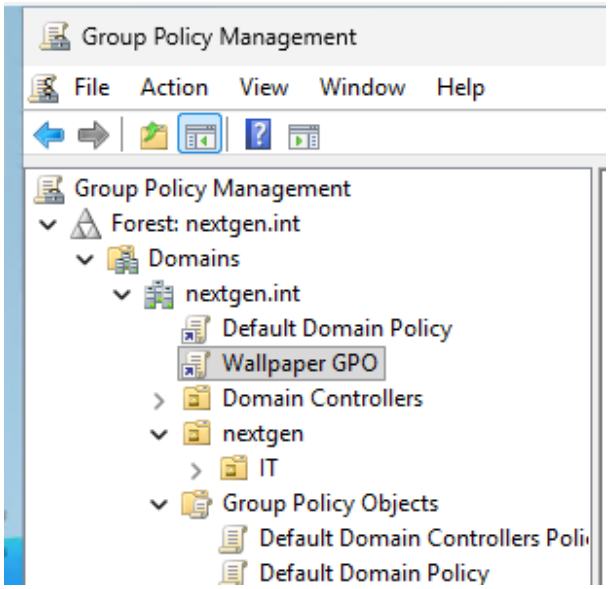
Step 3: How to Apply the Policy to a Group (OU)

Earlier, we had already created an OU called "nextgen." If we specifically wanted this wallpaper policy to apply only to that particular "nextgen" OU, and nowhere else, we could do this by right-clicking directly on that specific OU, then selecting "Link an Existing GPO," and then choosing our Wallpaper policy from the resulting list. This action would then assign that specific policy only to the users and computers contained within that selected OU.

However, in our particular lab example, we instead chose to assign this wallpaper policy to the entire domain as a whole, meaning it will apply to every single user across the entire nextgen.int domain.



As shown above, we right-click directly on the root domain itself (nextgen.int), and select "Link an Existing GPO," in order to apply our chosen policy across the entire domain.



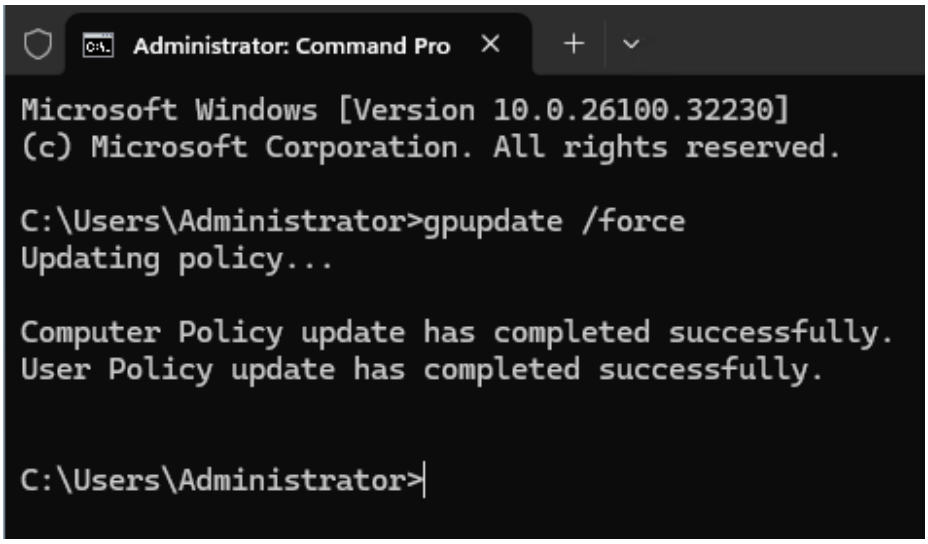
The screenshot above confirms that our "Wallpaper GPO" is now correctly listed, and successfully linked, directly under the nextgen.int domain.

Applying the Policy Immediately (Instead of Waiting)

Under completely normal circumstances, a newly created Group Policy will typically take up to around 90 minutes, before it automatically applies itself to all of the relevant computers across the network.

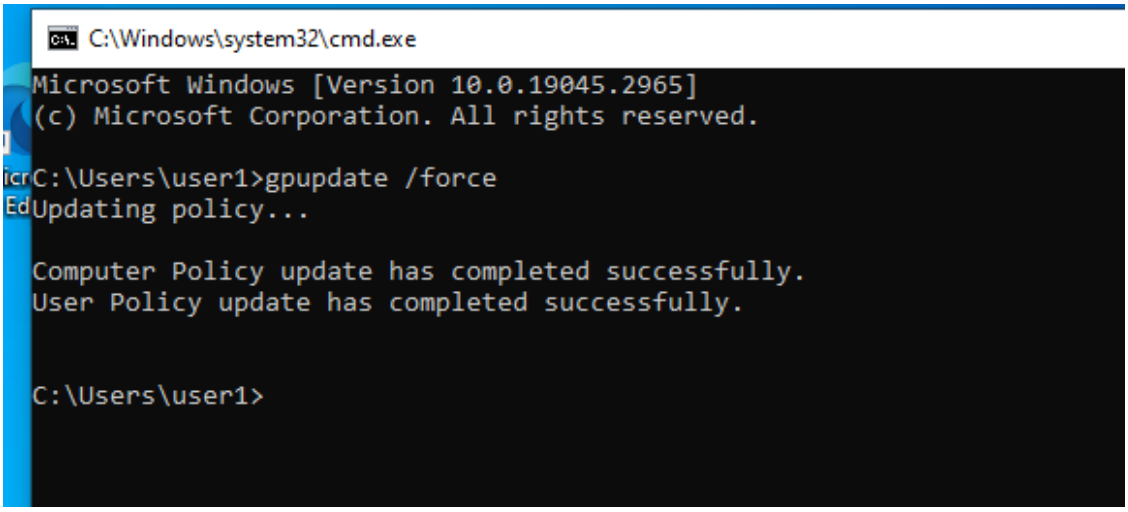
However, if we don't want to wait that long, we can instead manually force this update to happen immediately, using Command Prompt, on both the server side and the client side.

On the Server Side



We simply open Command Prompt on the server, and run the command gpupdate /force. As we can see in the result above, this command successfully updated both the Computer Policy and the User Policy immediately.

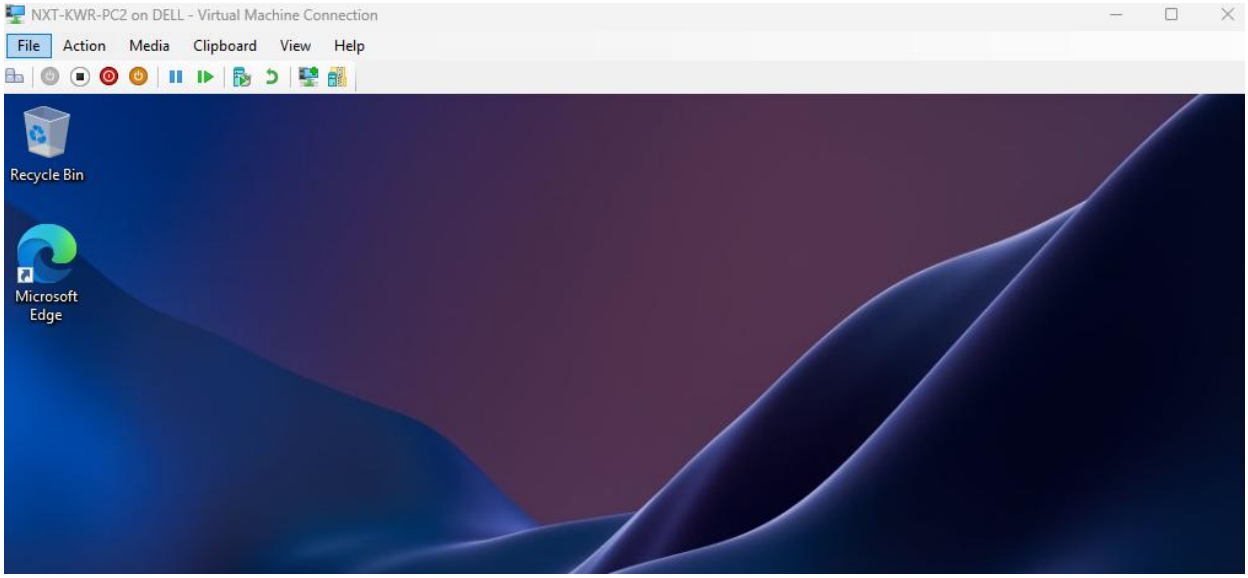
On the Client Side



We repeat this exact same process on the client computer as well, running gpupdate /force there too. As shown above, this was run successfully while logged in as our domain user, "user1."

After running this command on both sides, we then simply restart the client PC, in order for the new wallpaper setting to fully take effect.

Final Result



As we can clearly see in the screenshot above, after restarting, the client computer now correctly displays our chosen corporate wallpaper automatically, confirming that our entire Group Policy configuration worked successfully, exactly as intended.

Important Note: Changing Server or Client IP Addresses Later

If we ever need to change the IP address of either our AD server or our client computer, at some point in the future, we will also need to clear out some old, cached information, and provide the server with our updated details. To do this, we open Command Prompt, and run the following commands:

```
ipconfig /flushdns
ipconfig /registerdns
```

ipconfig /flushdns (Clearing the Old Cached Memory)

What this command does: any client machine usually keeps a small memory cache, of old IP addresses and server details, that it has previously connected to before. When we change our IP address, and then run this particular command, it completely wipes out that entire old cached memory, making the machine fully ready to correctly work with its brand new IP address instead.

ipconfig /registerdns (Giving the Server Our New Details)

What this command does: by running this particular command, the client machine actively contacts the DNS server, running on our Active Directory Domain Controller (DC), and essentially says: "Hey, this is actually my new IP address now, not my old one anymore. Please update your DNS records (your database) accordingly, with this brand new IP address, next to my computer's name."

NextGen Active Directory Structure (nextgen.int)

This document records the complete Organizational Unit (OU) structure, security groups, and user accounts that were built for the NextGen IT Lab, along with the exact PowerShell script used to create everything, and the commands used to verify or rebuild the structure.

1. Organizational Unit (OU) Structure

The following OU hierarchy was created under a parent OU named "NextGen", inside the nextgen.int domain. Each department was given its own OU, and inside each department OU, two further sub-OUs were created: one named "Users" (to hold user accounts), and one named "WKS" (to hold workstation/computer accounts). A separate "Groups" OU was also created, specifically to hold the security groups.

Parent OU	Department OU	Sub OUs
NextGen	IT	Users, WKS
NextGen	MKT	Users, WKS
NextGen	FIN	Users, WKS
NextGen	HR	Users, WKS
NextGen	Groups	(Security Groups only)

Full OU Tree

```
NextGen
├── IT
│   ├── Users
│   └── WKS
├── MKT
│   ├── Users
│   └── WKS
├── FIN
│   ├── Users
│   └── WKS
├── HR
│   ├── Users
│   └── WKS
└── Groups
```

2. Security Groups

All security groups are located under the following path:
OU=Groups,OU=NextGen,DC=nextgen,DC=int

Group Name	Scope	Type	Members

IT	Global	Security	ituser1, ituser2
MKT	Global	Security	mktuser1, mktuser2
FIN	Global	Security	finuser1, finuser2
HR	Global	Security	hruser1, hruser2

3. User Accounts

Full Name	First Name	Last Name	Logon Name	Department	Group
IT User1	IT	User1	ituser1	IT	IT
IT User2	IT	User2	ituser2	IT	IT
MKT User1	MKT	User1	mktuser1	MKT	MKT
MKT User2	MKT	User2	mktuser2	MKT	MKT
FIN User1	FIN	User1	finuser1	FIN	FIN
FIN User2	FIN	User2	finuser2	FIN	FIN
HR User1	HR	User1	hruser1	HR	HR
HR User2	HR	User2	hruser2	HR	HR

3.1 Password Policy

Setting	Value
Password	Kandage12@#12
Password Never Expires	Enabled (True)
Account Enabled	Yes
UPN Format	username@nextgen.int

Note: Setting "Password Never Expires" to true is convenient for a lab environment, but in a real production network, this is generally discouraged, since it removes the automatic password rotation that helps protect accounts over time.

4. PowerShell Setup Script

The entire AD structure above, meaning all of the OUs, all of the security groups, and all of the user accounts, was created using a single PowerShell script, run inside the "Active Directory Module for Windows PowerShell," as Administrator.

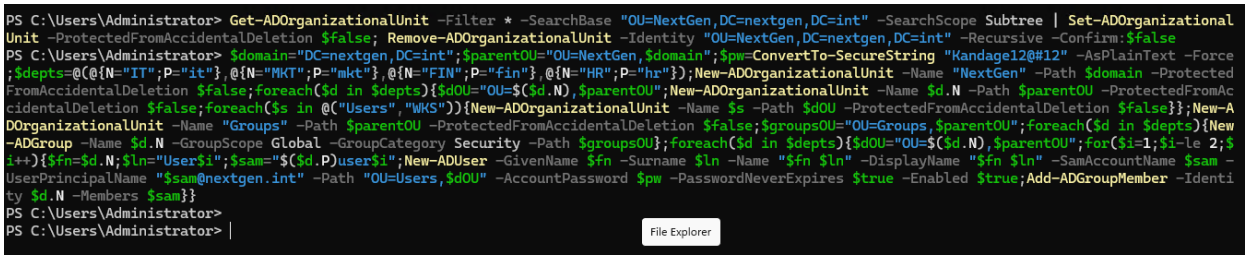
Step 1: Set the Execution Policy

Before running any custom script, Windows PowerShell needs to be given permission to run it. By default, Windows blocks scripts from running, as a security precaution. This command temporarily relaxes that restriction:

```
Set-ExecutionPolicy RemoteSigned -Force
```

Step 2: Build the Full AD Structure

The single command below builds everything at once: it creates the NextGen parent OU, creates each department OU (IT, MKT, FIN, HR) along with their Users and WKS sub-OUs, creates the Groups OU with one security group per department, and finally creates two users per department, placing each user inside their correct OU and adding them into their correct group.




```
$domain="DC=nextgen,DC=int";
$parentOU="OU=NextGen,$domain";
$pw=ConvertTo-SecureString "Kandage12@#12" -AsPlainText -Force;
$depts=@(
  @{N="IT";P="it"},
  @{N="MKT";P="mkt"},
  @{N="FIN";P="fin"},
  @{N="HR";P="hr"}
);
New-ADOrganizationalUnit -Name "NextGen" -Path $domain -ProtectedFromAccidentalDeletion
$false;
foreach($d in $depts){
  $dOU="OU=$($d.N),$parentOU";
  New-ADOrganizationalUnit -Name $d.N -Path $parentOU -ProtectedFromAccidentalDeletion
$false;
  foreach($s in @("Users","WKS")){
    New-ADOrganizationalUnit -Name $s -Path $dOU -ProtectedFromAccidentalDeletion $false
  }
};
New-ADOrganizationalUnit -Name "Groups" -Path $parentOU -ProtectedFromAccidentalDeletion
$false;
$groupsOU="OU=Groups,$parentOU";
foreach($d in $depts){
  New-ADGroup -Name $d.N -GroupScope Global -GroupCategory Security -Path $groupsOU
};
foreach($d in $depts){
  $dOU="OU=$($d.N),$parentOU";
  for($i=1;$i -le 2;$i++){
    $fn=$d.N; $ln="User$i"; $sam="$($d.P)user$i";
    New-ADUser -GivenName $fn -Surname $ln -Name "$fn $ln" -DisplayName "$fn $ln" `
      -SamAccountName $sam -UserPrincipalName "$sam@nextgen.int" `
      -Path "OU=Users,$dOU" -AccountPassword $pw -PasswordNeverExpires $true -Enabled
$true;
    Add-ADGroupMember -Identity $d.N -Members $sam
  }
}
```

Breaking Down What This Script Actually Does

- **$domain and $parentOU:** These two variables simply store the base path of the domain (DC=nextgen,DC=int), and the path of our new parent OU (OU=NextGen,...), so we don't have to retype these long paths over and over again throughout the script.
- **$pw:** This variable converts our plain-text password into a Secure String, which is the special, encrypted format that Active Directory commands require, whenever a password needs to be set.
- **$depts:** This variable stores a list of our four departments, along with a short prefix for each one (it, mkt, fin, hr), which is later used to automatically build each department's logon names, such as ituser1.
- **First foreach loop:** This loop goes through each department one at a time, creates that department's OU, and then creates the Users and WKS sub-OUs inside it.
- **Second foreach loop:** This loop goes through each department again, and creates one Global Security Group for that department, inside the Groups OU.
- **Third foreach loop (with a nested for loop):** This loop goes through each department one more time, and for each one, it creates two user accounts (User1 and User2), places them inside that department's Users OU, and then immediately adds each new user into their matching department security group.

5. Why We Used a PowerShell Script Instead of Manual (GUI) Configuration

This entire structure could technically have been built by hand, meaning by right-clicking through Active Directory Users and Computers, one item at a time. However, a single PowerShell script was used instead, for several important, practical reasons.

1. Efficiency (Saving Time)

Building this structure manually would require creating 6 OUs (NextGen, IT, MKT, FIN, HR, Groups), 8 sub-OUs (Users and WKS, under each of the 4 departments), 4 security groups, and 8 user accounts, one at a time, through the graphical interface. That adds up to more than 20 separate manual actions, each one requiring several clicks and typed fields.

Running the single script instead completes all of these same actions in only a few seconds. This is the core meaning of "increasing efficiency": achieving the exact same end result, in a fraction of the time.

2. Consistency and Avoiding Human Error

When typing information manually, over and over again, it becomes easy to make small mistakes, for example, typing "itUser1" instead of "ituser1", or accidentally assigning a Finance user into the Marketing group. Because the script uses loops to repeat the exact same logic for every department, every object is created using the exact same consistent pattern, which removes the opportunity for this kind of manual, human error.

3. Repeatability

As shown later in Section 7 (Delete and Rebuild Commands), the entire structure can be deleted, and then rebuilt from scratch, simply by running the same script again. If this had been built manually, recreating the entire structure after a deletion would mean repeating all 20+ manual steps all over again.

4. Scalability

In this lab, there are only 4 departments, with 2 users each. However, if a real company needed 20 departments, each with 100 users, the script could be adjusted simply by changing a few values, and it would still run in only a few seconds. Attempting the same scale of work manually, through the GUI, would become extremely time-consuming, and impractical.

5. Documentation and Auditability

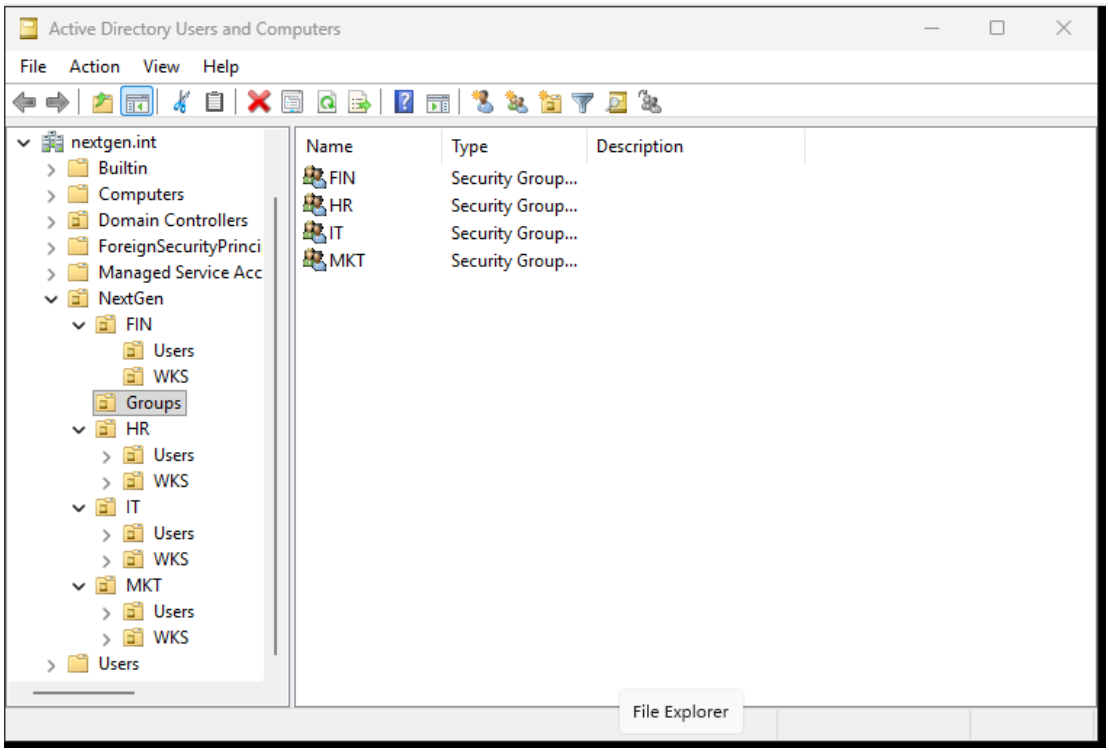
A script also serves as a written, exact record of precisely what was configured. With manual GUI changes, it can become difficult to later recall or prove exactly what settings were applied. Since the script itself lists out every single step, it makes the entire configuration much easier to review, audit, or hand over to another administrator later on.

6. Verify Commands

After running the setup script, we can use the following commands to confirm that everything was created correctly.

Command	**Purpose**
Get-ADOrganizationalUnit -Filter * | Select-Object DistinguishedName	List all OUs
Get-ADUser -Filter * -SearchBase "OU=NextGen,DC=nextgen,DC=int" | Select-Object Name,SamAccountName	List all users
foreach($g in @("IT","MKT","FIN","HR")){Get-ADGroupMember -Identity $g | Select-Object Name}	Check group members

Confirming the OU Tree in the GUI



The screenshot above shows the same OU tree, this time viewed directly inside Active Directory Users and Computers. This confirms visually that the NextGen parent OU was created correctly, containing IT, MKT, FIN, HR, and Groups, with Users and WKS sub-OUs correctly nested inside each department.

Listing All OUs

```
PS C:\Users\Administrator> Get-ADOrganizationalUnit -Filter * | Select-Object DistinguishedName

DistinguishedName
-----------------
OU=Domain Controllers,DC=nextgen,DC=int
OU=NextGen,DC=nextgen,DC=int
OU=IT,OU=NextGen,DC=nextgen,DC=int
OU=Users,OU=IT,OU=NextGen,DC=nextgen,DC=int
OU=WKS,OU=IT,OU=NextGen,DC=nextgen,DC=int
OU=MKT,OU=NextGen,DC=nextgen,DC=int
OU=Users,OU=MKT,OU=NextGen,DC=nextgen,DC=int
OU=WKS,OU=MKT,OU=NextGen,DC=nextgen,DC=int
OU=FIN,OU=NextGen,DC=nextgen,DC=int
OU=Users,OU=FIN,OU=NextGen,DC=nextgen,DC=int
OU=WKS,OU=FIN,OU=NextGen,DC=nextgen,DC=int
OU=HR,OU=NextGen,DC=nextgen,DC=int
OU=Users,OU=HR,OU=NextGen,DC=nextgen,DC=int
OU=WKS,OU=HR,OU=NextGen,DC=nextgen,DC=int
OU=Groups,OU=NextGen,DC=nextgen,DC=int


PS C:\Users\Administrator>
```

Running Get-ADOrganizationalUnit -Filter * | Select-Object DistinguishedName lists out the full distinguished name of every single OU that exists under the domain, confirming the entire structure was created exactly as planned.

Listing All Users

```
PS C:\Users\Administrator> Get-ADUser -Filter * -SearchBase "OU=NextGen,DC=nextgen,DC=int" | Select-Object Name,SamAccountName

Name      SamAccountName
----      --------------
IT User1  ituser1
IT User2  ituser2
MKT User1 mktuser1
MKT User2 mktuser2
FIN User1 finuser1
FIN User2 finuser2
HR User1  hruser1
HR User2  hruser2


PS C:\Users\Administrator>
```

Running Get-ADUser -Filter * -SearchBase "OU=NextGen,DC=nextgen,DC=int" | Select-Object Name,SamAccountName confirms that all 8 users (two per department) were successfully created, with the correct display names and logon names.

Checking Group Members

```
PS C:\Users\Administrator> foreach($g in @("IT","MKT","FIN","HR")){Get-ADGroupMember -Identity $g | Select-Object Name}

Name
----
IT User1
IT User2
MKT User1
MKT User2
FIN User1
FIN User2
HR User1
HR User2


PS C:\Users\Administrator>
```

Running the foreach loop against each of the four groups (IT, MKT, FIN, HR) confirms that each department's two users were correctly added as members of their matching security group.

7. Delete and Rebuild Commands

If we ever need to clean out this entire structure, and start fresh from scratch, we can use the following three steps.

Step 1: Unprotect

Since every OU in our setup script was created with protection from accidental deletion turned off, this step is mainly a safeguard, in case that protection was ever manually turned back on. This command finds every OU under NextGen, and makes sure deletion protection is switched off for all of them.

```
Get-ADOrganizationalUnit -Filter * -SearchBase "OU=NextGen,DC=nextgen,DC=int" -SearchScope
Subtree | Set-ADOrganizationalUnit -ProtectedFromAccidentalDeletion $false
```

Step 2: Delete

This command deletes the entire NextGen OU, along with everything inside it (all department OUs, all sub-OUs, all groups, and all users), in one single action.

```
Remove-ADOrganizationalUnit -Identity "OU=NextGen,DC=nextgen,DC=int" -Recursive -
Confirm:$false
```

Step 3: Rebuild

Once the old structure has been completely deleted, we simply run the full setup script from Section 4 again, to rebuild the entire OU structure, groups, and users from scratch.

Deep Dive into AD Server: Backup, Restore, Inheritance, Enforcement, Restricted Groups, and WMI Filtering

Best Practices: Why Keep One Physical Domain Controller

Nowadays, many companies virtualize their entire network, including their Active Directory Domain Controllers, meaning these Domain Controllers run as virtual machines rather than on dedicated physical hardware. However, best practice still recommends keeping at least one Domain Controller running on actual, physical hardware, rather than virtualizing every single one.

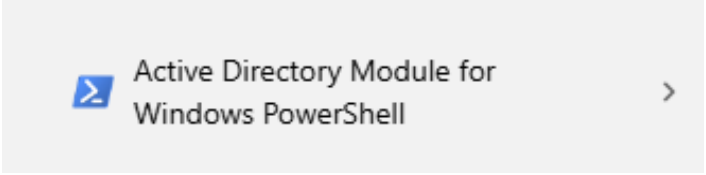
Here is the reasoning behind this recommendation: if the entire virtualization cluster (the host servers running all of our virtual machines) were to go down completely, for example due to a power failure or hardware crash, the virtual cluster itself cannot restart or authenticate itself properly, without an already active Active Directory server available to handle authentication and cluster services. If every single Domain Controller was also virtualized, and living on that same crashed cluster, there would be no working AD server available anywhere, to help the cluster restart and authenticate itself again. Keeping one physical, standalone Domain Controller avoids this particular chicken-and-egg problem entirely.

Reverse Lookup Zones for Multiple Departments

If a company has many different departments, each potentially using a different IP address range, then a separate Reverse Lookup Zone needs to be created for each of these individual IP ranges, so that reverse (IP-to-name) lookups work correctly across every department's network.

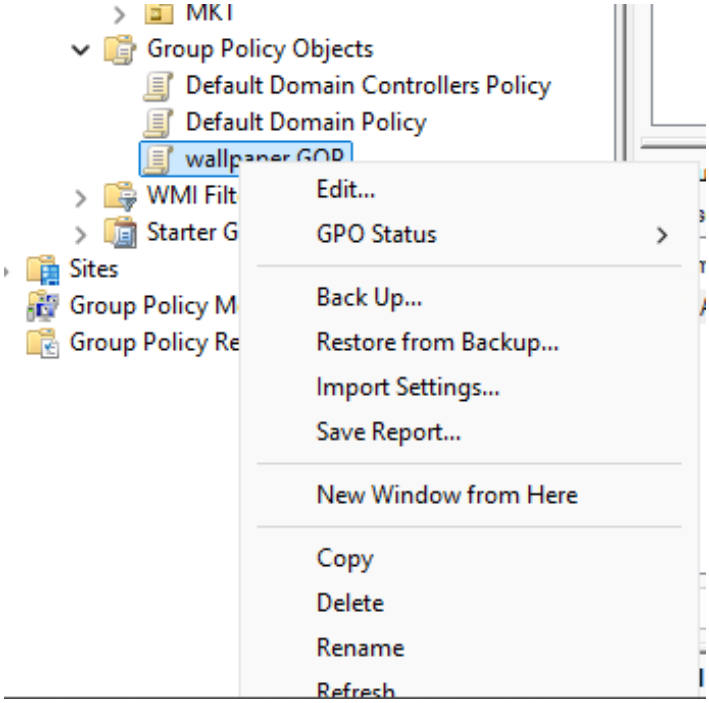
Creating OUs with PowerShell

It's important to remember that Organizational Units can indeed be created using PowerShell, but this cannot be done using a normal, plain PowerShell window. Instead, this must be done specifically using the "Active Directory Module for Windows PowerShell," which loads the special AD-related cmdlets (such as New-ADOrganizationalUnit) that a regular PowerShell session does not have access to by default.

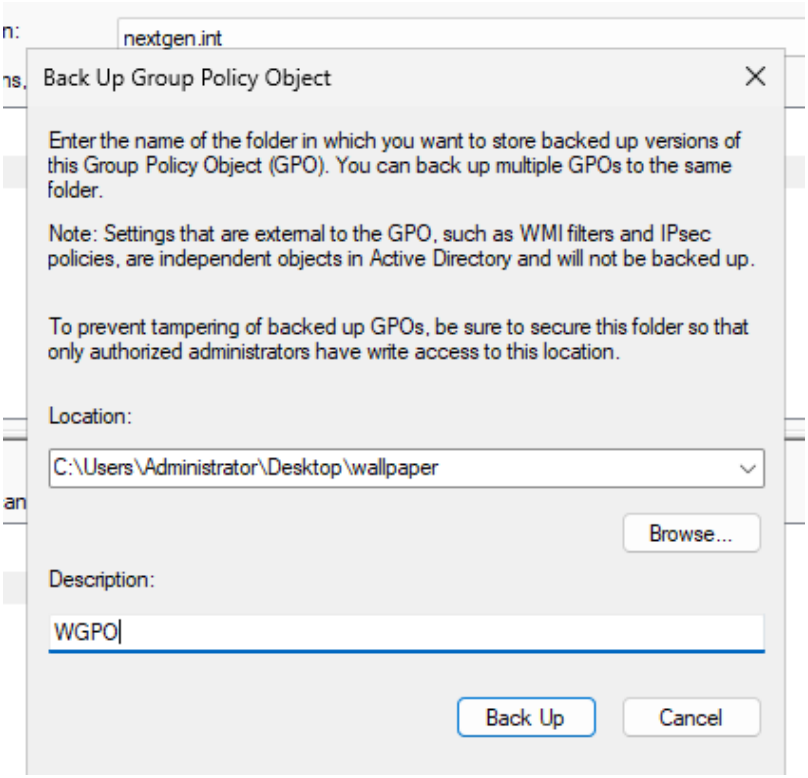


How to Back Up a Group Policy

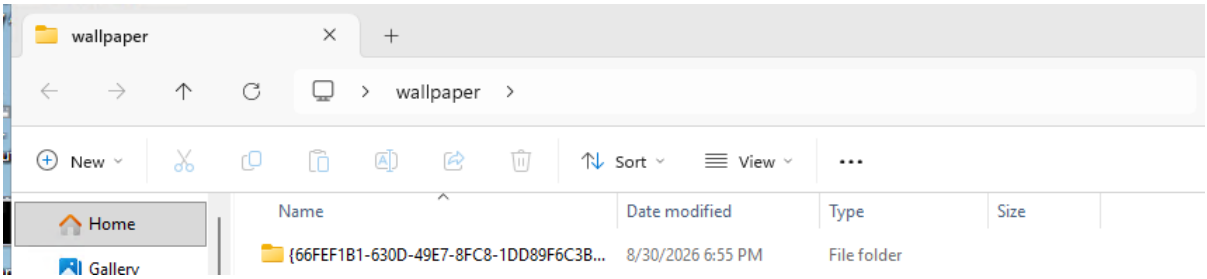
Group Policy Objects (GPOs) can be backed up at any time, which is very useful in case a policy is later accidentally changed, deleted, or broken.



To do this, we right-click on the specific policy we want to back up (in this case, the "wallpaper GPO"), and select "Back Up..." from the menu.



A window then appears, asking us to provide a Location, meaning the folder where we want this backup to be saved, along with a Description, which helps us remember what this particular backup contains, later on. Once we click "Back Up," the policy's settings are saved to that folder.



After the backup completes, we can see a new folder appear inside our chosen backup location, named with a long, unique identifier (a GUID). This folder contains all of the actual backed-up settings and data for that GPO.

How to Restore a Policy

Imagine that our GPO file was accidentally deleted, and now we want to bring it back. It's important to understand that this is not simply done using a "restore" action alone; we actually need to understand the difference between "Restore" and "Import," since they serve two very different purposes.

Restore GPO (Fixing What Is Broken)

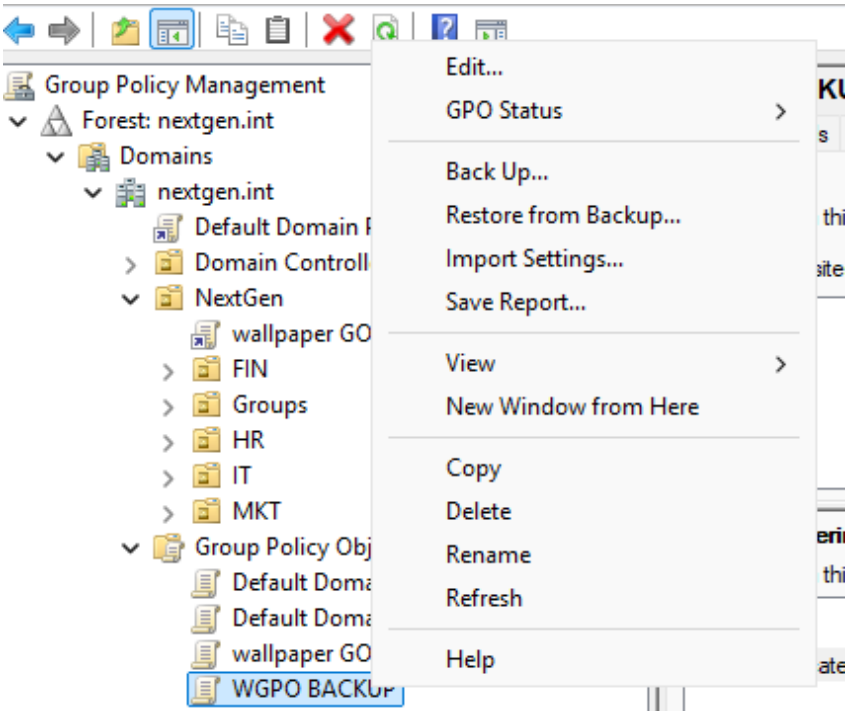
What it is: This brings a deleted or broken GPO back to life, exactly the way it was before it was deleted or broken.

- **How it works:** It uses the backup taken from the same domain, and recreates the GPO using its original, exact ID (its GUID).
- **When to use it:** Use this specifically for Disaster Recovery, for example, when we accidentally delete a working GPO, and want to undo that mistake.

Import Settings (Copying Settings Into a New GPO)

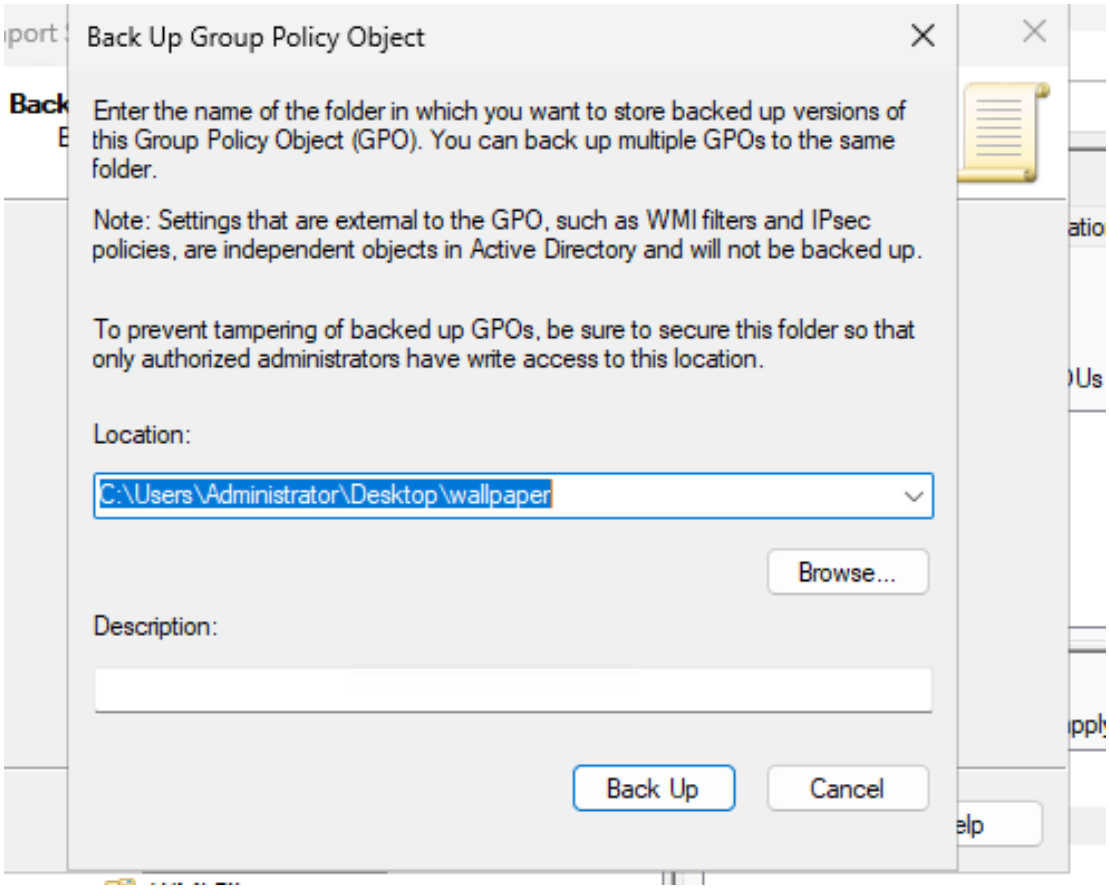
What it is: This copies the rules and settings from a backup, and pastes them into a brand-new, completely empty GPO.

- **How it works:** It does not care about the original GPO's ID at all. It simply reads the settings from the backup, and injects them into whichever new GPO we are working with.
- **When to use it:** Use this specifically for Reusability, for example, when we want to copy a useful GPO configuration from a friend's lab, or from a completely different domain, into our own lab environment.

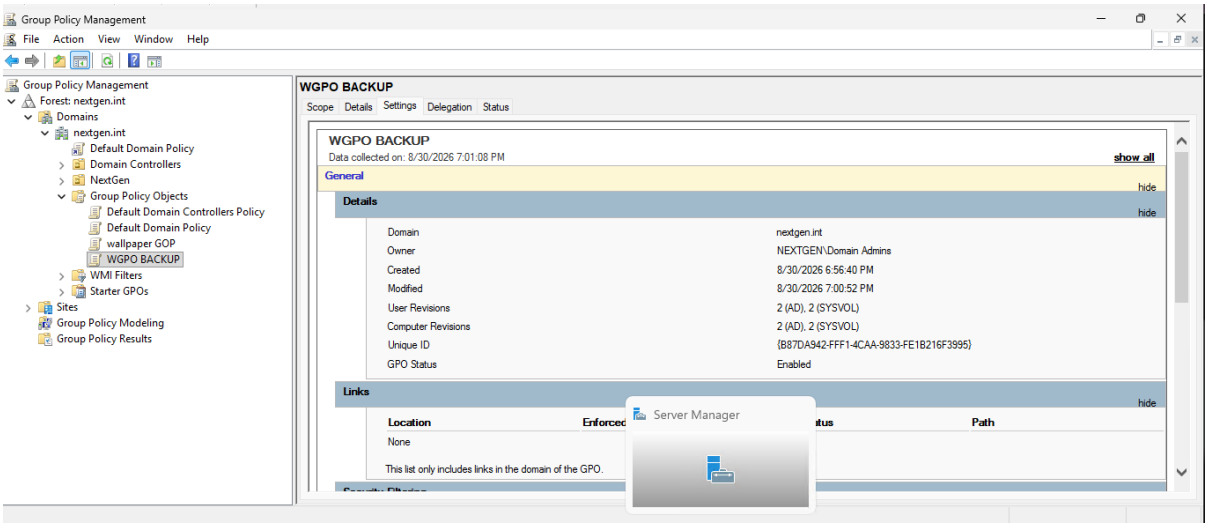


Practical Example: Importing a Backed-Up GPO

In this example, we created a brand-new, empty GPO named "WGPO BACKUP." We then right-clicked on it, and selected "Import Settings...", to bring in the previously backed-up wallpaper GPO settings.

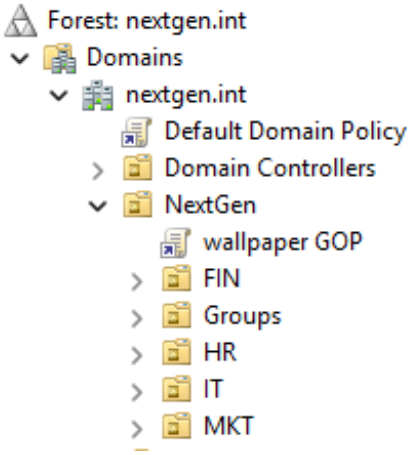


The wizard then asks us to select the folder where our backup is stored. We select the correct folder, and click Next.



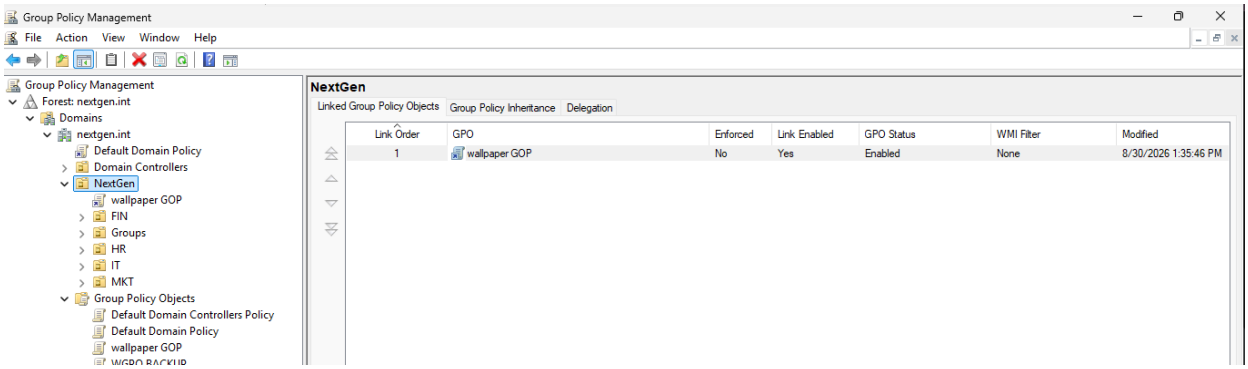
After clicking Next, and finally clicking Finish, the import process completes. As shown above, we can now see the "WGPO BACKUP" listed in our Group Policy Objects folder, containing an exact copy of the wallpaper GPO's original settings, but as a completely separate, brand-new GPO with its own new Unique ID (visible in the Details section of this report). This confirms that Import truly does create a new object, rather than restoring the original one.

Policy Inheritance



In our lab, nextgen.int is the root domain of our Domain Controller. If we link (add) a GPO at this root domain level, it will automatically be applied down to all of the OUs that exist underneath it. This automatic "flow-down" behavior is what we call Inheritance.

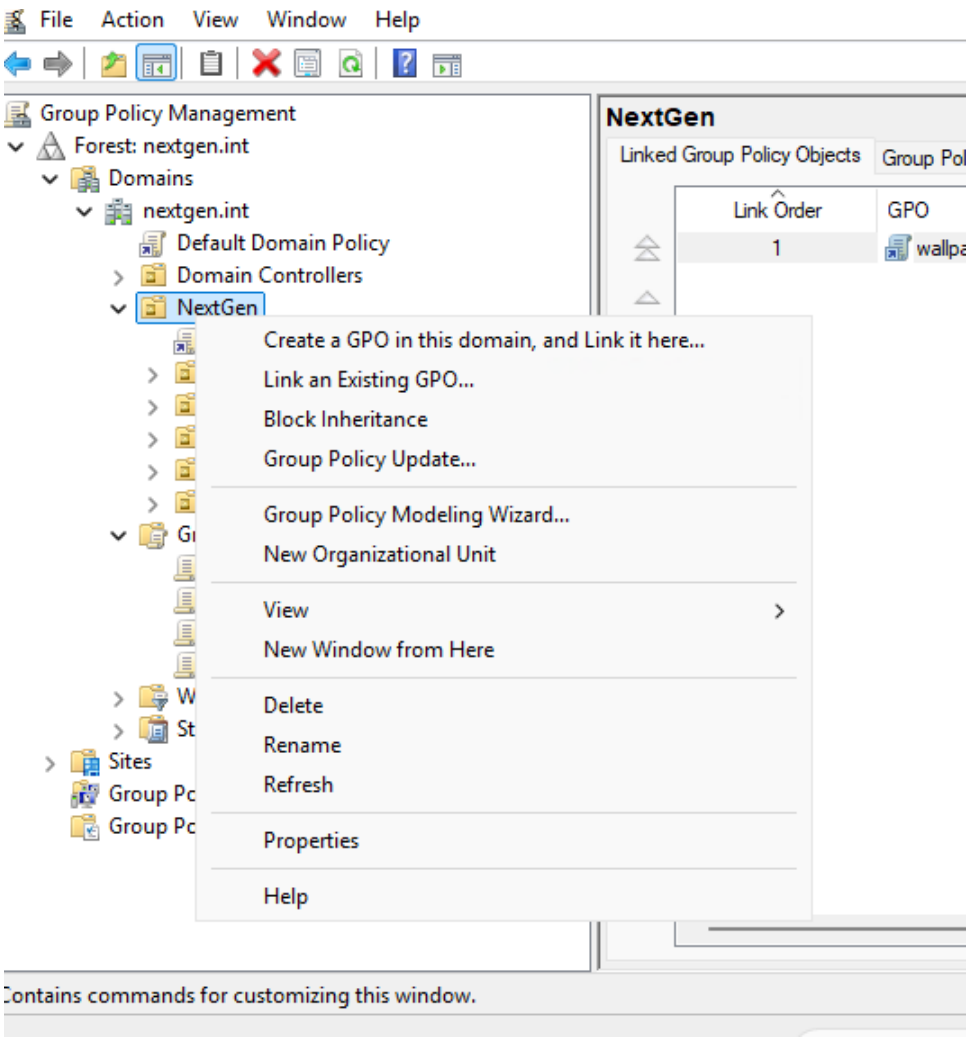
In the example below, we can see that the "Wallpaper GPO" is linked directly to the NextGen OU, meaning its settings automatically flow down and apply to every single OU that exists underneath NextGen (such as FIN, HR, IT, and MKT), without needing to link the policy separately to each one of those individual OUs.



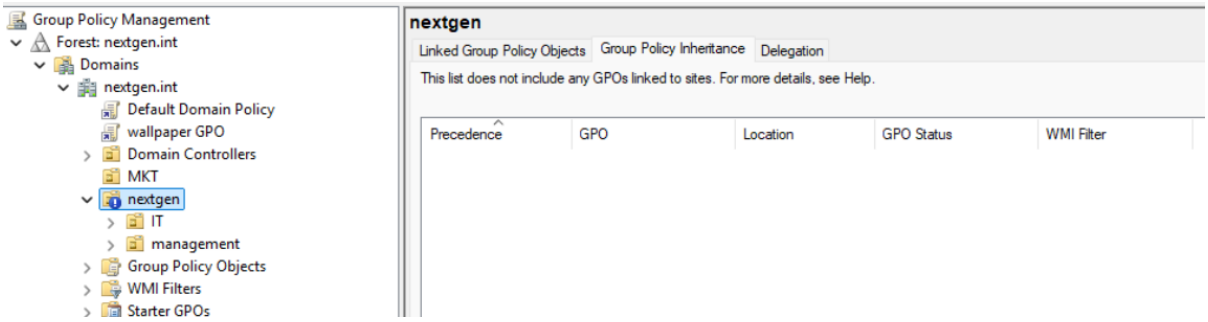
Blocking Inheritance

Sometimes, we may not want a particular child OU to automatically receive policies from its parent OU above it. We can achieve this using an option called Block Inheritance.

To do this, we simply right-click on the specific OU where we want to stop inheritance (in this case, the "nextgen" OU), and select "Block Inheritance" from the menu.



After applying Block Inheritance, if we check that OU's Group Policy Inheritance tab, we will see that no GPOs are being applied to it anymore at all, even ones that were previously being inherited automatically from its parent.

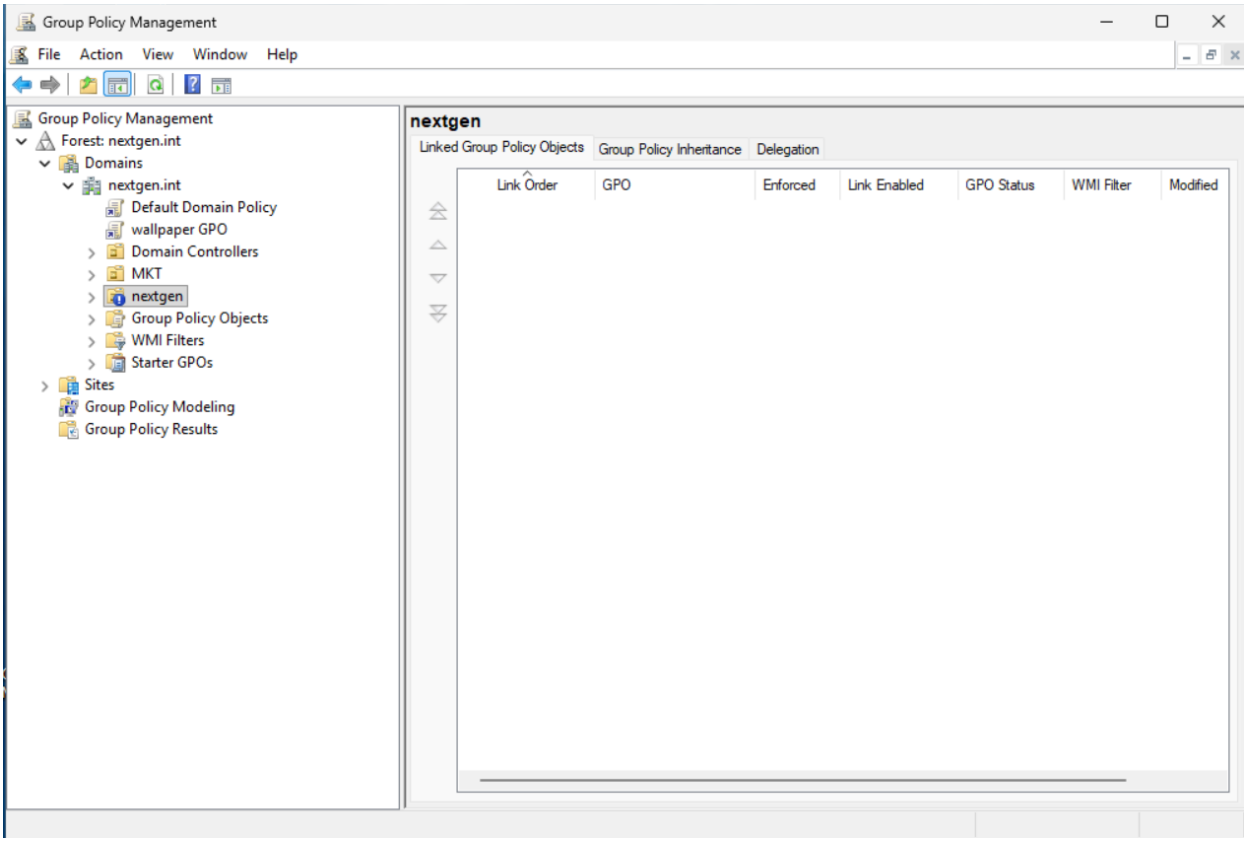


Policy Enforced

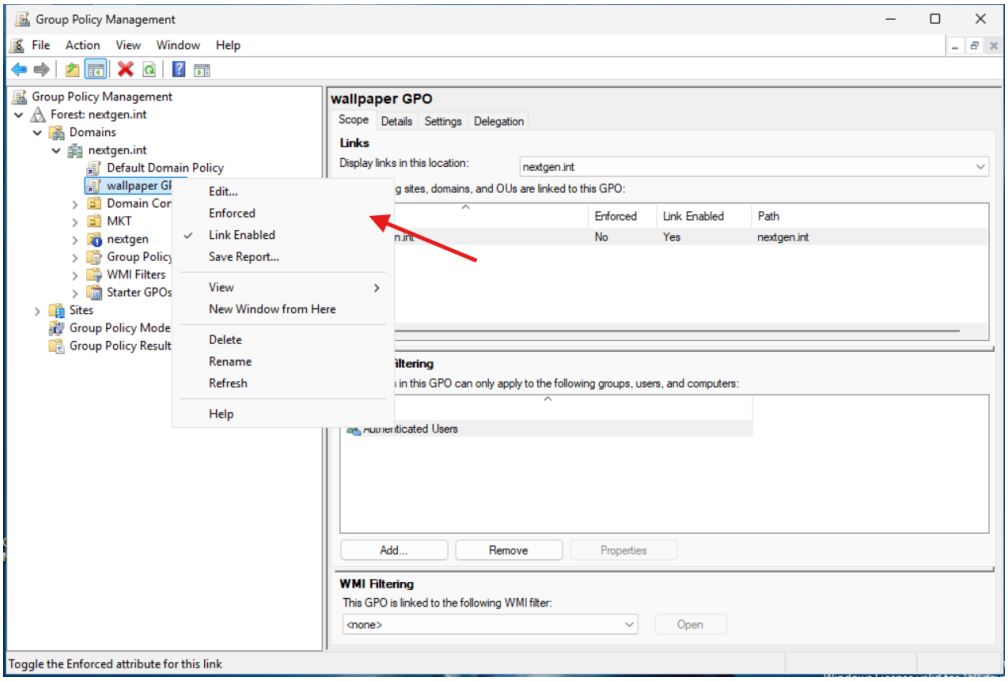
When we use the "Block Inheritance" option on an OU, that particular OU no longer receives any policies from above it, as we just saw. However, using an option called "Enforce," we can still force a specific policy to apply to that OU, even when that OU has Block Inheritance turned on.

In other words, the "Enforce" option allows a specific policy to bypass Block Inheritance completely, and apply to all of its child OUs regardless.

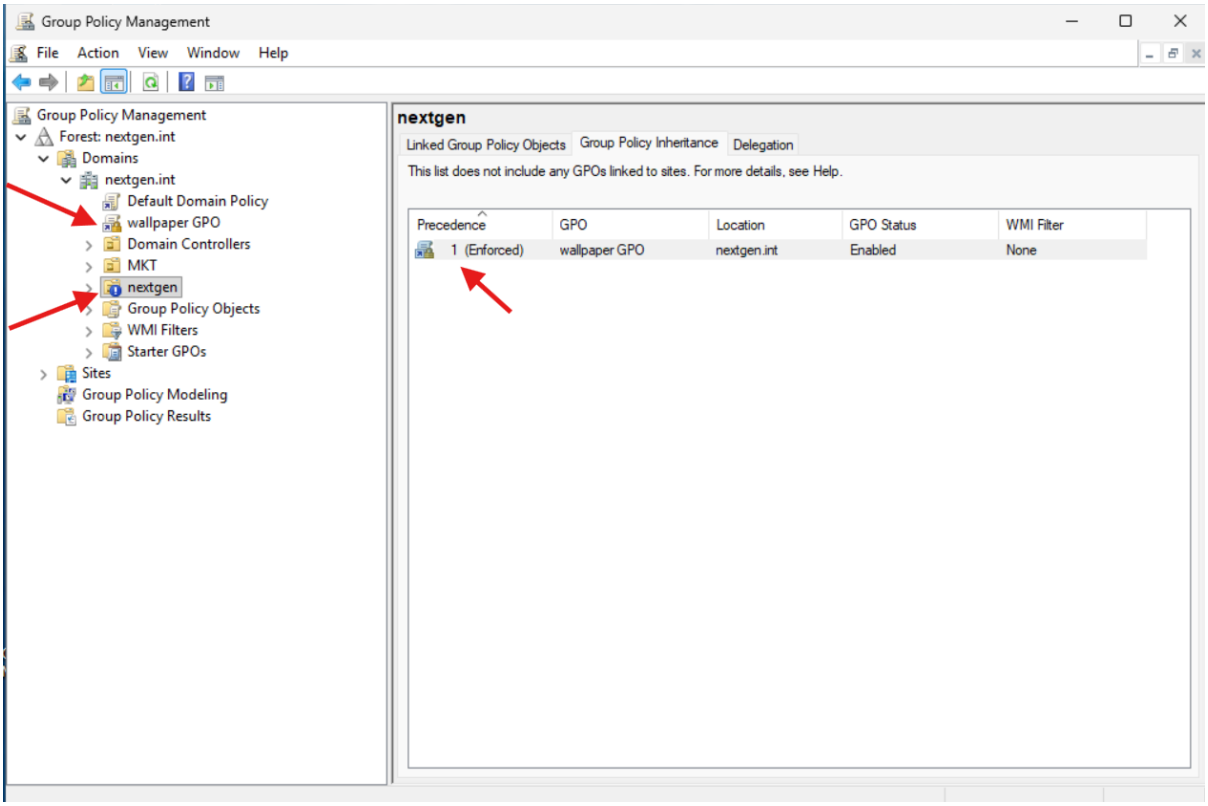
Let's consider our wallpaper GPO example again. We had already applied "Block Inheritance" to the nextgen OU, meaning it currently has no policies applied to it at all.



Now, we go back to the Wallpaper GPO itself, and select the "Enforced" option from its right-click menu.



Once we enable "Enforced" on the Wallpaper GPO, this particular policy is now automatically applied to all of its child OUs again, even though those child OUs still have Block Inheritance status turned on. As shown below, the wallpaper GPO now appears in the nextgen OU's inheritance list, clearly marked as "1 (Enforced)."

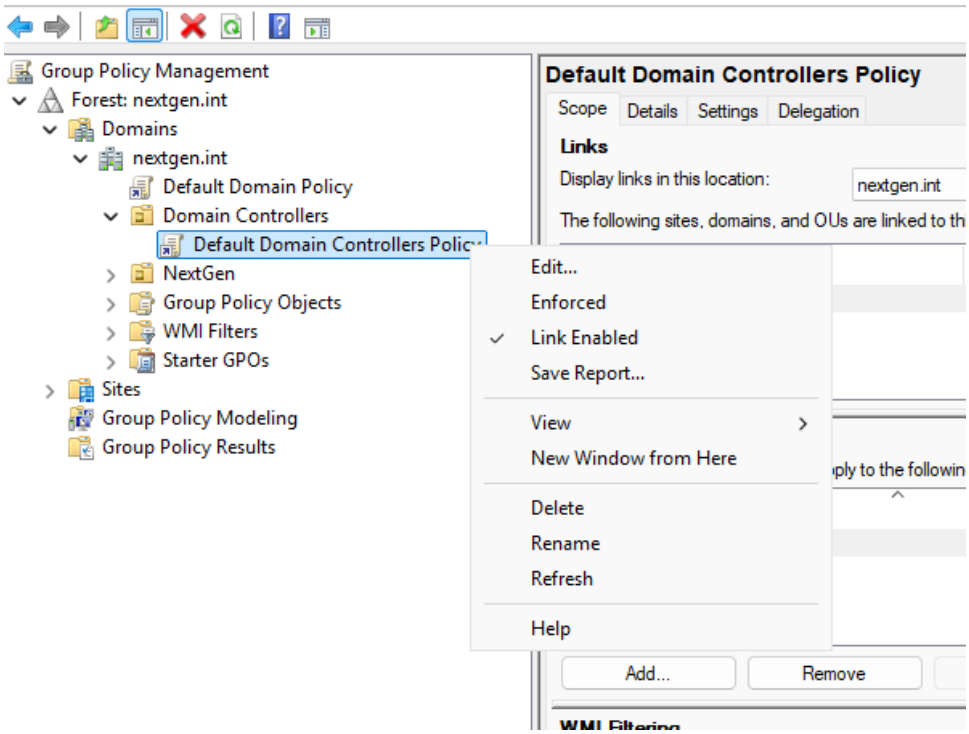


Difference Between Local Policy and Group Policy

- **Local Policy:** This applies only to one specific, individual computer. It is managed locally, directly on that machine, using a tool called gpedit.msc. This is typically used for standalone computers that are part of a workgroup, rather than a domain.
- **Group Policy (GPO):** This applies to multiple users and computers at once, across an entire domain, or across specific OUs within that domain. It is managed centrally, from the Domain Controller, using a tool called gpmc.msc (Group Policy Management Console).

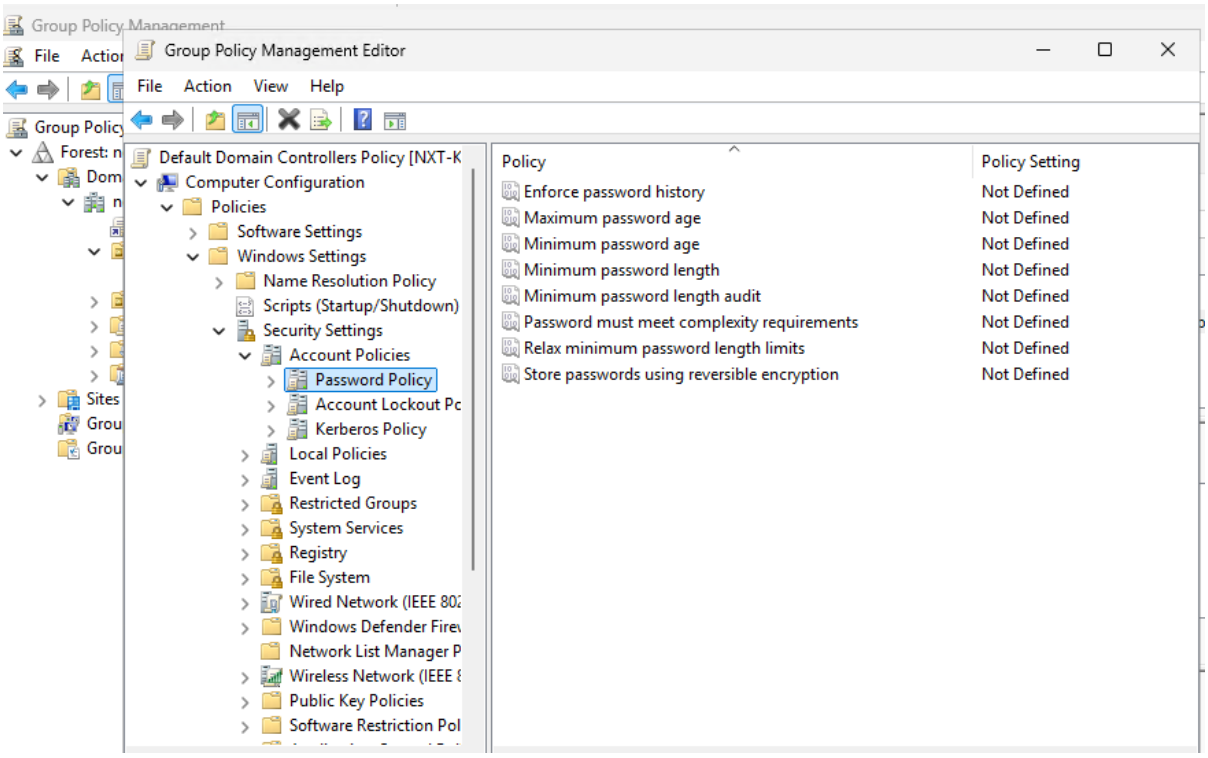
Practical Example: Changing the Password Policy in the Default Domain Policy

Let's walk through an example of changing the password policy, directly within the Default Domain Policy. We start by right-clicking on the Default Domain Policy, and selecting Edit.

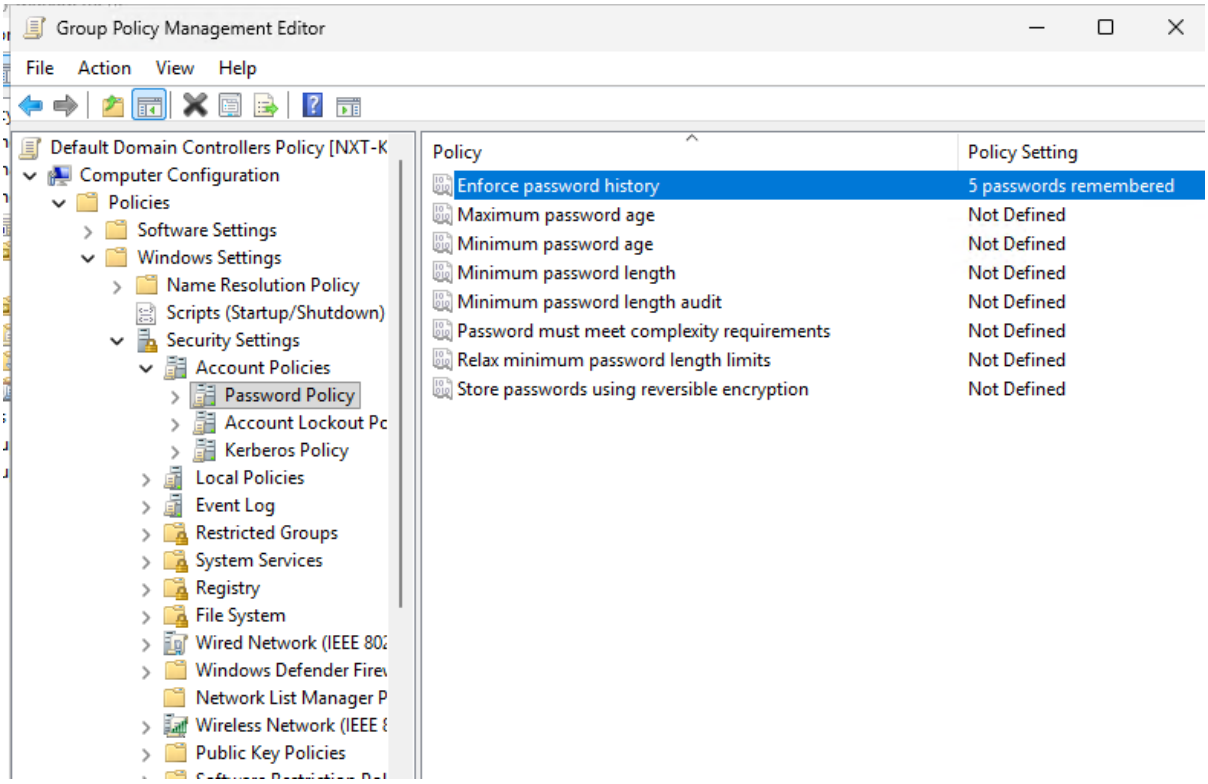


We then navigate to this specific path, inside the policy editor:

```
Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Password Policy
```



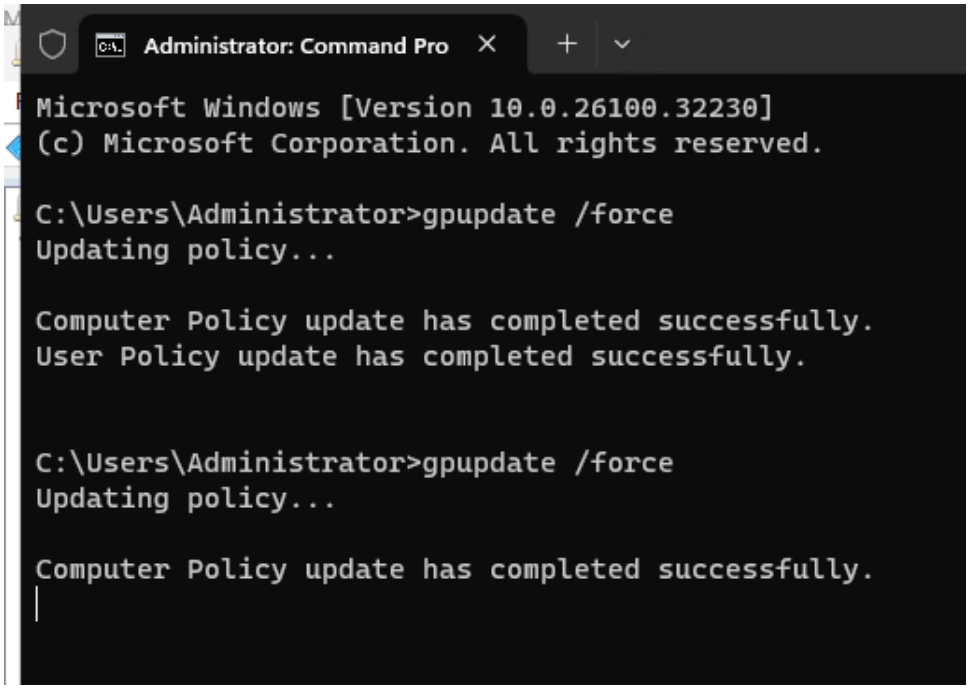
At this point, we can see all of the various password-related settings, currently showing as "Not Defined." We then go ahead and change a few of these settings to our desired values.



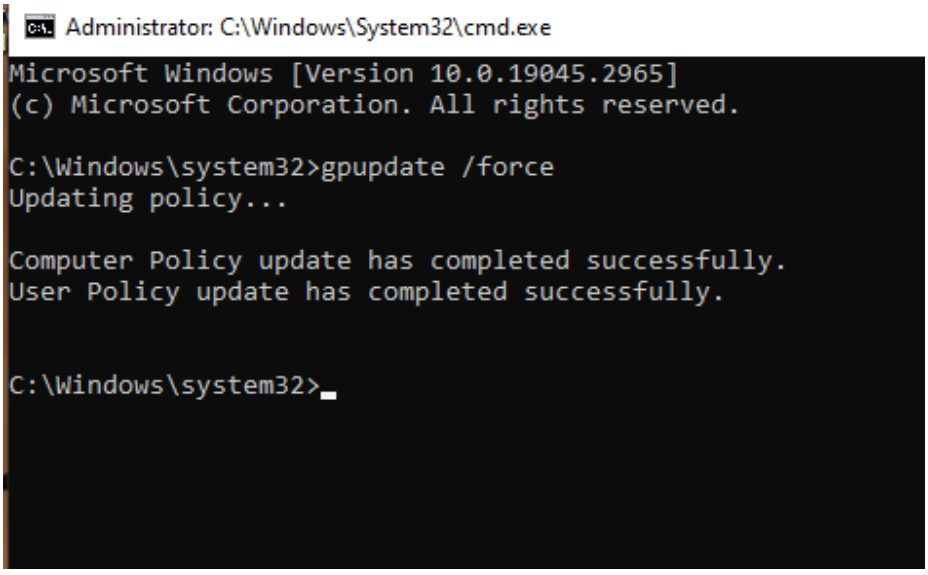
In our example, we changed the "Enforce password history" setting to "5 passwords remembered," meaning the system will remember the last 5 passwords used by each account, and will not allow a user to immediately reuse any of them.

Applying the Policy Update via Command Prompt

Next, we update the policy immediately, using the gpupdate /force command, run directly on the server.



Now, let's check whether this same policy change has correctly applied to a client PC as well. We open Command Prompt as Administrator on the client PC, and run the same update command there.



Checking the Applied Policy Using RSoP

Next, to check whether this password policy change has actually taken effect on the client PC, we can type "rsop" into Command Prompt on that client machine. RSoP stands for Resultant Set of Policy, and it shows us a clear report of exactly which policies are currently active on that specific machine, and where each one came from.



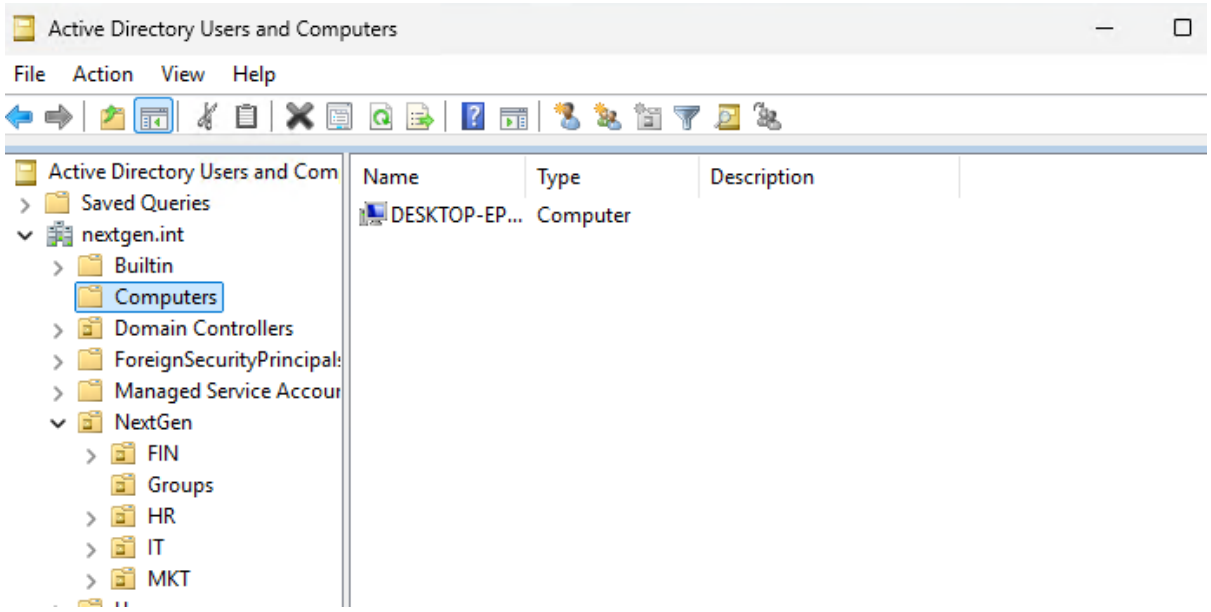
Looking at this report, we notice something interesting: the "Enforce password history" setting is now showing as "12 passwords remembered," not the "5" value we had just set. This happens because we made our change inside the Default Domain Policy, which is linked directly at the root of the entire domain. Since this policy is applied at the very top of the domain, it naturally takes effect for every single computer within that domain, including our client PC, this is expected, and not a problem at all; it simply confirms that a root-level policy change applies domain-wide, as intended (with the displayed value here reflecting whatever value was actually saved during editing).

Applying Policies to Other OUs (Not the Root Domain)

When we want to apply a GPO to a specific OU, rather than the entire root domain, there is an important detail to be aware of, for Computer Configuration policies specifically: if the target client computer's own computer object is not physically located inside that specific OU, the policy will not be applied to it, no matter how correctly the GPO itself is configured.

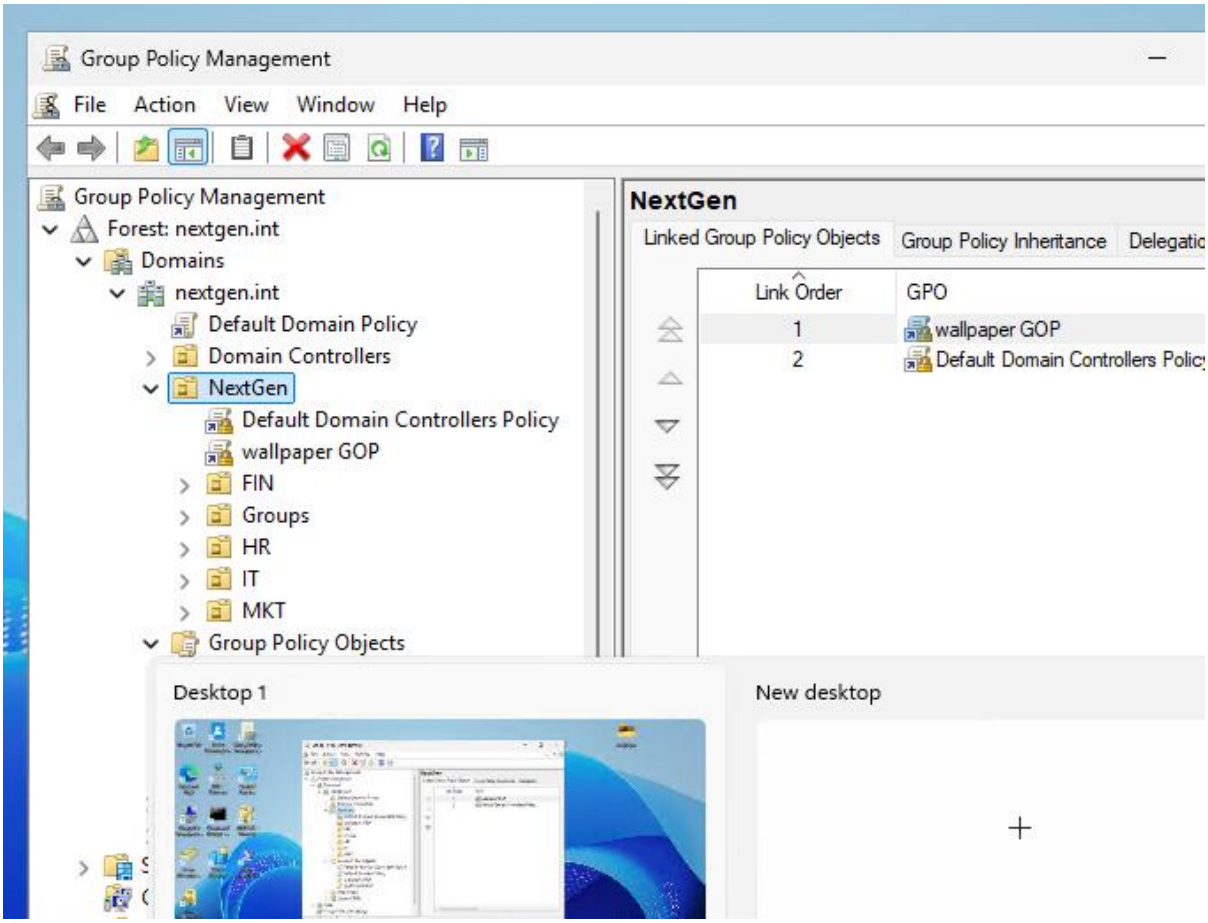
Because of this, before assigning a Computer Configuration policy to a particular OU, we must first move the actual computer object into that relevant OU.

Let's check where our client PC's computer object currently exists, inside Active Directory Users and Computers.

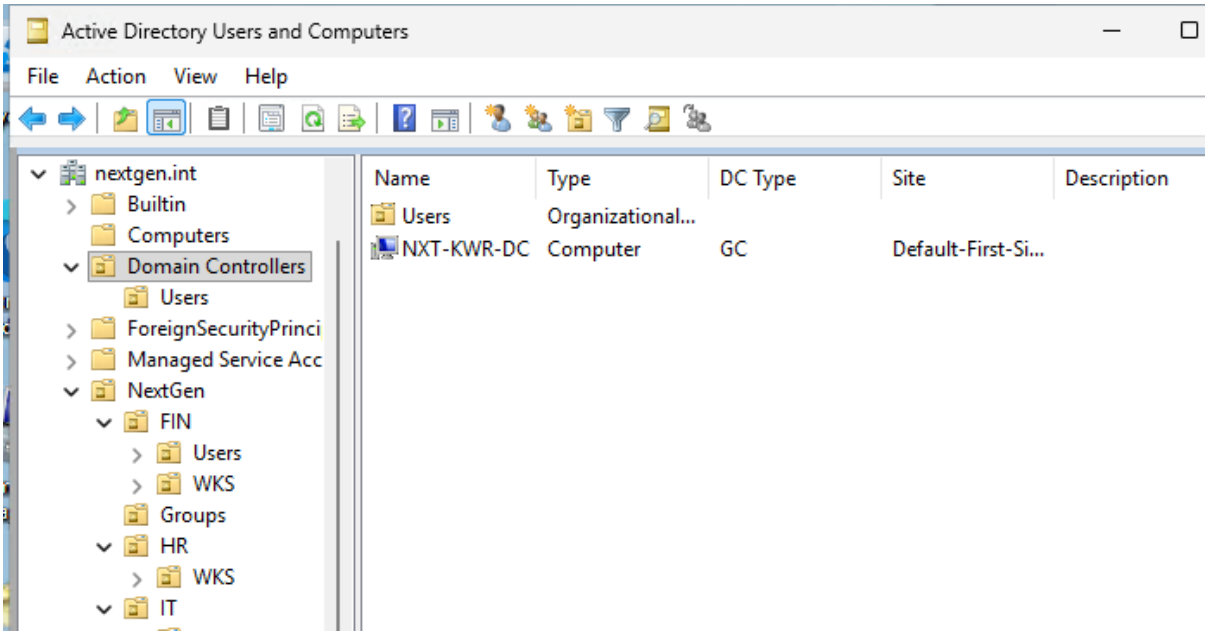


As shown above, our client computer currently exists inside the default "Computers" container, rather than inside any of our custom departmental OUs (such as IT, HR, FIN, or MKT).

Because of this, if we look inside the Group Policy Management console, our custom OUs will not show this particular computer as being affected by department-specific policies, since the computer object itself has not yet been moved into the correct OU.



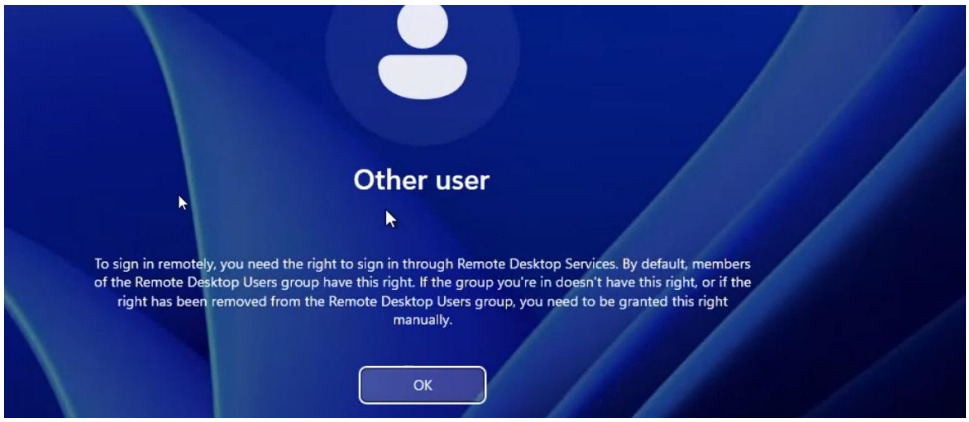
To fix this, we simply drag and drop the computer object, from the default Computers container, into the correct, relevant OU (for example, moving it into the WKS sub-OU, under the correct department).



An Important Clarification: User Policies vs Computer Policies

- **User Policies will work perfectly,** even without moving the computer object, because our user accounts are already correctly created and stored inside our own custom OUs (such as IT or Management).
- **When a user logs in,** Active Directory applies GPOs based on that specific user's OU location, completely regardless of where the computer object itself happens to be located.
- **We only need to drag and drop the computer object** specifically when we want to apply Computer Configuration Policies to that particular machine.

Understanding a Common Hyper-V Login Error



This is a very common error that comes up when working inside Hyper-V virtual machines. The error means: the specific user account we are trying to log in with does not have permission to log in through Remote Desktop Services (RDP).

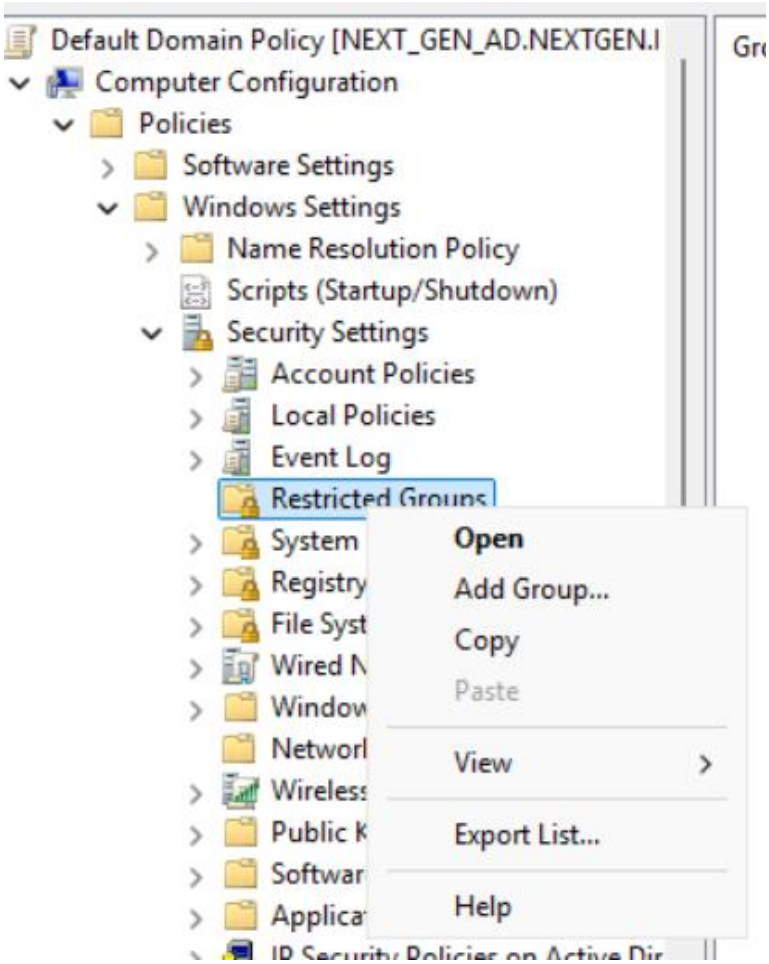
Fixing This: Granting Remote Desktop Permission via Group Policy

Let's walk through how to properly grant this permission, to an entire OU or group at once, using Group Policy, instead of configuring it manually on each individual machine.

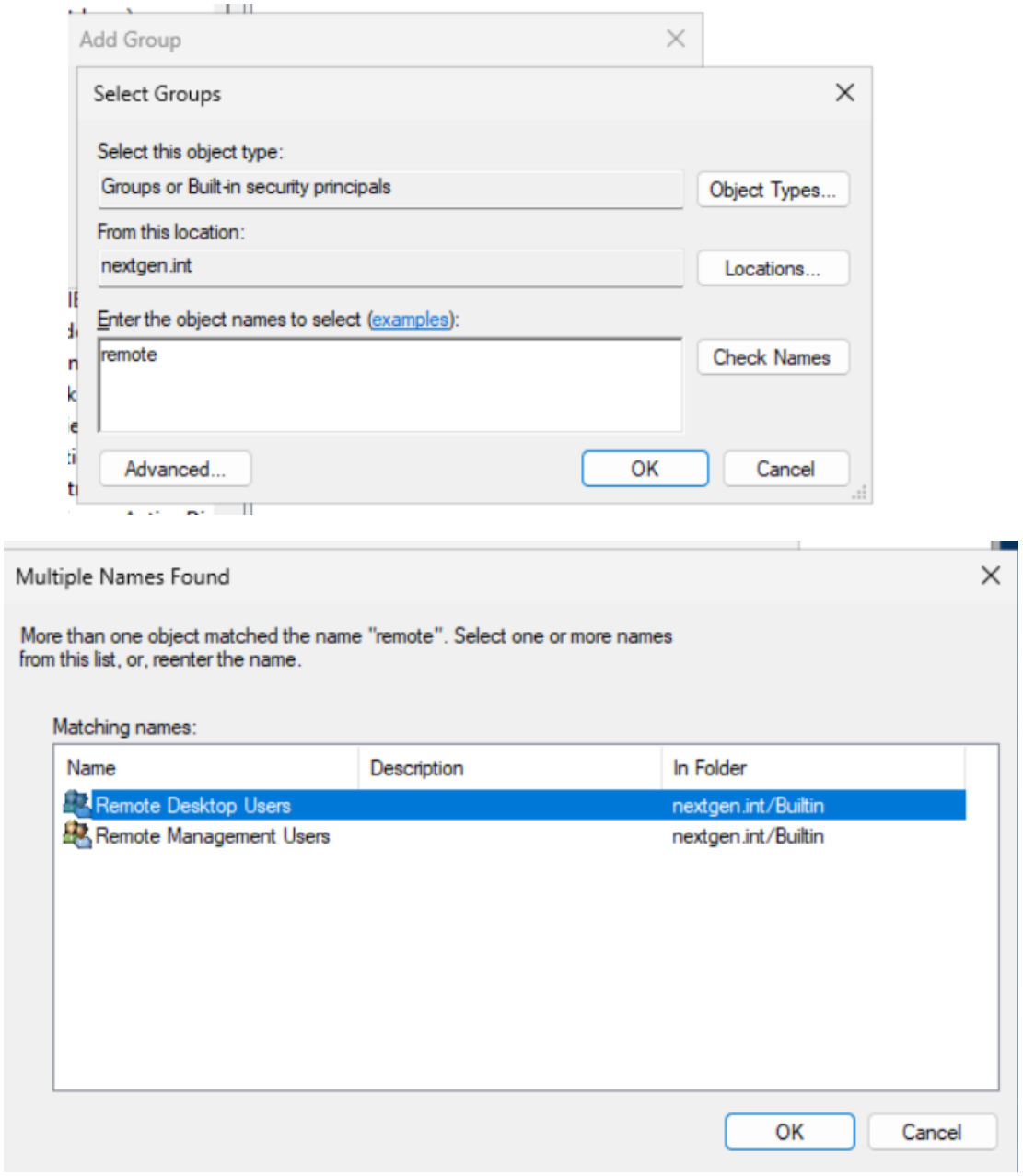
We start by right-clicking the Default Domain Policy, selecting Edit, and navigating to this path:

```
Computer Configuration > Policies > Windows Settings > Security Settings > Restricted Groups
```

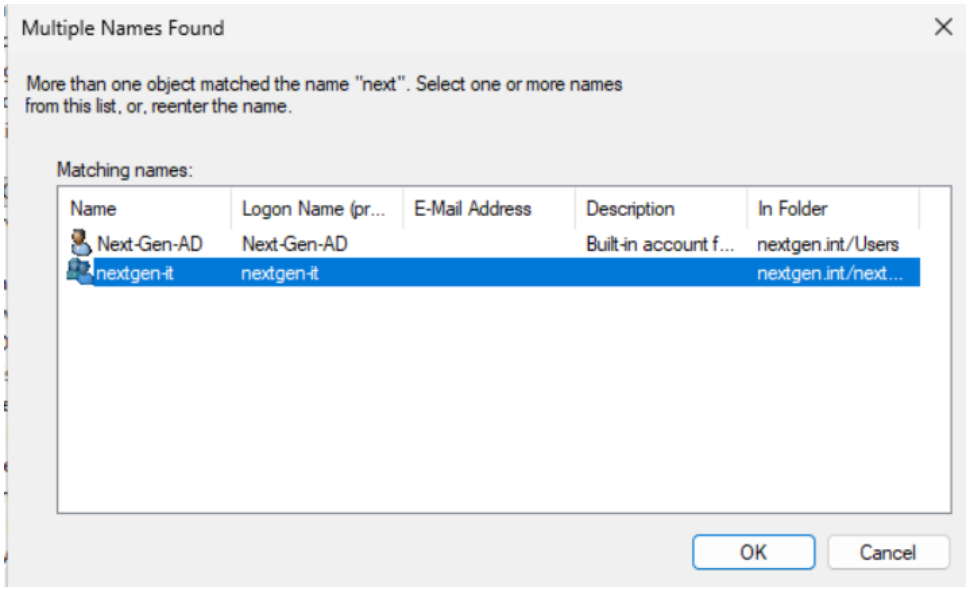
We then right-click on "Restricted Groups," and select "Add Group..."



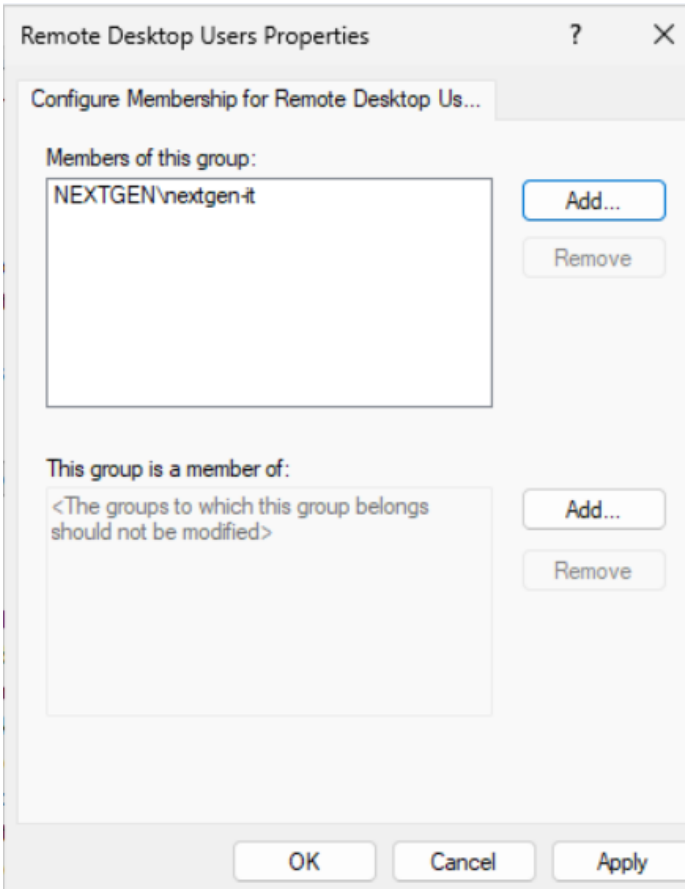
In the window that appears, we click Browse, and select the built-in "Remote Desktop Users" group.



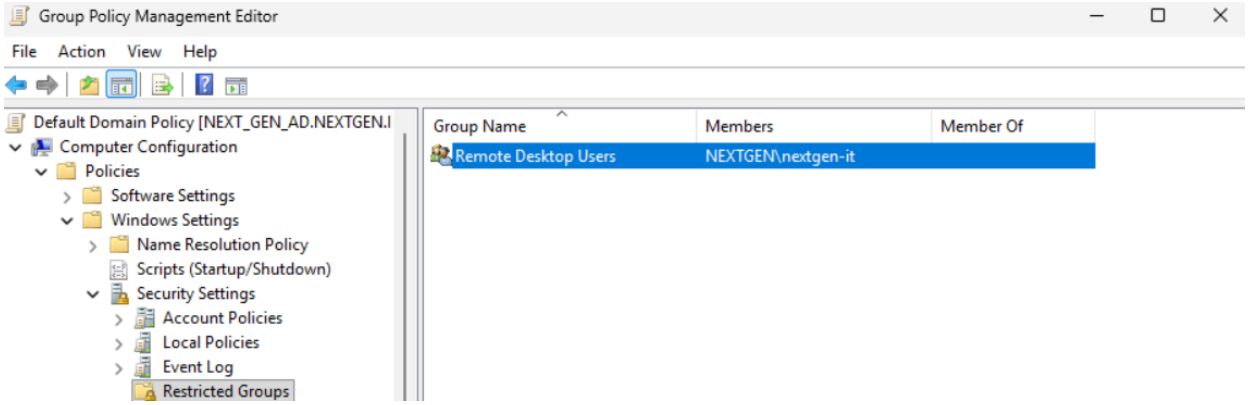
After clicking OK, we now need to add our own specific group as a member of this Remote Desktop Users group. We click "Add..." under the "Members of this group" section, then click Browse, and select the group we want to grant this permission to (in this case, our "nextgen-it" security group).



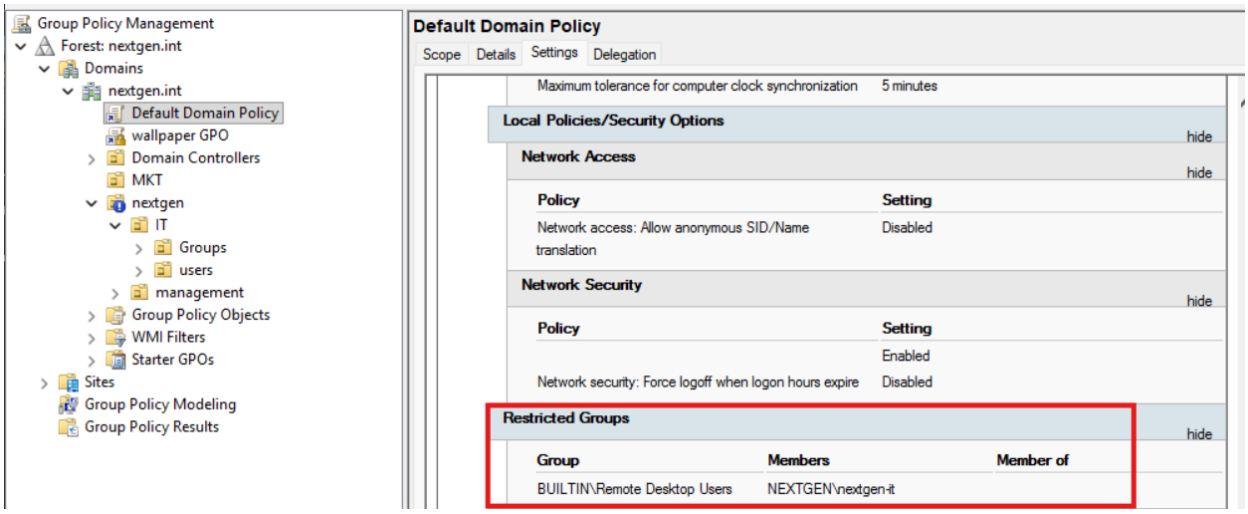
After selecting our group, we click Apply, and then OK.



As shown in the final result below, our "nextgen-it" group (shown as NEXTGEN\nextgen-it) is now correctly listed as a member of the Remote Desktop Users group.

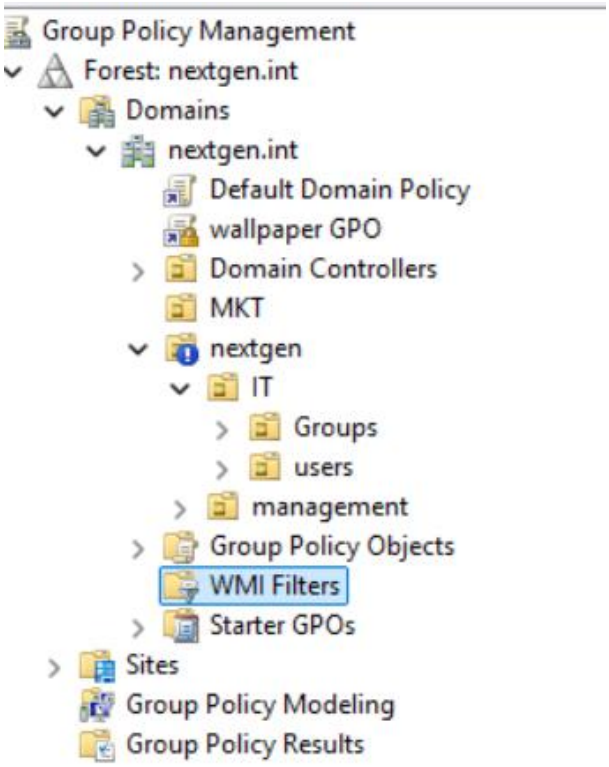


Finally, exactly as before, we apply this policy change using the gpupdate /force command, run on each of the relevant computers, so that the new Remote Desktop permission takes effect immediately.



The settings report above confirms our final configuration: within the Default Domain Policy, under Restricted Groups, the built-in "Remote Desktop Users" group now correctly lists "NEXTGEN\nextgen-it" as one of its members, meaning every user inside that security group is now permitted to log in remotely, via RDP, without encountering the earlier permission error.

WMI Filtering



WMI (Windows Management Instrumentation) Filtering is an advanced feature within Active Directory Group Policy, which allows administrators to dynamically target Group Policy Objects, based on specific hardware or software attributes of the target computer.

Unlike standard GPO linking, which simply targets entire OUs, WMI filtering actually evaluates the machine's own environment, before the policy is applied at all. This means a policy can be set up to only apply to computers that meet very specific technical conditions, regardless of which OU those computers happen to sit inside.

Practical Use Cases in Enterprise Environments

Administrators commonly use WMI filtering to separate out different policies, without needing to create messy, overly complicated OU structures. A few common examples include:

1. Operating System Filtering

This applies specific software settings only to machines running a particular operating system version, for example, applying a setting only to Windows 11 machines, or only to Windows Server 2022 machines.

Example WQL query (targets Windows 10 or 11 only):

```
SELECT * FROM Win32_OperatingSystem WHERE Version LIKE "10.%"
```

2. Chassis Type Filtering (Laptops vs Desktops)

This applies power management or Wi-Fi profile settings strictly to mobile devices (laptops), by specifically checking for the presence of a battery in that machine.

Example WQL query:

```
SELECT * FROM Win32_Battery WHERE BatteryStatus > 0
```

3. Hardware Architecture Filtering

This makes sure that 64-bit software packages are only ever deployed to machines running a 64-bit operating system, avoiding compatibility issues on older, 32-bit systems.

Example WQL query:

```
SELECT * FROM Win32_Processor WHERE AddressWidth = "64"
```

Managing Shared Folders and Software with Group Policy

A Beginner's Guide to Drive Mapping, WMI Filtering, Software Deployment, and Browser Management in Active Directory

1. What Is Drive Mapping?

Imagine your company keeps important files, like HR documents or IT files, on a central server. Normally, to open those files, an employee would have to type in a long computer address (called a network path), such as \\server\folder. This is slow and confusing for most users.

Drive Mapping solves this problem. It takes that long network address and turns it into a simple drive letter, like N: or H:, that shows up under "This PC" on the employee's computer — just like a USB drive or the C: drive. The employee can then click on that letter to open the shared folder, without ever needing to know the real address.

This is done using a Windows Active Directory feature called Group Policy Preferences (GPP). Active Directory is the system that manages all the computers and users in a company network, and Group Policy is the tool used to push out settings — like drive mappings — to many computers at once, automatically.

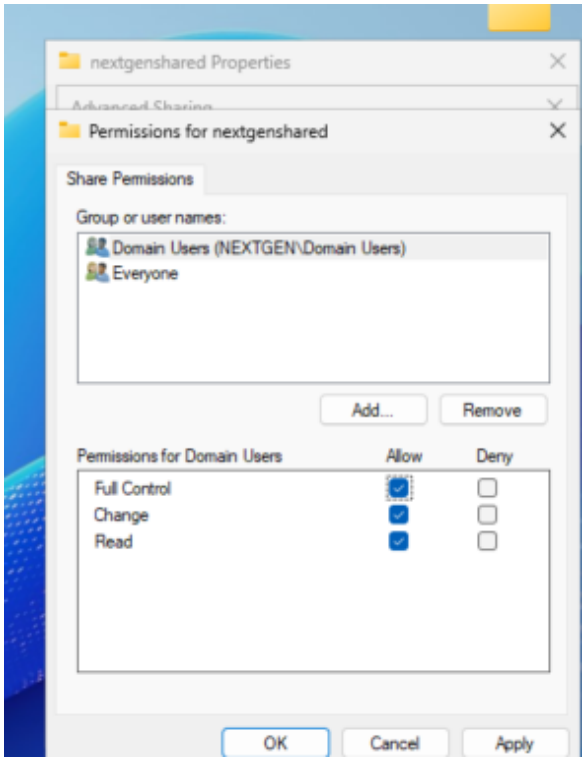
Why Companies Use Drive Mapping

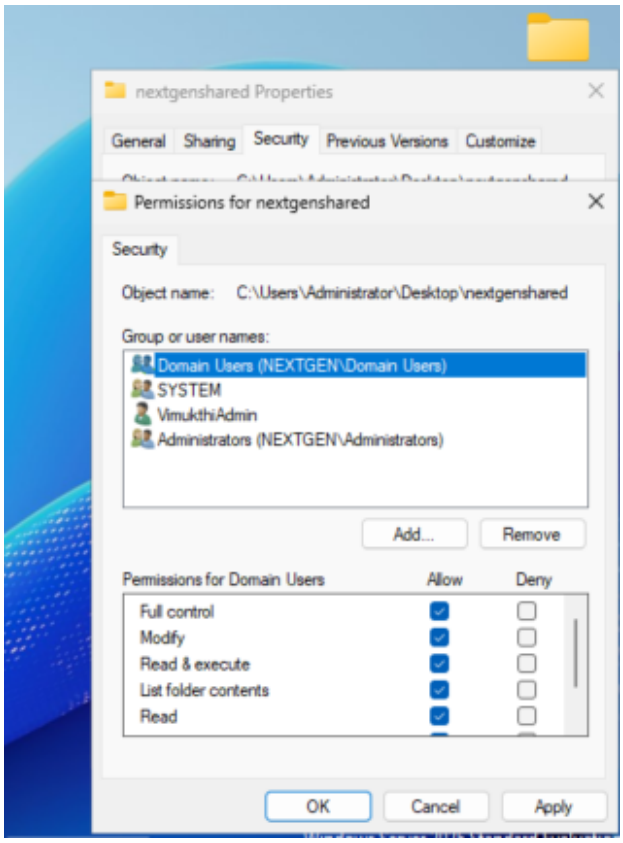
- **Easy access:** Staff simply click a drive letter instead of typing a network address.
- **Different drives for different teams:** For example, the HR team might get an H: drive, while the IT team gets an I: drive. Each team only sees the folder meant for them.
- **Automatic setup:** The drive appears or disappears on its own, based on Group Policy rules and who is logged in. Admins do not need to visit every computer by hand.

Step 1: Share the Folder

Before we can map a drive to a folder, that folder must first be shared on the network, and we must decide who is allowed to access it. This is done using "Advanced Sharing" and the folder's Security settings (permissions).

In the example below, only the HR group is given access, instead of giving access to every user in the company ("Domain Users"). This keeps the folder private to the people who should see it.

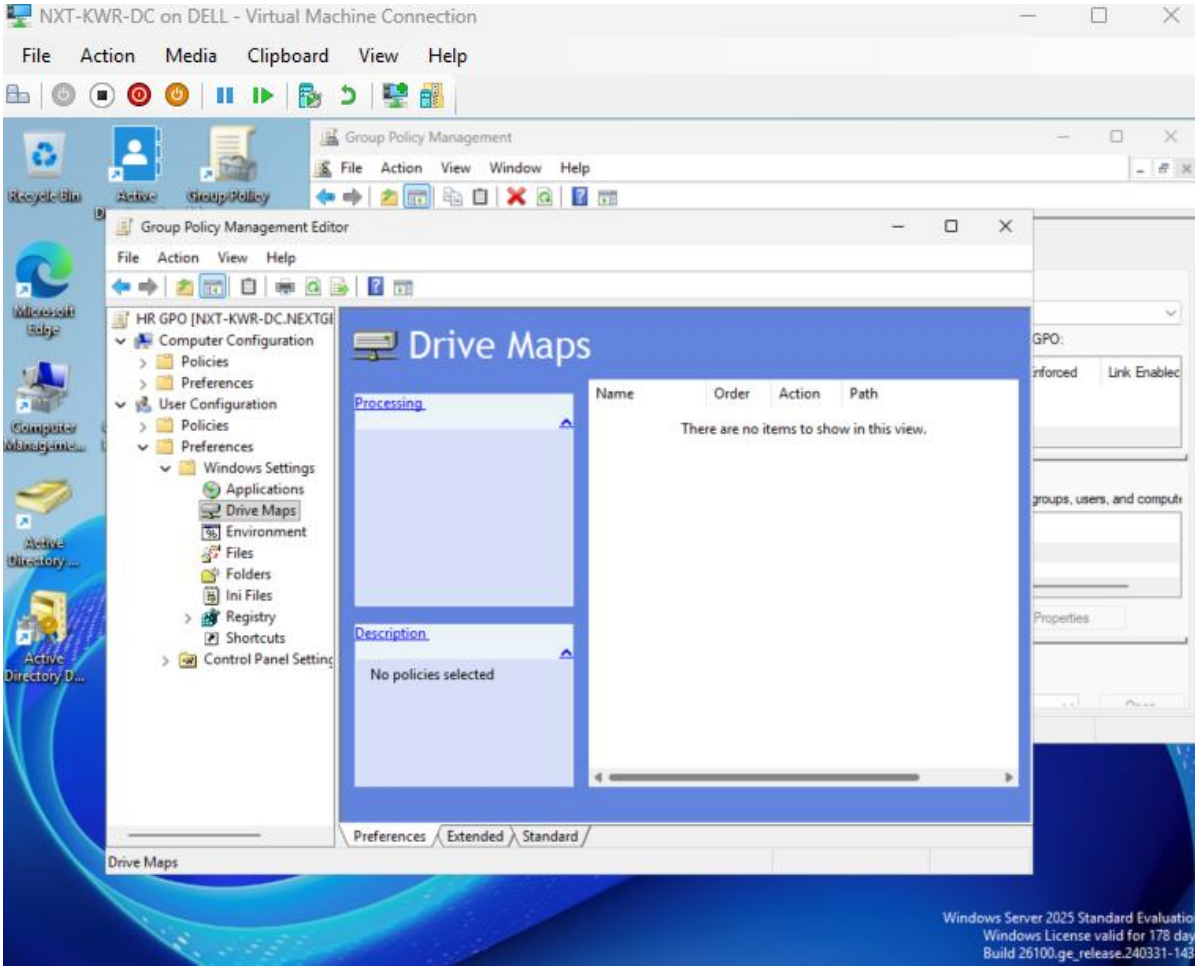




Step 2: Create the Group Policy Object (GPO)

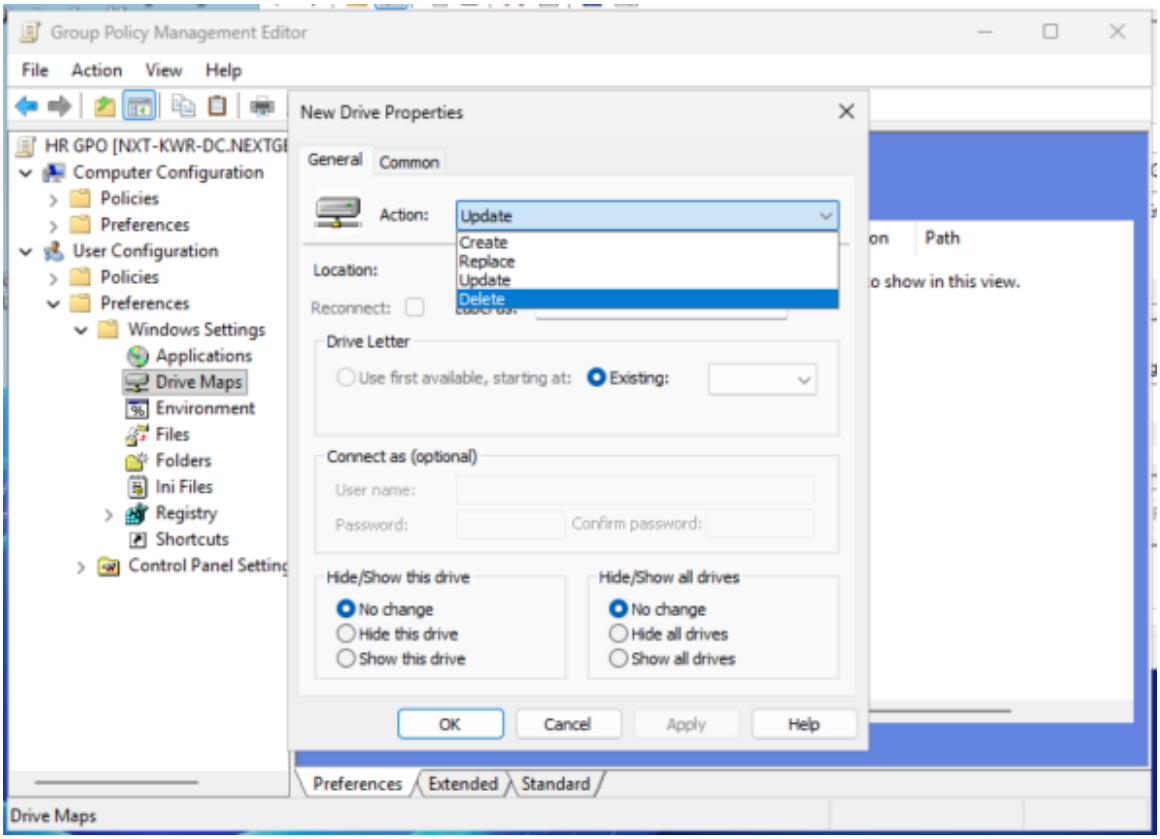
A GPO (Group Policy Object) is like a rulebook that Windows applies automatically to a group of computers or users. To map a drive for the HR team, we create a GPO named something like "HR GPO".

Inside the GPO editor, we go to: User Configuration → Preferences → Windows Settings → Drive Maps. This is the section where we tell Windows which drive letter to create and which shared folder it should point to.



Step 3: Add a New Mapped Drive

Inside the Drive Maps section, we click the small green plus (+) icon in the toolbar to add a new mapped drive. This opens a settings window where we choose exactly how the drive should behave.



2. Understanding the Drive Map Settings

The settings window has several important options. Here is what each one means, explained simply.

Action Type (What Should Happen to the Drive)

This tells Windows what to do with the drive letter — should it be added, changed, or removed? These four choices come from a common computing idea called CRUD (Create, Read, Update, Delete):

- **Create:** Adds the drive only if it does not already exist. If it already exists, nothing happens.
- **Replace:** Deletes the old mapped drive completely, then creates a brand-new one with the current settings. This is a more forceful reset.
- **Update (Recommended):** The safest option. It changes the existing drive's settings (for example, renaming its label) without removing it. If the drive doesn't exist yet, it will create it.
- **Delete:** Removes a mapped drive that was set up earlier.

Connection Settings

- **Location:** This is the network path (called a UNC path) of the shared folder, written like \\servername\foldername (for example, \\nextgen.int\nextgen1s). This tells Windows exactly where the shared folder lives on the network.
- **Reconnect:** When turned on, the drive automatically connects again every time the user logs in, so they don't have to set it up more than once.

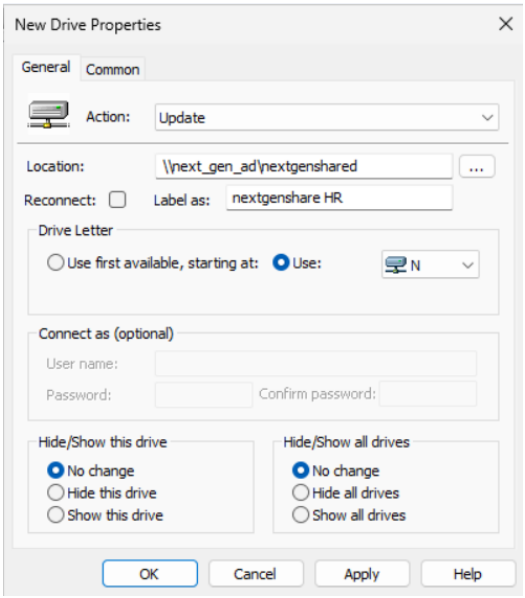
- **Drive Letter:** The letter (such as H:, M:, or Z:) that will represent this shared folder inside "This PC" on the user's computer.

Visibility Settings

- **Hide/Show this drive:** Controls whether this one specific drive is visible in File Explorer.
- **Hide/Show all drives:** Controls whether every mapped network drive on the computer is visible at once.

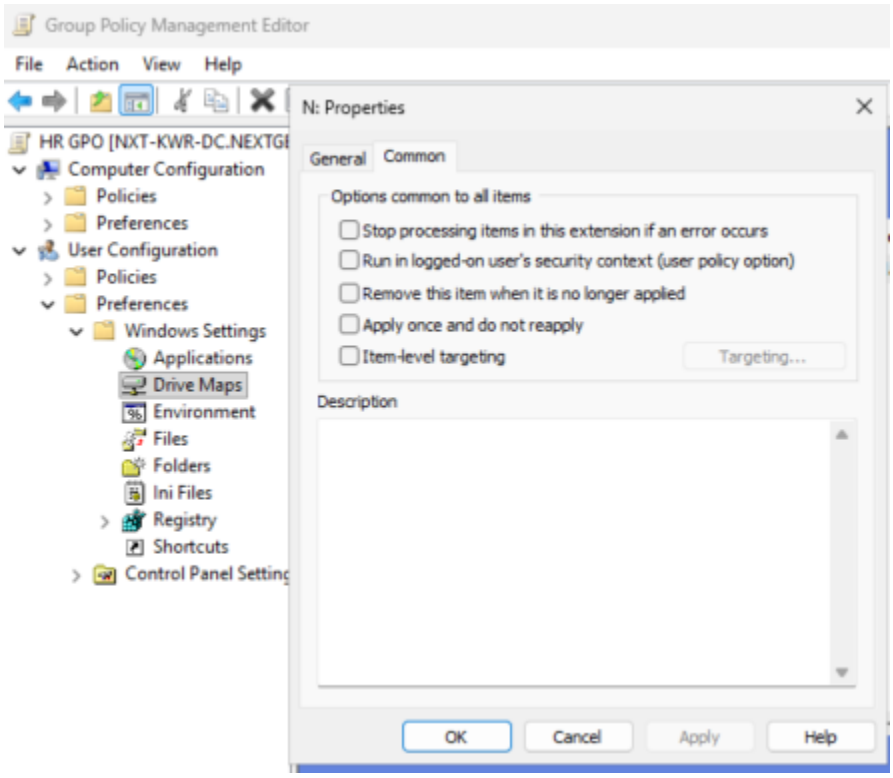
In most everyday situations, we leave visibility settings on "No change," so drives behave normally and stay visible.

> *Note: When typing the Location, you must paste the exact network path — even a small typing mistake will stop the drive from connecting.*



3. The "Common" Tab — Extra Behaviour Settings

Next to the main drive settings, there is a tab called "Common." This tab has extra checkboxes that control how the Group Policy item behaves. Here is what each one means and when to use it.



1. Stop Processing Items in This Extension if an Error Occurs

What it means: If something goes wrong while mapping this particular drive, Windows will immediately stop and skip any other drives still waiting to be mapped in the list.

When to use it: Only tick this if you have several drives (for example, 5 or 6) and you want the entire process to stop the moment the first one fails. Most of the time, this box is left unticked, so that even if one drive fails, the others still get mapped.

2. Run in Logged-On User's Security Context (User Policy Option)

What it means: This decides whose permission is used to connect the drive — the actual person who is logged in, or a hidden background account called SYSTEM.

When to use it: It is strongly recommended to tick this box for network drives. To open a network path like \\nextgen.int\nextgen1s, Windows needs the real username and password of the logged-in person. Ticking this box makes sure the drive connects correctly under that person's own account.

3. Remove This Item When It Is No Longer Applied

What it means: If the GPO is deleted later, or the user is removed from the group the policy applies to, the mapped drive will automatically disappear from their computer.

When to use it: Tick this if you want the drive to vanish automatically the moment someone leaves, for example, the HR department. If left unticked, the drive will stay on the computer until someone removes it by hand.

4. Apply Once and Do Not Reapply

What it means: The drive is created only the very first time the user logs in. After that, Windows will not try to set it up again, even on future logins.

When to use it: Use this if you want to let users freely change or delete the drive letter afterwards. Since most network admins actually want the drive to always stay in place, this box is normally left unticked.

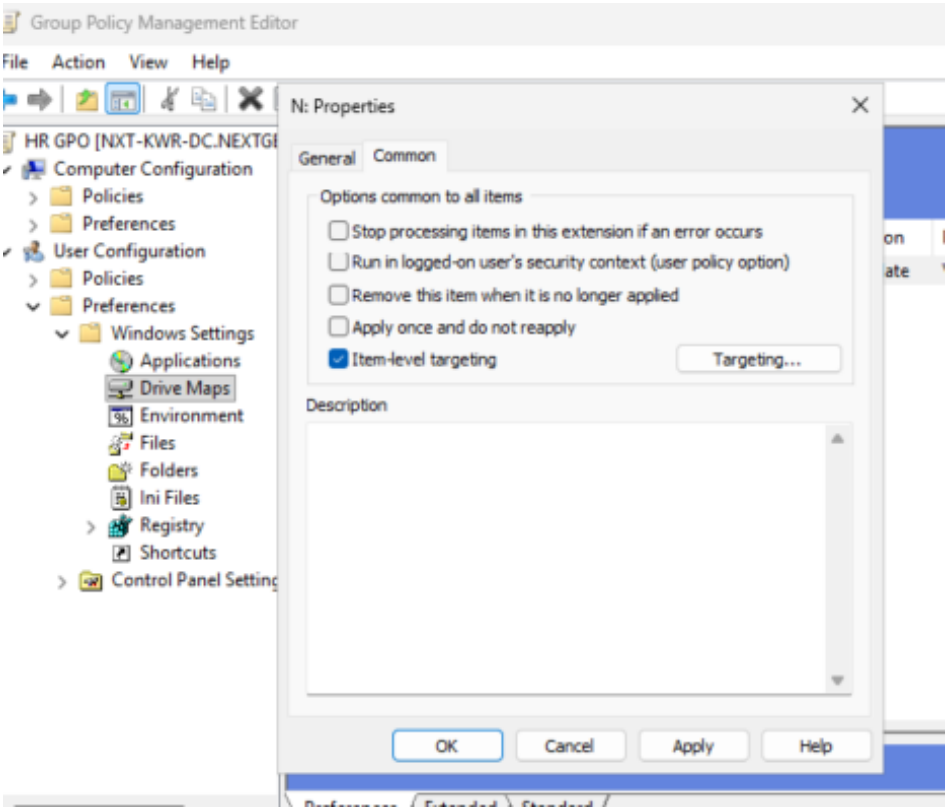
5. Item-Level Targeting

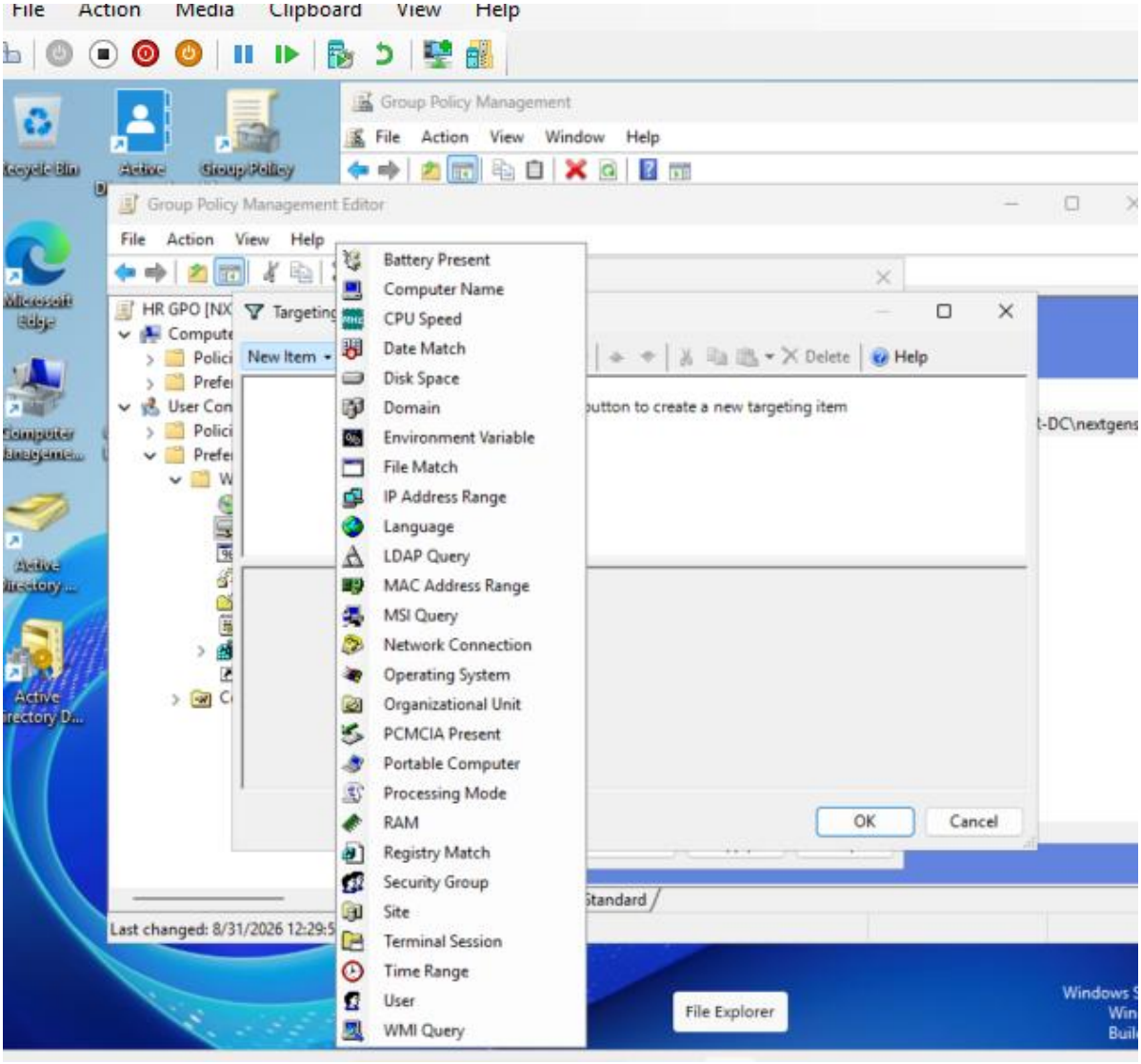
What it means: A powerful option that lets you choose precisely who receives this mapped drive, based on conditions you set — such as membership in a specific security group.

When to use it: Tick this box, then click the "Targeting…" button, and choose a condition such as "Security Group = nextgen-HR." This makes sure that even if the GPO is linked to the whole company, only people in the HR group will actually get the drive. Everyone else, including IT staff, will not see it.

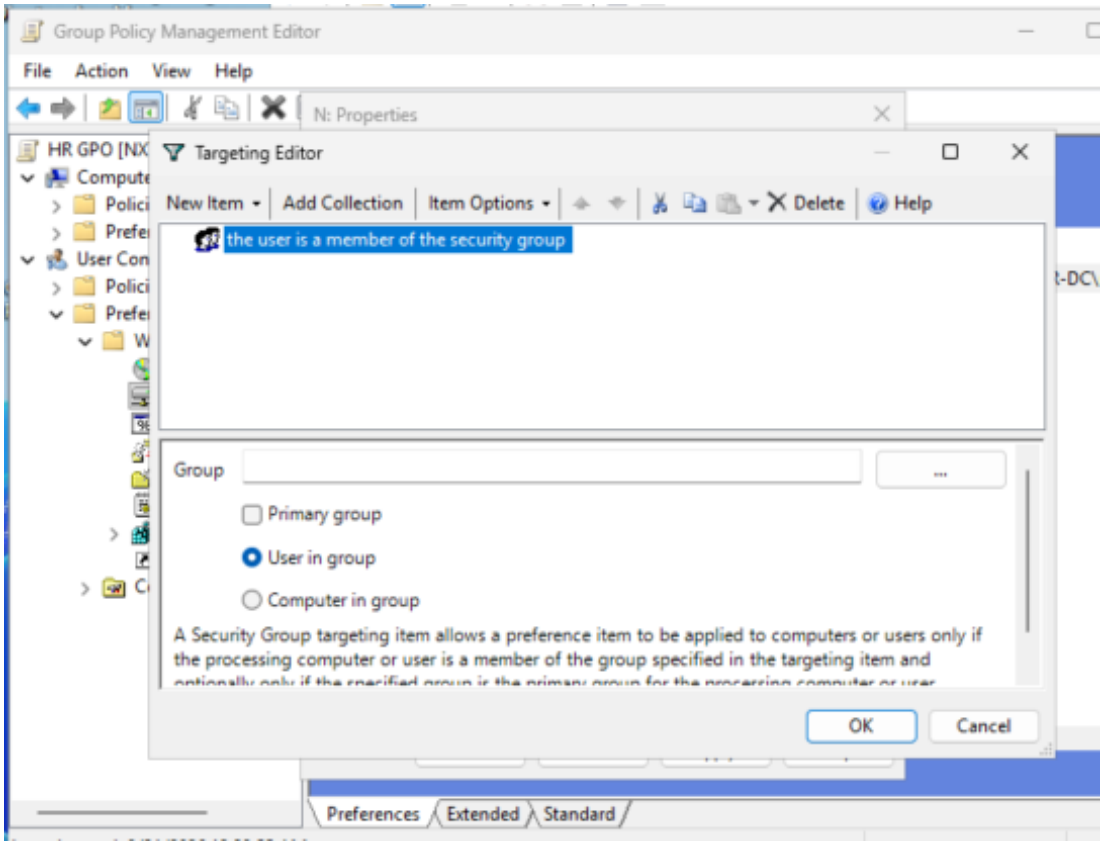
Setting Up Item-Level Targeting

In this example, item-level targeting is used so that only the correct group receives the mapped drive. After ticking the box, we open the Targeting editor and choose the group we want to target.

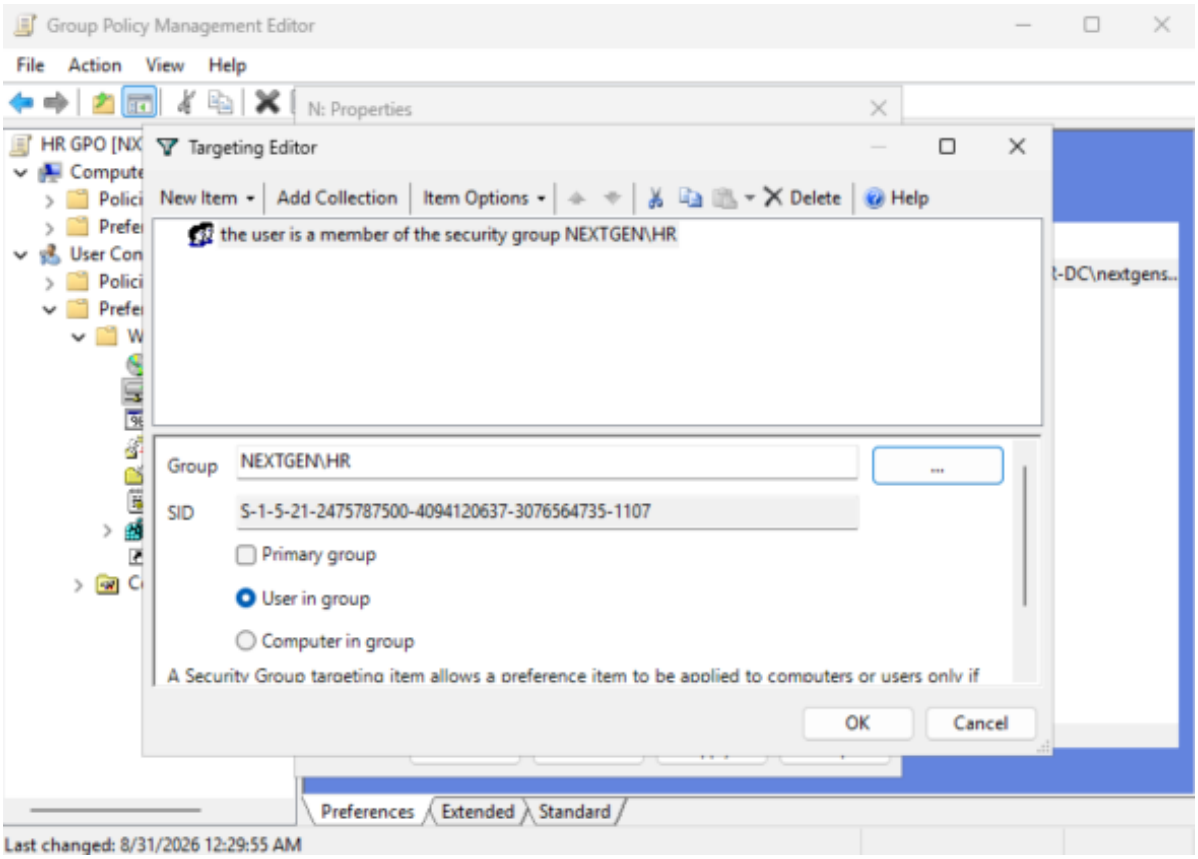




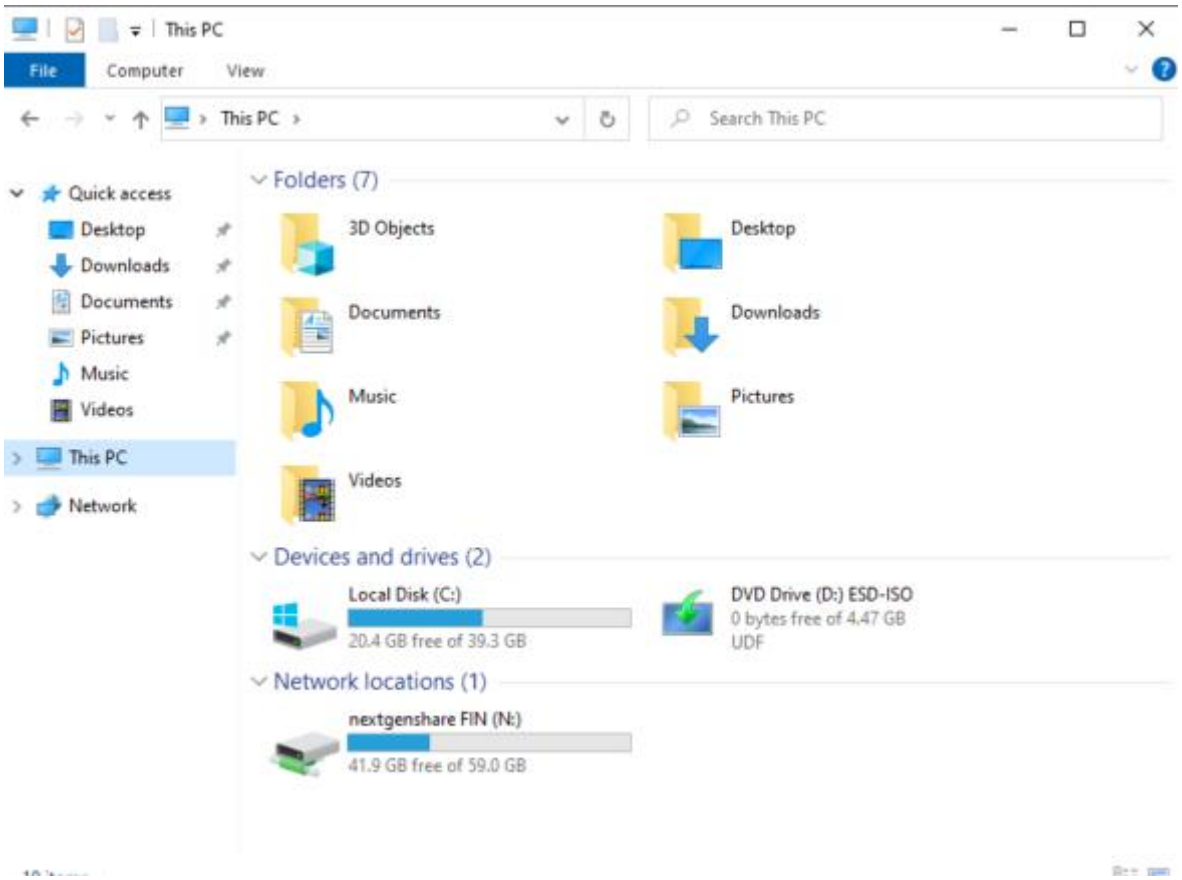
Next, we select the specific security group that should receive this drive.



The final targeting rule looks like this before saving:



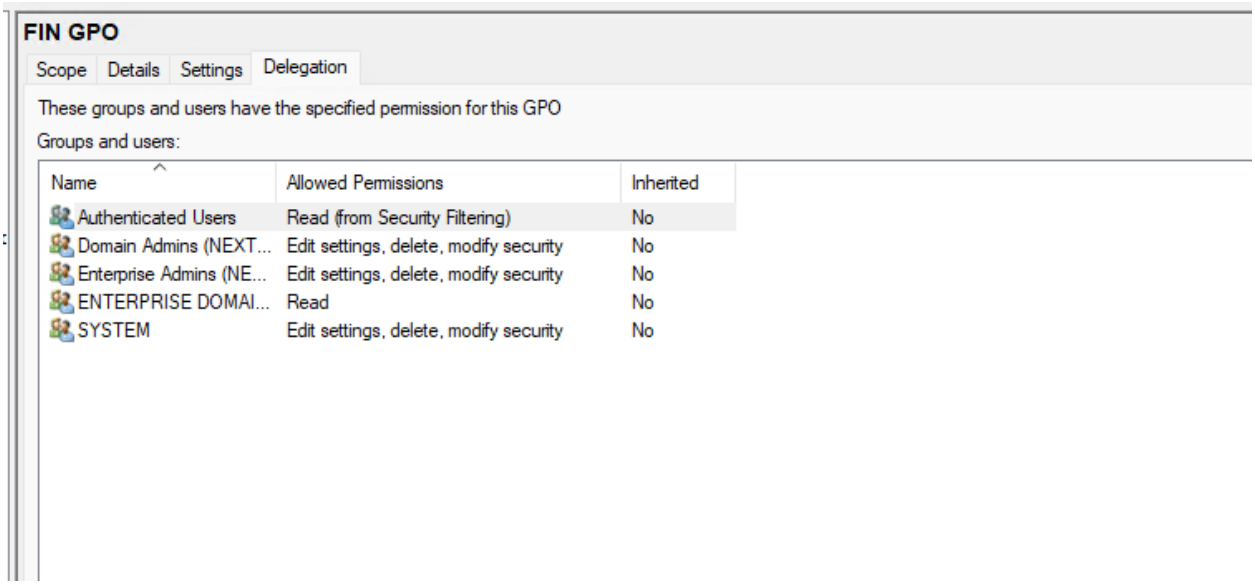
After clicking OK, the policy is saved. When the correct user (in this example, a user called finuser1) logs into their computer, the new drive appears automatically.



> *Note: If you want this same mapped drive to apply to every user in the domain, instead of only one group, simply untick Item-Level Targeting.*

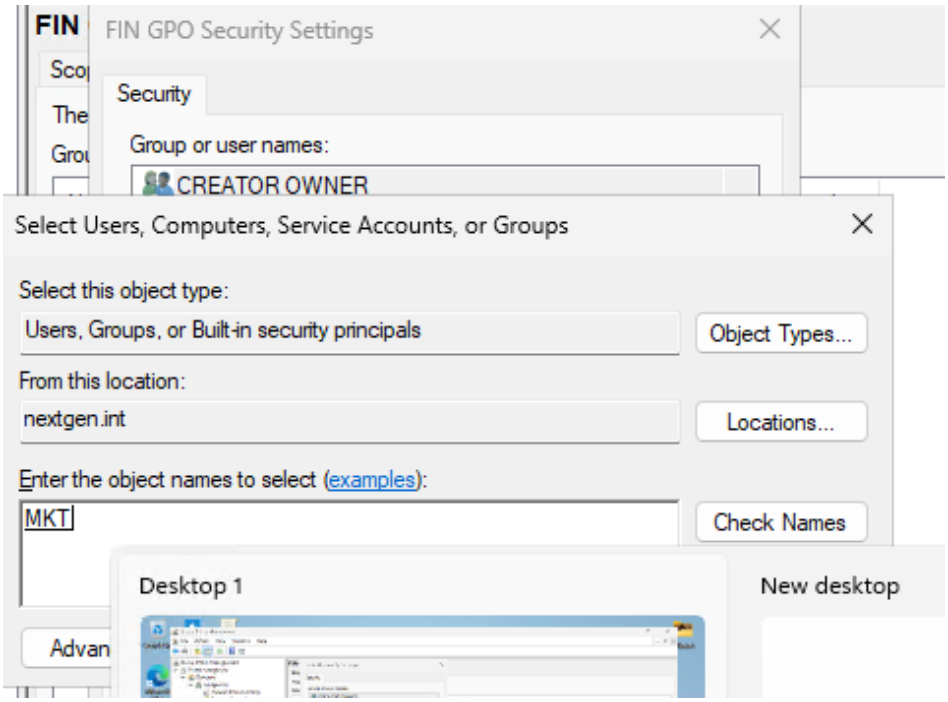
4. Blocking a Policy from Applying (Delegation / Deny)

Sometimes, instead of giving access, you need to do the opposite — completely stop a specific person or group from ever receiving a certain policy. This is done through the Delegation tab of the GPO.

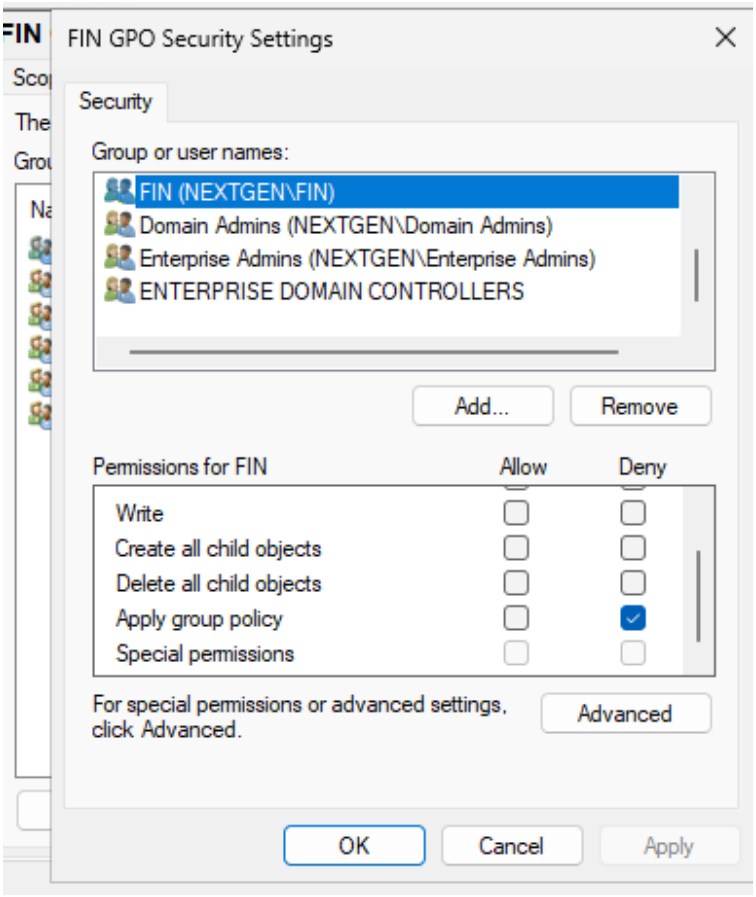


How to Deny a Policy

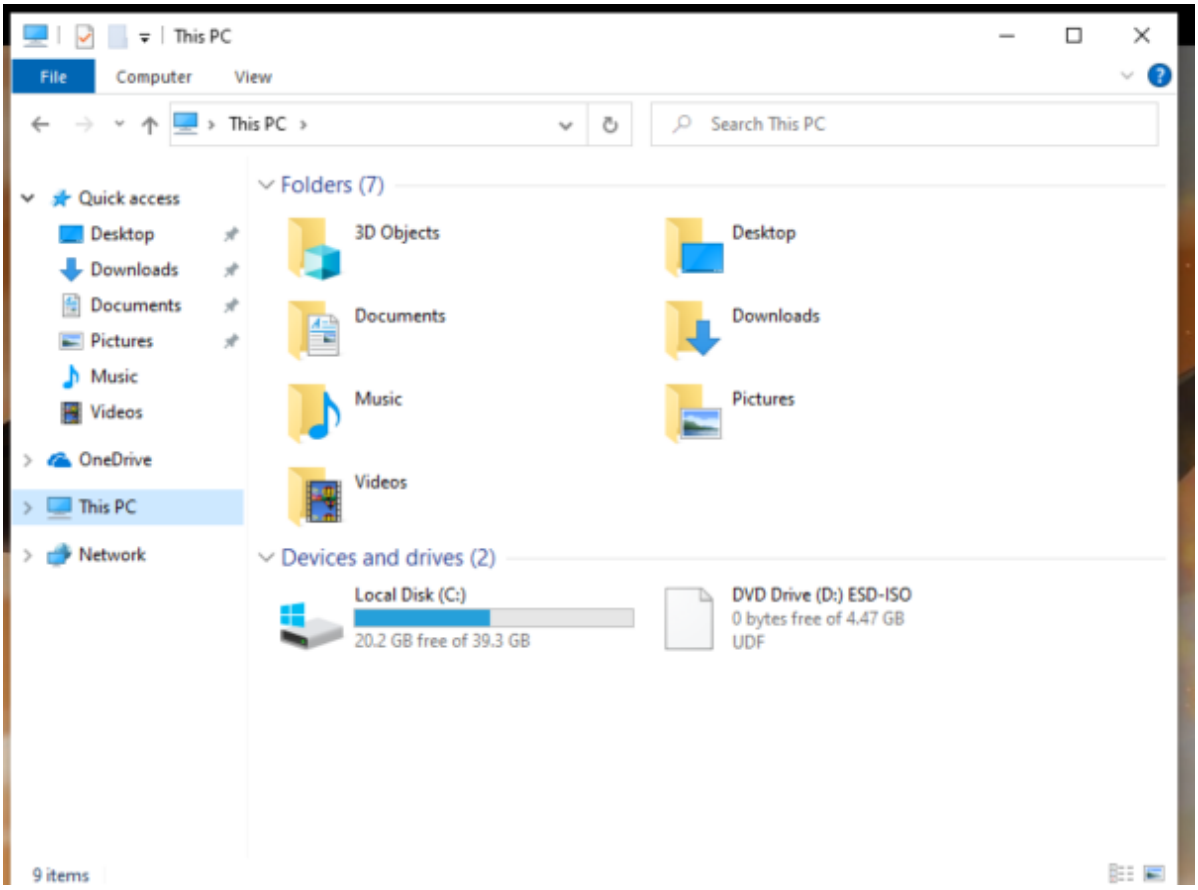
1. Open the GPO and click the Delegation tab.
2. Click the Advanced button to see detailed permissions.
3. Find the permission called "Apply Group Policy" and tick the Deny checkbox next to it.



Finally, click Advanced and tick Deny next to "Apply Group Policy":



Once this is done, the affected users will no longer be able to see the shared folder, even if they belong to a group the policy is normally linked to.



5. WMI Filtering — Applying Policies by Computer Condition

What Is WMI Filtering?

WMI stands for Windows Management Instrumentation. WMI Filtering is a feature in Active Directory that lets an administrator apply a Group Policy only when a computer meets a certain condition — for example, only if it is running a specific type of operating system.

This is different from normal targeting, which is based on users or Organizational Units (OUs, which are folders used to organise users and computers). A WMI filter instead looks directly at the technical details of the computer itself — such as its operating system or hardware — before deciding whether to apply the policy.

How WMI Filtering Works

Every time a computer starts up, or a user logs in, the computer checks the WMI query attached to the GPO:

- **If the result is True:** the computer matches the condition, so the GPO is applied normally.
- **If the result is False:** the computer does not match, so the GPO is completely ignored on that machine.

Example: Targeting Only Domain Controllers

A Domain Controller (DC) is a special server that manages logins and security for the whole network — it is different from an ordinary member computer or workstation. The example below shows a WMI query that applies a GPO only to domain controllers, skipping regular servers and workstations.

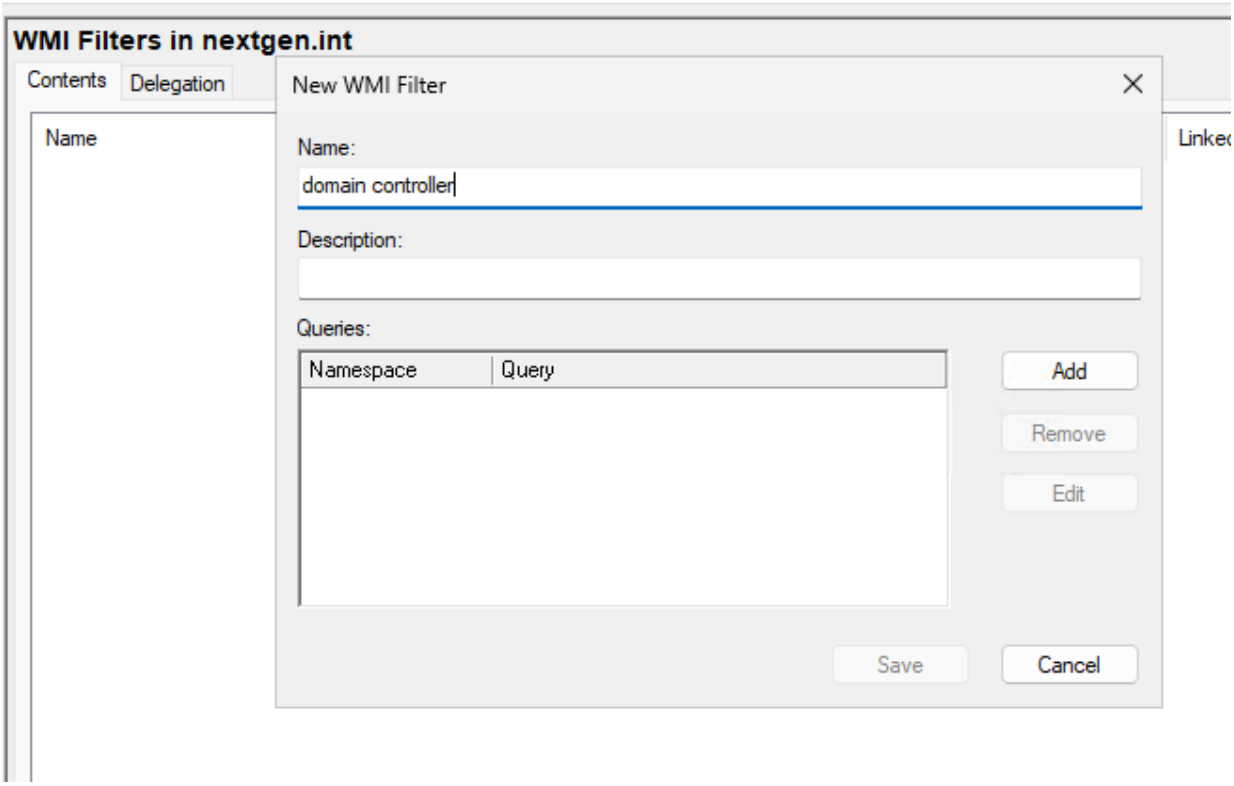
This works by asking the computer's operating system a direct question, written in a simple query language (similar to how you would search a database):

```
SELECT * FROM Win32_OperatingSystem WHERE ProductType = "2"
```

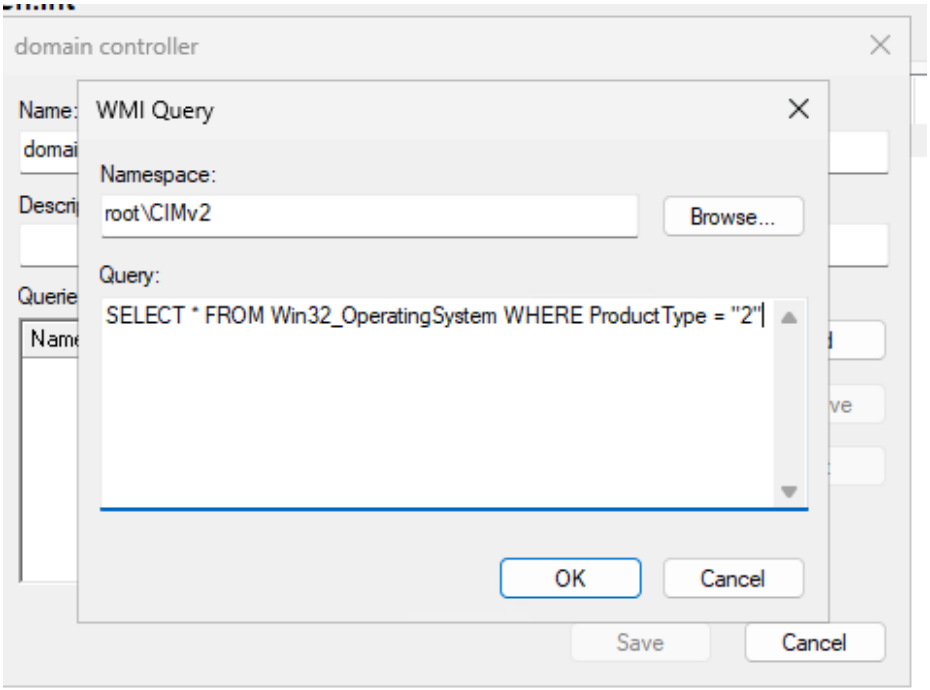
In plain terms, this query asks: "What type of Windows operating system is this?" and only continues if the answer marks the machine as a domain controller (ProductType = 2).

Creating a WMI Filter

1. Right-click on the WMI Filters section in Group Policy Management, then choose New.
2. Give the filter a clear name so you can recognise it later.
3. Click Add, then paste the query into the box.

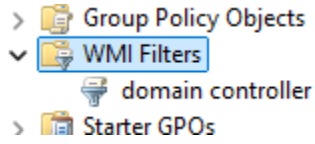


Click OK to save the new filter.

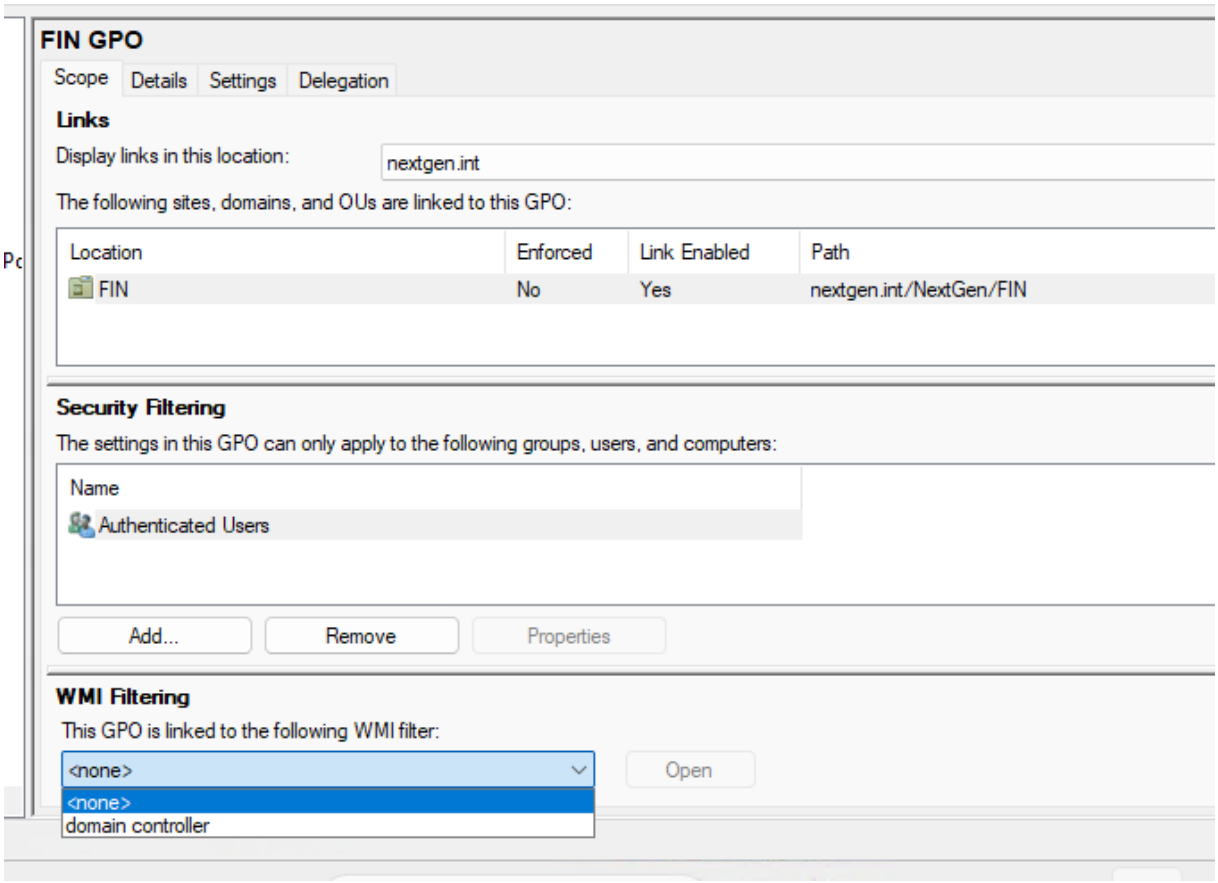


Attaching the WMI Filter to a GPO

Once the filter is created, it needs to be attached to the GPO you want it to control.



1. Select the GPO (for example, the HR GPO).
2. In the WMI Filtering section, choose the filter you just created — in this case, the one that targets domain controllers.



Once this filter is attached, the mapped drive (N:) will only appear on domain controllers. Ordinary user computers will no longer see the shared folder, because they do not match the WMI condition.

6. Deploying Software Automatically with Active Directory

Beyond mapping drives, Group Policy can also be used to install software automatically on many computers at once — without an administrator having to visit each machine.

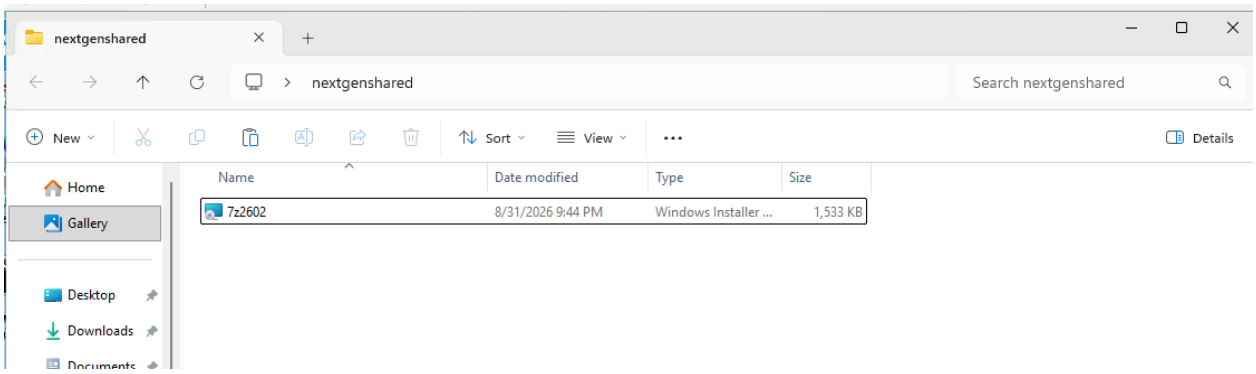
What You Need Before You Start

- **An MSI installer file (.msi):** Group Policy Software Installation only understands the Microsoft Installer (.msi) format. Ordinary .exe installer files cannot be used directly for this method.
- **A shared network folder:** The .msi file must sit inside a shared folder (for example, \\server\software) that everyone has at least Read permission to. This lets client computers reach the installer file when they need it during startup.

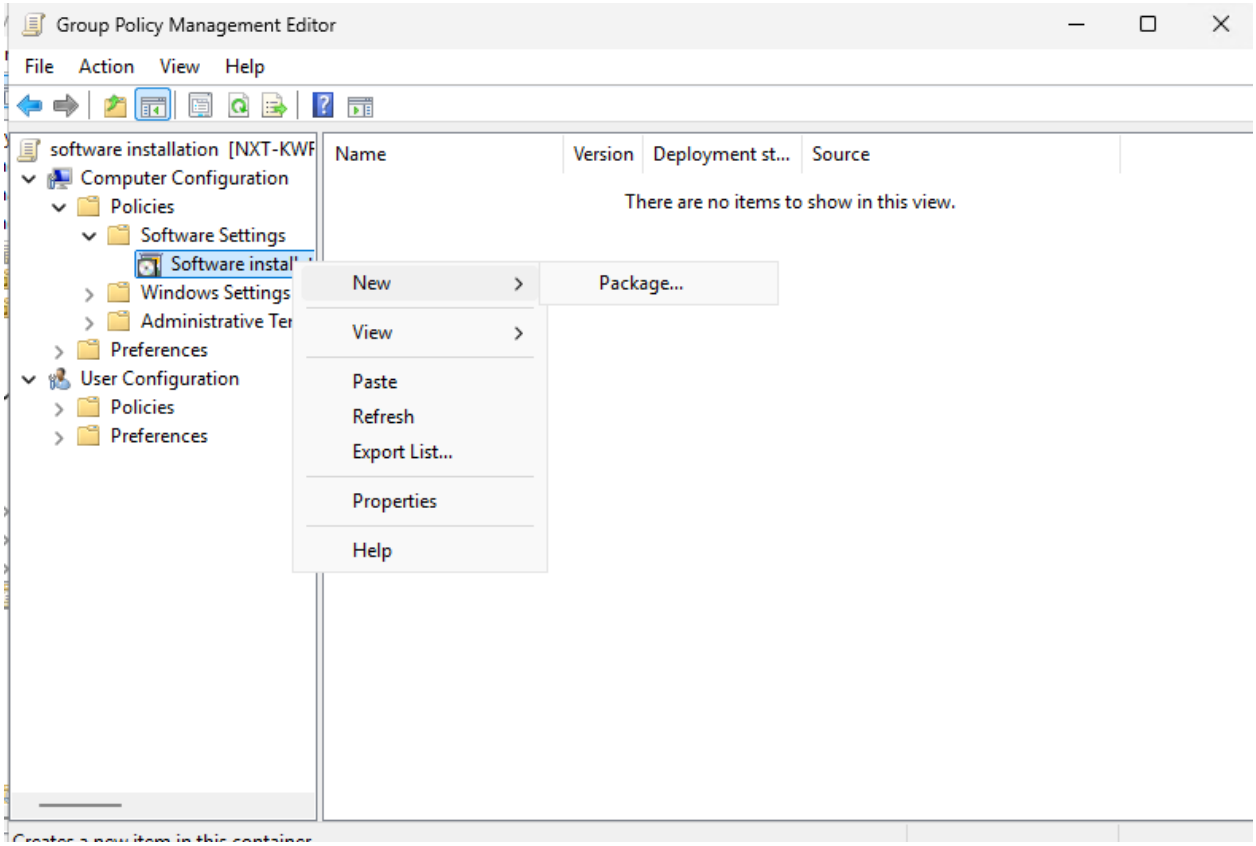
Example: Deploying 7-Zip

This example uses the free tool 7-Zip (in its .msi version) to demonstrate the process.

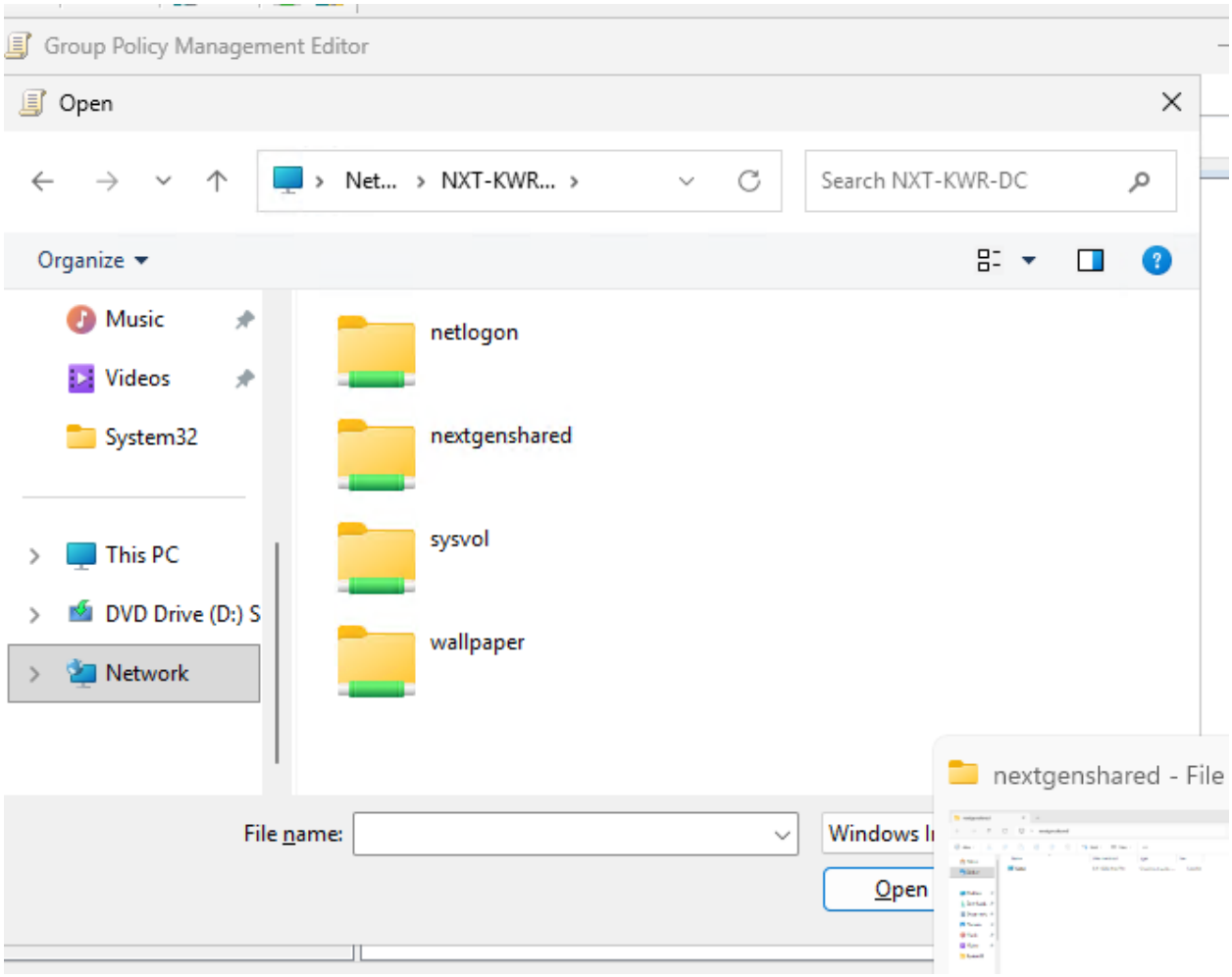
1. First, place the 7-Zip .msi installer file inside the shared software folder.



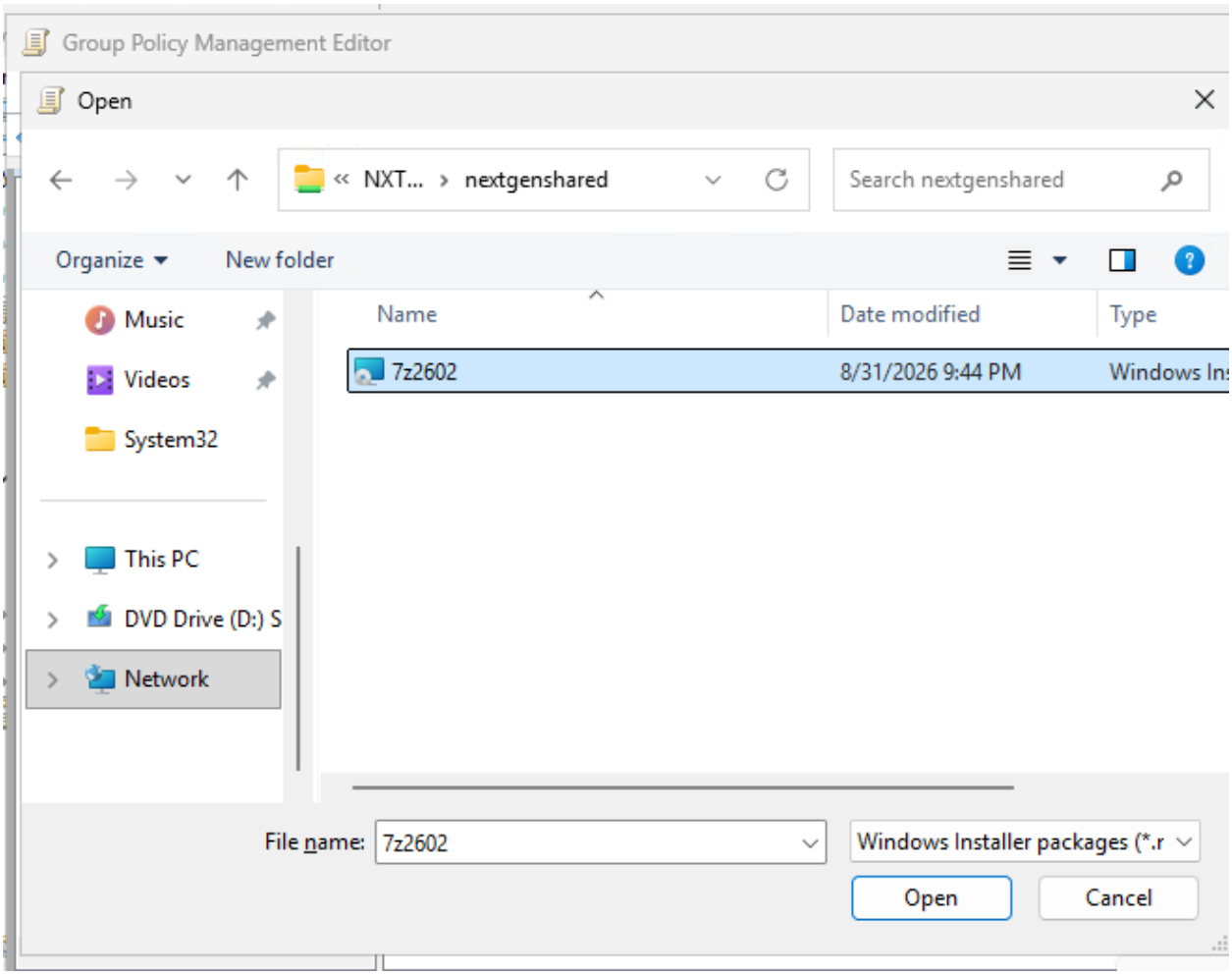
2. Create a new GPO for software installation. Open it for editing, then go to Computer Configuration → Policies → Software Settings, right-click Software Installation, choose New, then Package.



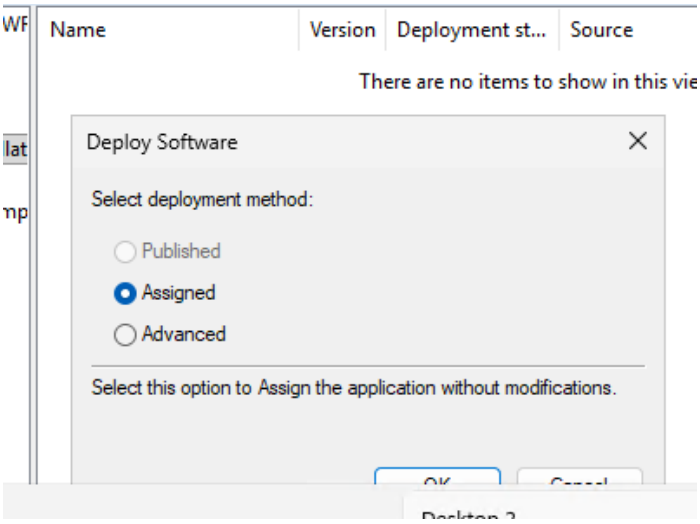
3. Type in the name of the domain controller and browse to select the installer file.



4. Choose the .msi file from inside the shared folder.



After clicking Open, a small pop-up window will appear, asking how you want to deploy the software:



Choosing a Deployment Method

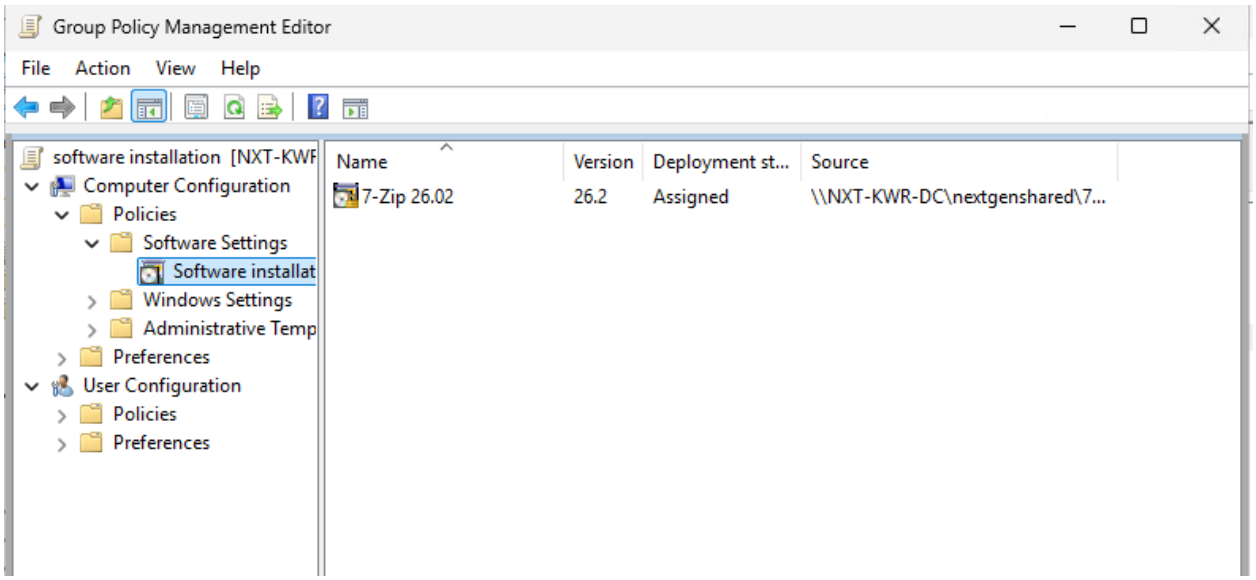
- **Published:** The software is optional — it is not forced onto the computer. Instead, the user can find and install it themselves by going to Control Panel → Programs and Features →

"Install a program from the network." This option is best for software that is useful but not required for everyone.

Note: Because this package is being set up under Computer Configuration, the "Published" option will appear greyed out. This is because software can only be published to users, not to computers.

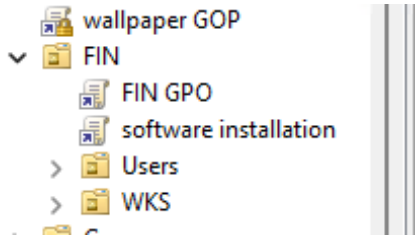
- **Assigned (used in this example):** The installation is mandatory. Since it is configured under Computer Configuration, the software installs silently in the background while the computer is starting up, before anyone even logs in. This is the best choice for essential software that everyone in the company must have, such as antivirus software, a web browser, or Microsoft Office.
- **Advanced:** This lets you assign the software while also opening extra configuration options. Choosing this and clicking OK immediately opens the software's Properties window, where you can make further changes, such as switching the language or adding a custom patch file (.mst).

Click OK to confirm the deployment method.



Applying the Policy to a Specific Group of Computers

An Organizational Unit (OU) is a way of grouping computers or users together in Active Directory, usually by department. In this example, the GPO is linked only to the Finance (FIN) OU, so only computers in that department will receive the software.



After the policy has been updated on the domain controller, the client computer will pick it up and begin installing the software the next time it restarts. This may take a short while, depending on the size of the software.

> *Note: You can check whether Group Policy has applied correctly on a computer by running the command gpresult /r in the Command Prompt (CMD). This shows a report of every policy currently applied to that machine.*

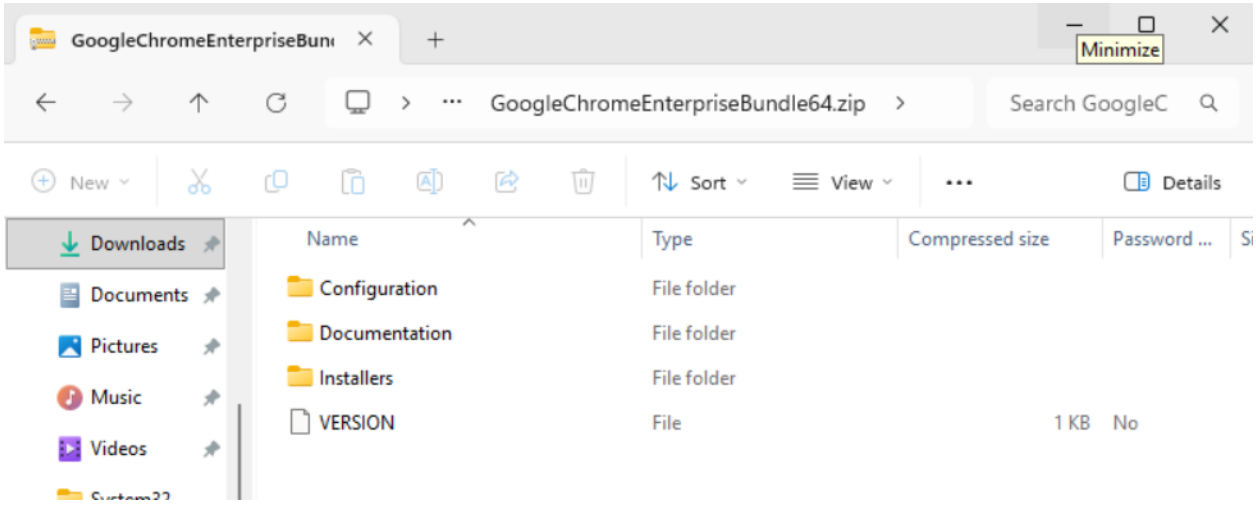
7. Managing Web Browser Settings with Active Directory

Active Directory can also be used to control settings for web browsers, such as Google Chrome, Microsoft Edge, and Firefox, across every company computer, all from one central location. This includes things like setting a default homepage, blocking certain extensions, and configuring network proxy settings.

By default, standard Group Policy does not include the ability to manage third-party browsers like Chrome or Firefox. To manage them, we must first install special template files, called Administrative Templates (ADMX templates), onto the domain controller.

What Are ADMX and ADML Files?

ADMX files describe the actual settings that a program supports (for example, all the Chrome settings Group Policy can control). ADML files provide the text labels for those settings, in a chosen language — for example, English (en-US).



Step 1: Copy the ADMX Files

1. Open the Configuration folder.
2. Inside it, open the admx folder.
3. Copy the two files named chrome.admx and google.admx.
4. Paste both files into C:\Windows\PolicyDefinitions\ on your server.

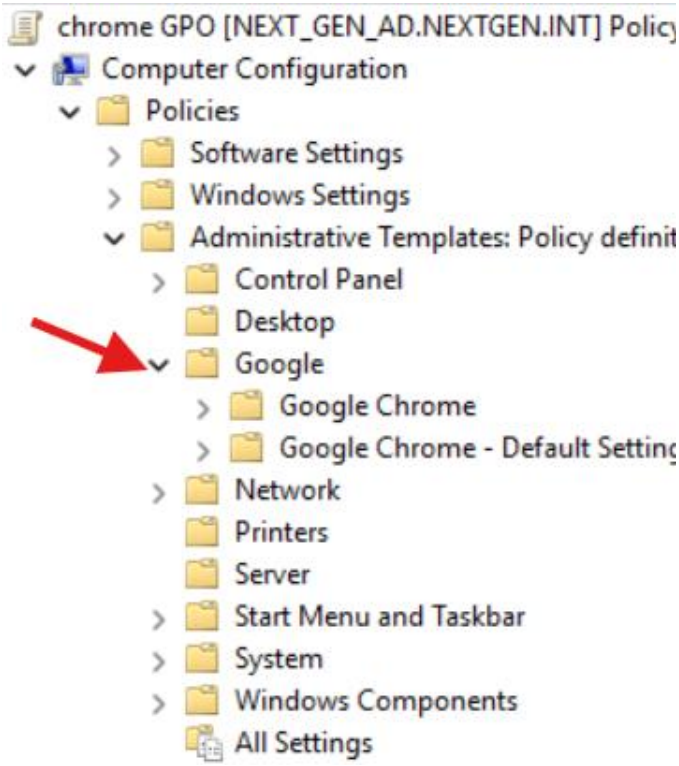
Step 2: Copy the ADML (Language) Files

1. Inside the same admx folder, open the en-US folder.
2. Copy the two files named chrome.adml and google.adml.

3. Paste both files into C:\Windows\PolicyDefinitions\en-US\ on your server.

Step 3: Confirm the New Settings Appear

Once these files have been copied, open Group Policy Management on the server. When you create a new GPO and click Edit, a new Google Chrome section will now appear under Administrative Templates.

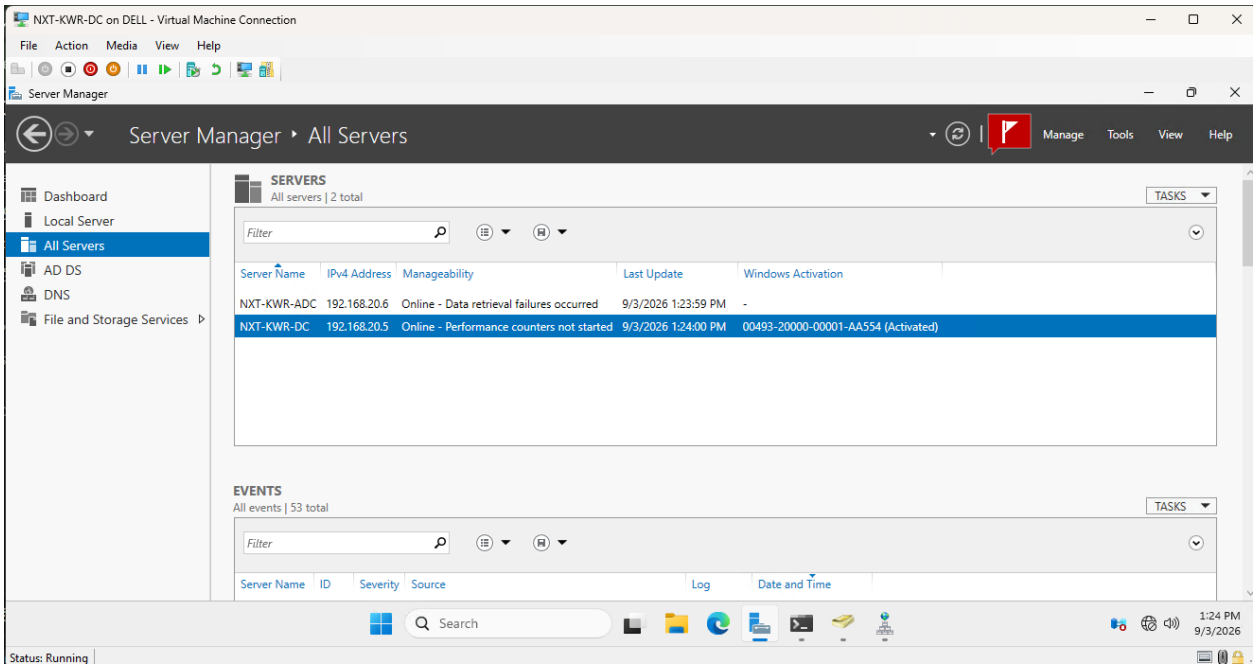


From here, you can create and apply policies to control Chrome's behaviour across every computer the GPO is linked to.

Additional Domain Controller (ADC)

Deploy and Manage an Additional Domain Controller (ADC), DC Health Check

An Additional Domain Controller (ADC) is a supplementary server added to the same network in addition to the existing primary Active Directory Domain Controller (PDC/Root DC).



Power shell command creates OUs

```
# ============================================================
# NextGen AD Structure - Build Script
# Run this in "Active Directory Module for Windows PowerShell"
# on your DC, as Administrator.
# ============================================================

# Step 1 -- Set Execution Policy (allows this script to run)
Set-ExecutionPolicy RemoteSigned -Force

# Step 2 -- Basic variables
$domain    = "DC=nextgen,DC=int"
$parentOU  = "OU=NextGen,$domain"
$pw        = ConvertTo-SecureString "Kandage12@#12" -AsPlainText -Force

# Departments: Name (N) and short prefix used in usernames (P)
$depts = @(
  @{N="IT";  P="it"},
  @{N="MKT"; P="mkt"},
  @{N="FIN"; P="fin"},
  @{N="HR";  P="hr"}
)

# Step 3 -- Create the top-level "NextGen" OU
New-ADOrganizationalUnit -Name "NextGen" -Path $domain -ProtectedFromAccidentalDeletion $false

# Step 4 -- For each department, create its OU + Users/WKS sub-OUs
foreach ($d in $depts) {
  $dOU = "OU=$($d.N),$parentOU"
  New-ADOrganizationalUnit -Name $d.N -Path $parentOU -ProtectedFromAccidentalDeletion $false

  foreach ($s in @("Users","WKS")) {
    New-ADOrganizationalUnit -Name $s -Path $dOU -ProtectedFromAccidentalDeletion $false
  }
}

# Step 5 -- Create the "Groups" OU
```

In here you can see DC=NEXTGEN, DC=INT — this represents the root domain name of our server, like NEXTGEN.INT (do not consider capital / simple letters).

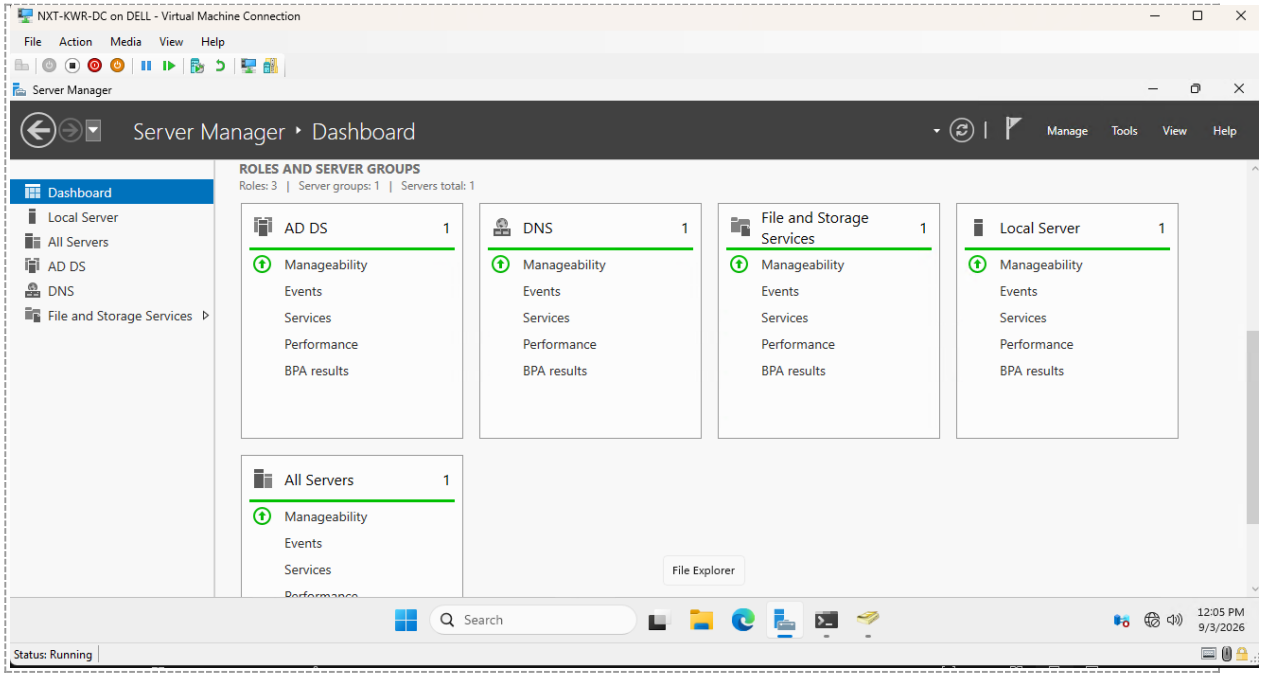
> 💡 *DC=NEXTGEN,DC=INT is just the Active Directory way of writing the domain name nextgen.int. Each part of the domain name before a dot becomes its own "DC=" component.*

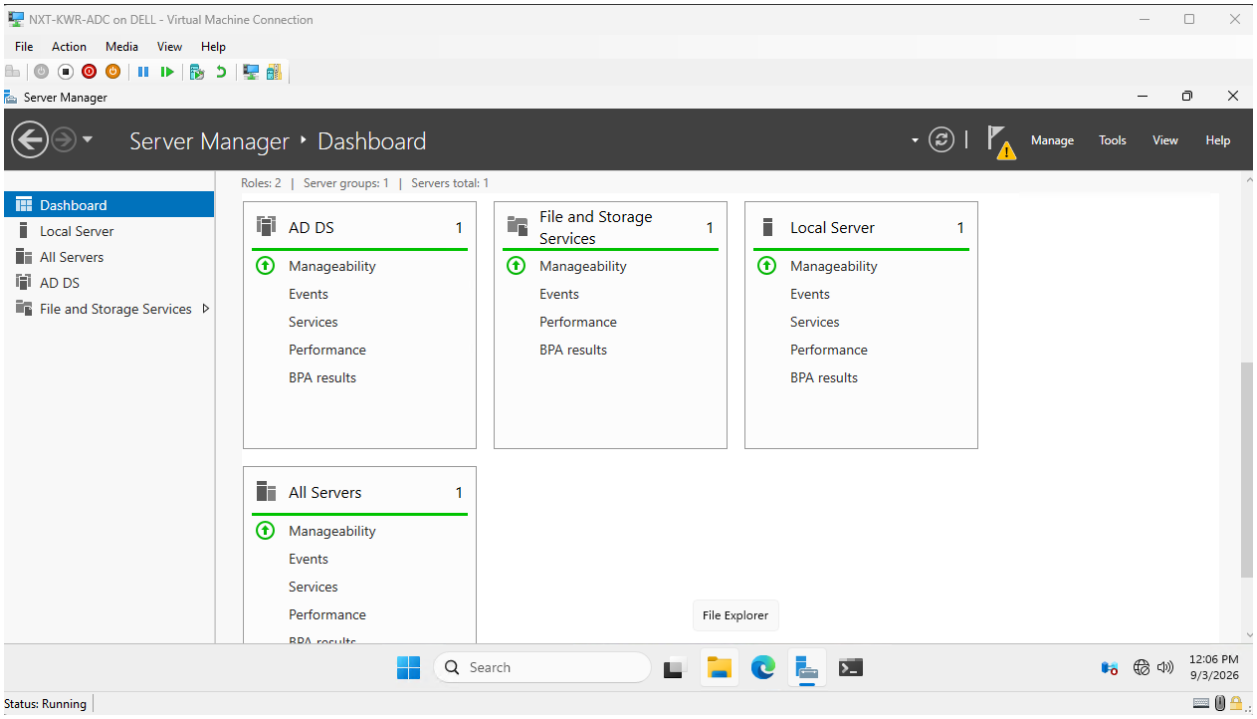
Let's add an additional domain controller to our existing DC

First, we want to add the Active Directory Domain role on the other server. When we configure the ADC domain server, we want to apply all default settings and check the IP address and check DNS.

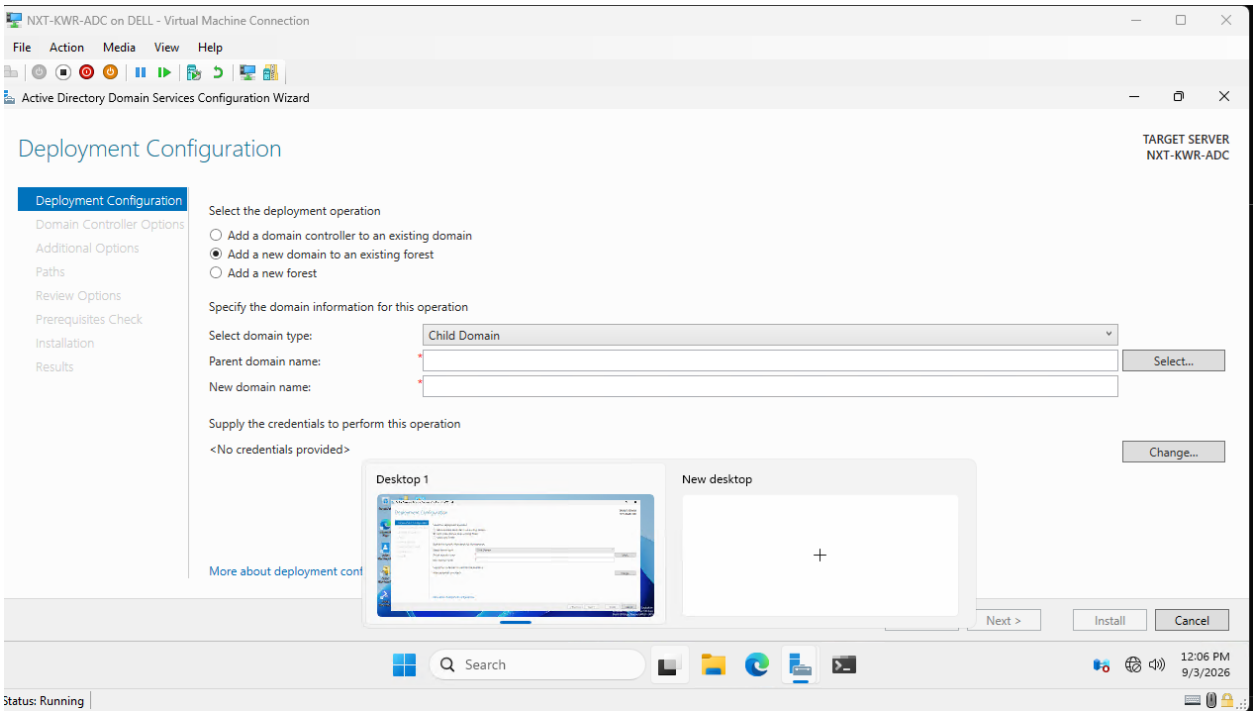
Ensure our existing DC has no errors.

In a DC, errors usually persist for up to 48 hours after Active Directory is promoted.





In the Promote section, when we are promoting the server we should select the "Add domain controller to existing domain" option.



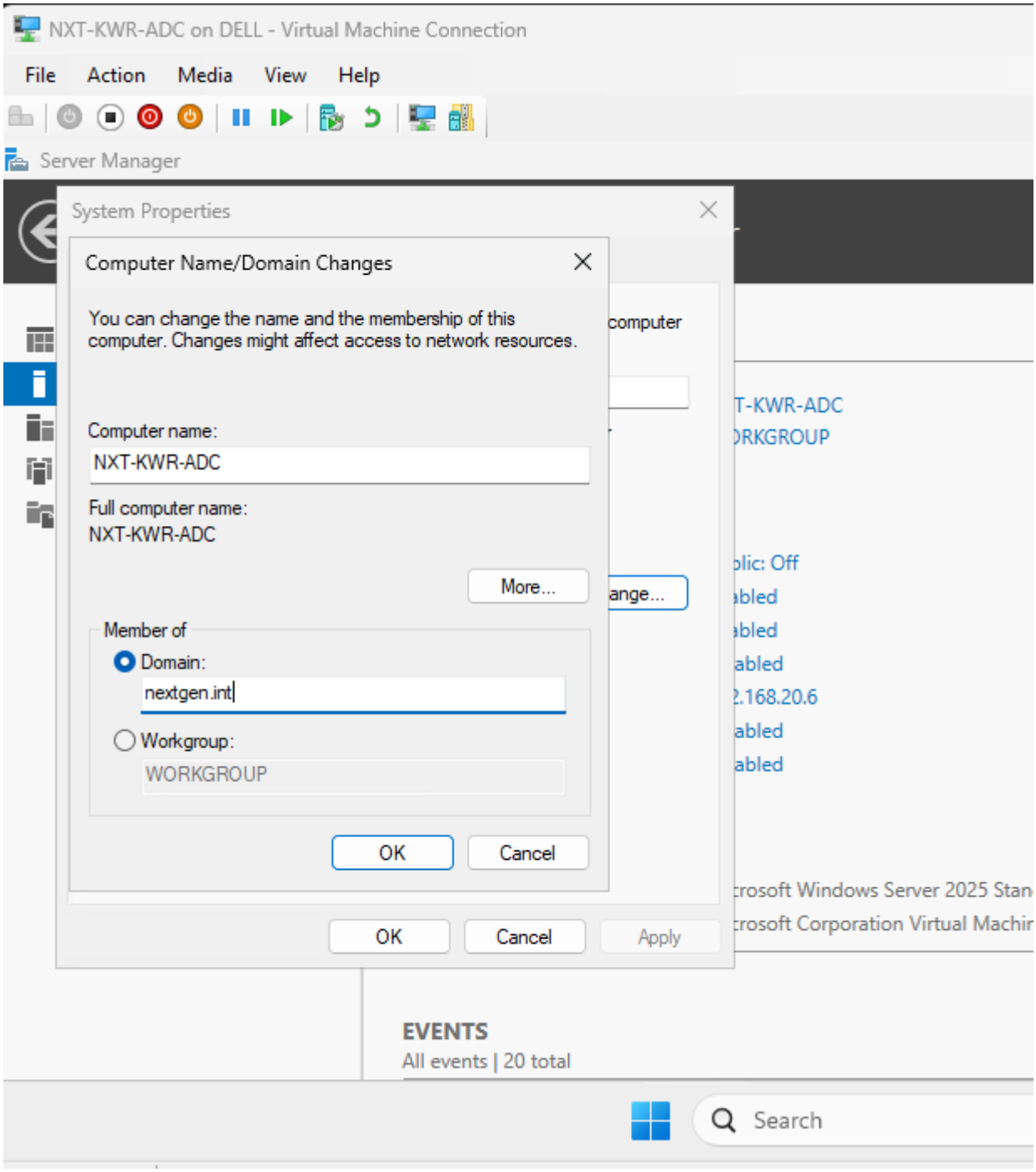
Important

1st step

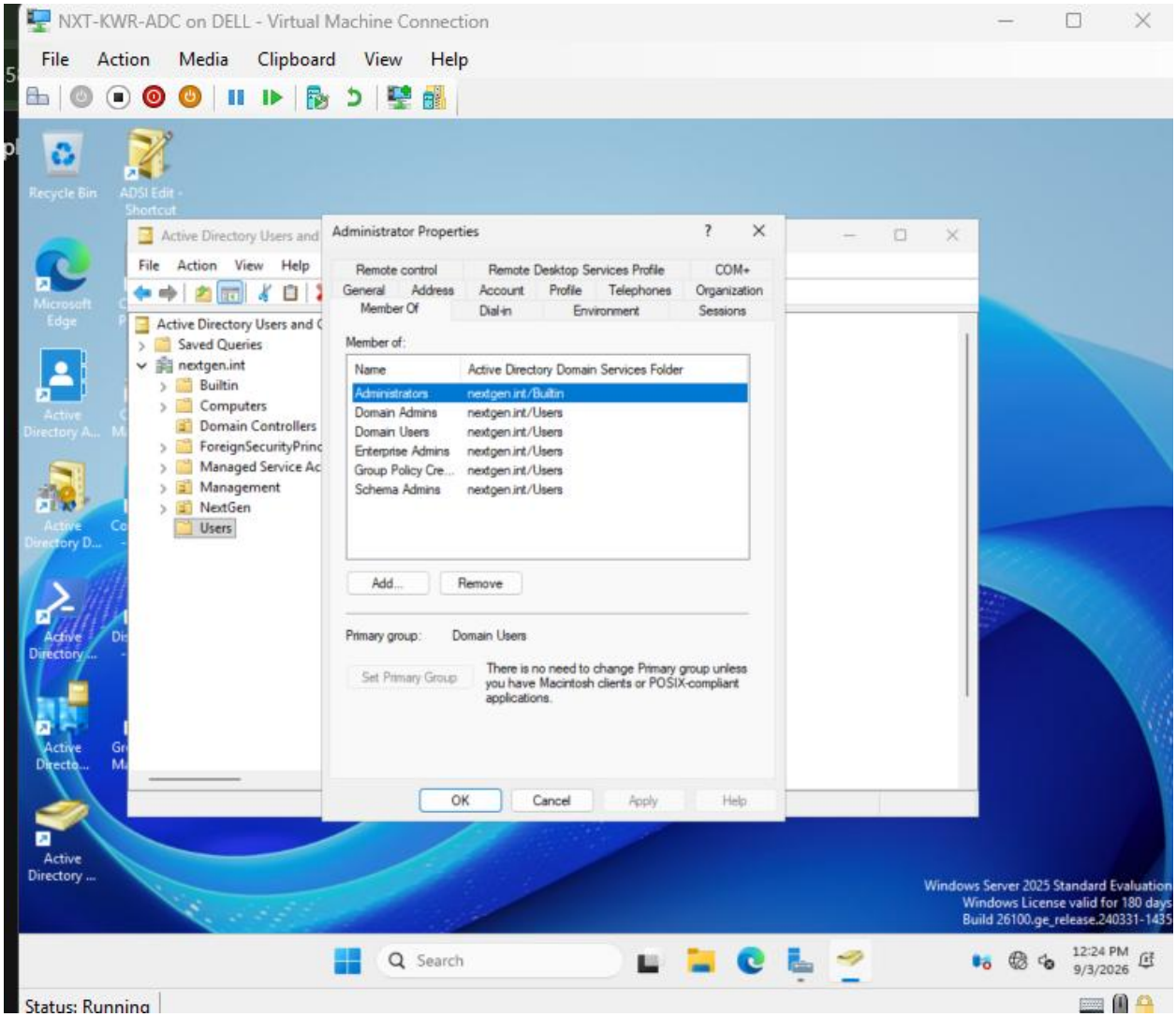
When doing this task, we want to add this VM as a domain member in the existing Domain Controller.

Right click This PC → Properties → Computer name → Change → tick Domain in the "Member of" section and give the username and password of the existing DC, then restart.

> *This step is what actually joins the new server to the domain before it can even be promoted — without this, the promotion wizard will not see the existing domain at all.*



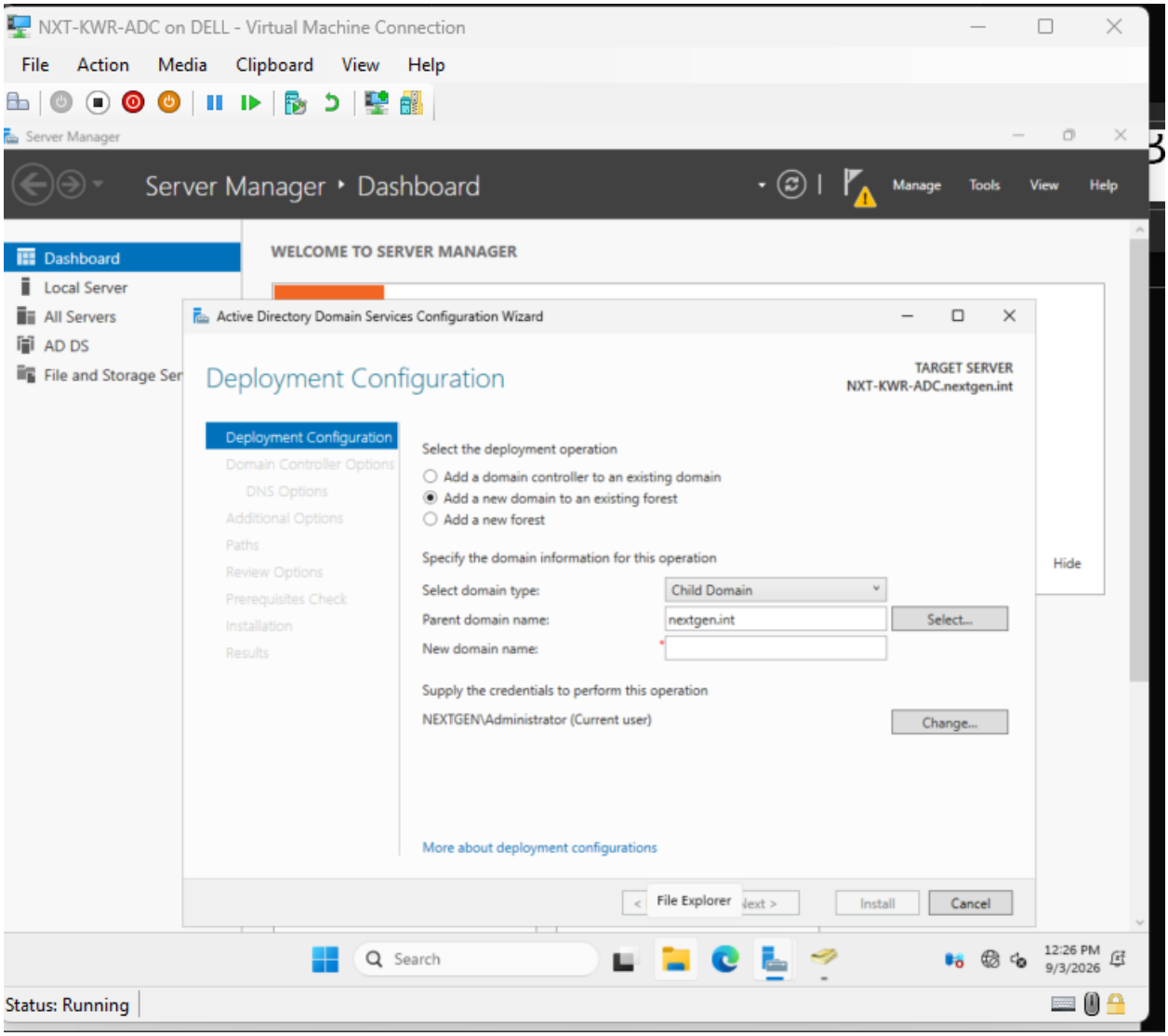
When promoting a new server to become an Additional Domain Controller (ADC) within an existing infrastructure, the user account performing this operation must explicitly possess Domain Admin, Schema Admin and Enterprise Admin group privileges. (This is added by default.)



2nd step

Let's promote the AD server — here we should choose the first option, "Add a new domain controller to an existing domain".

What is the 2nd option, "Add a new domain to an existing forest"?



This option is used to add a domain to an existing forest.

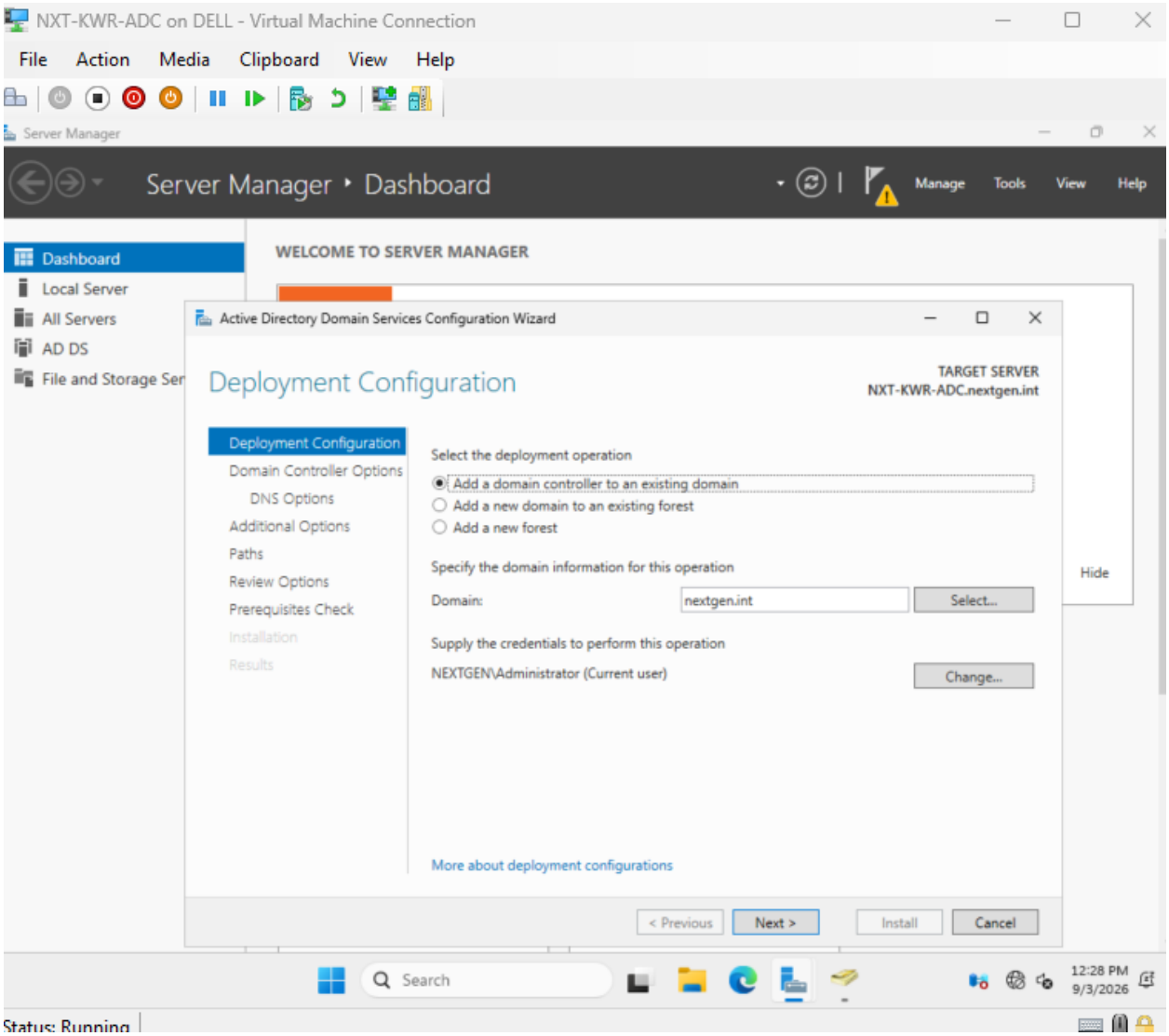
The Two Types You Can Create

When you choose this option, you can create two types of new domains:

- Child Domain: A domain that sits directly under your main domain and shares its ending name.
    - Example: If your main domain is nextgen.int, a child domain would be sales.nextgen.int or uk.nextgen.int.
- Tree Domain: A domain inside the same forest that has a completely different ending name.
    - Example: If your company buys another business and you want to name it brandtwo.com but keep it managed under the same IT team.

Okay, let's do that — now choose "Add a domain controller to an existing domain" and click Next.

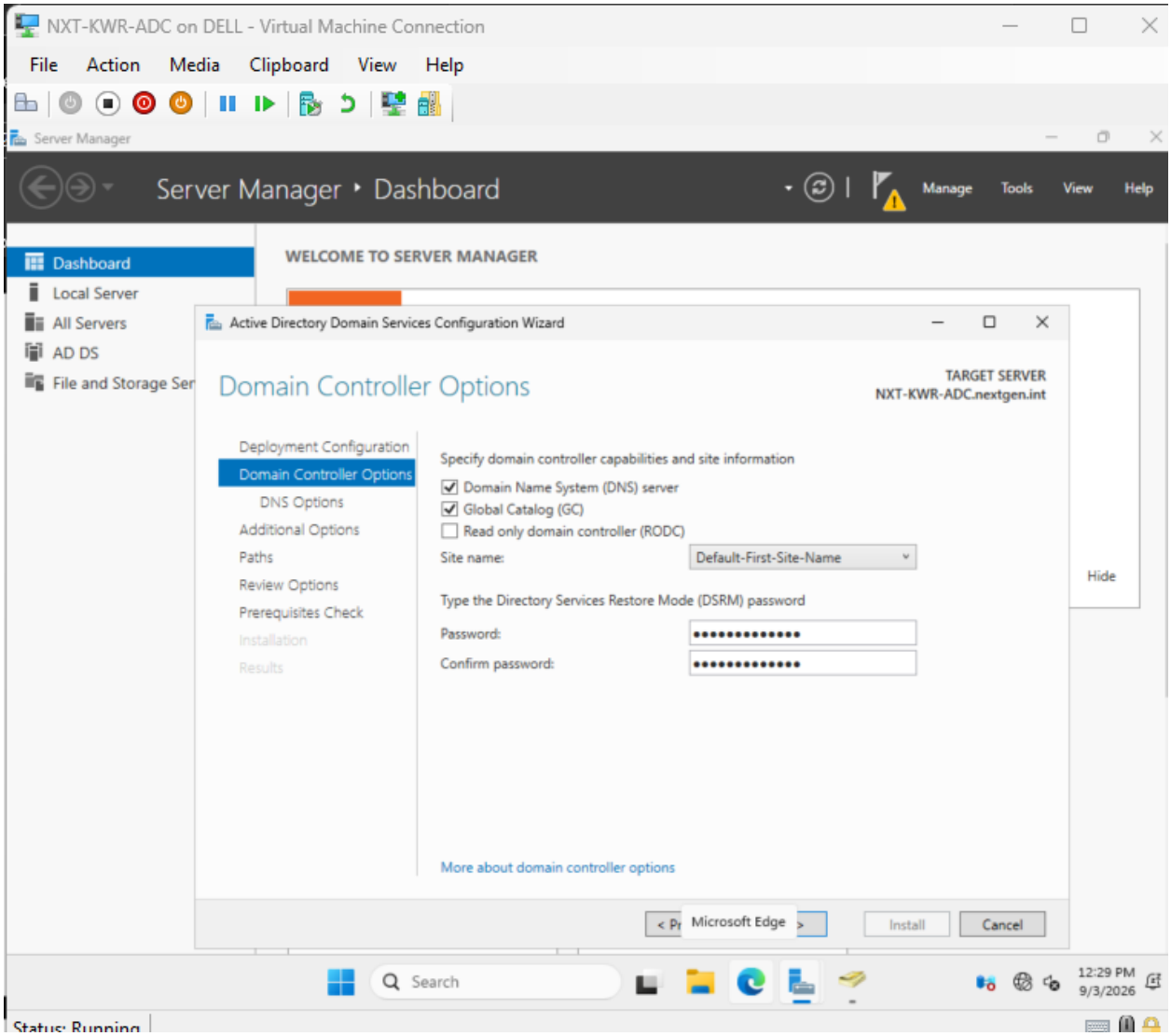
(The domain will automatically fill in because we already joined this VM as a domain member in the AD server.)



Here we use the site name to connect the ADC to a different site. Example: connect the ADC to a DR (Disaster Recovery) site.

In AD we can maintain sites. Click Active Directory Sites and Services and here we can create sites. If we want to add a site, first we create the site in AD and then we can choose that site for the ADC.

> 💡 *For a single-lab setup you can leave the site as "Default-First-Site-Name" — separate sites are mainly used when the DCs are physically in different locations (like different branches or a DR data centre).*



When creating an AD, we cannot choose these options:

Domain Name System (DNS) Server

- In Simple Terms: This option installs a copy of the DNS server role directly onto your new Additional Domain Controller (ADC).
- Why It Matters: Active Directory cannot function without DNS. If the Primary Domain Controller's DNS server goes down or crashes, this secondary DNS server takes over immediately to help client computers find network resources (High Availability).
- For the Lab: This option must always be checked (✓).

2. Global Catalog (GC)

- In Simple Terms: This acts like a central index or a master directory list that holds the main details of every object (every User, Computer, and Group) across the entire Active Directory Forest.